Repetition, similarity, context: Clarifying how feature overlap across experience organizes memory search in free recall

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For my thesis, I propose an investigation into how repetition and similarity across study events influence memory search in free recall tasks, focusing on the role of feature overlap across experiences within the framework of retrieved context theory (RCT) (Howard & Kahana, 2002; Kahana et al., 2022). Implementing variants of the Context Maintenance and Retrieval (CMR) model of memory search, I'll conduct and analyze a series of simulations and behavioral experiments to explore how these factors interact to shape the organization and retrieval of memories. This work will clarify the mechanisms underlying repetition and similarity effects in memory, providing a more nuanced and integrated account of how overlapping features in experience influence memory search.

The first chapter introduces and evaluates an instance-based specification of the CMR model, where individual memory traces are encoded for each event and its associated context. Models based on RCT typically use a connectionist approach, where the relationships between items and their contexts are represented by adjustable weights in a network, influencing how activation flows within it (Howard & Kahana, 2002; L. J. Lohnas et al., 2015; Polyn et al., 2009). In contrast, instance models capture each event as a distinct memory trace within an expandable memory matrix, showing versatility across various memory domains like category learning and semantic memory (Cox & Criss, 2020; Hintzman, 1984, 1988; Jamieson et al., 2018; Shiffrin & Steyvers, 1997; Turner, 2019). Here we bridge these methodologies by introducing an instance-based version of the Context Maintenance and Retrieval (CMR) model, which constructs individual memory traces for each item and its contextual associations, driving recall by activating these traces based on their context (Polyn et al., 2009). Our initial simulations confirm that both the instance-based and traditional connectionist versions of CMR are functionally equivalent and equally explain benchmark phenomena in free recall tasks. For my thesis, I propose to further adapt the model to explain semantic contiguity effects and illustrate how RCT can be integrated with other instance-based models. We discuss the implications of these findings for understanding the dynamics of memory search in free recall tasks and the potential for integrating different modeling traditions in cognitive science (Newell, 1973).

In the second chapter, the focus shifts to repeated experience. To investigate the structure of memory for repeated experiences, this study examines how the lag-contiguity effect – the tendency for items positioned close together in a study list to be retrieved successively during free recall (Kahana, 1996) – manifests for items with multiple study positions. Participants will study lists containing both once- and twice-presented items, with the proximity of each presentation of a repeated item to the beginning or end of the study lists varied systematically to probe the influence of serial position effects (Murdock Jr, 1962) on memory trace accessibility. Using a modified lag-conditional response probability (lag-CRP) analysis, we will compare transition probabilities from repeated items to items studied near either their first or second presentation. Preliminary analyses across two archival datasets suggest that, even after accounting for serial position effects, the temporal contiguity effect is more pronounced relative to the first presentation of a repeated item than to the second. Initial results also indicate that the influential Context Maintenance and Retrieval (CMR) model (Polyn et al., 2009) may not accurately predict these differences, pointing to potential limitations in its current framework (L. Lohnas & Kahana, 2014). Further data collection, follow-up analyses, and proposed modeling aim to clarify these findings, enhancing our understanding of memory search and refining theoretical models of memory for repeated experiences.

The final chapter investigates whether the mechanisms underlying repetition and similarity effects in free recall are fundamentally distinct or reflect a continuum over a single underlying process. Despite the intuitive appeal of the continuum hypothesis, an apparent contradiction arises from the contrasting effects of spacing on memory performance: while repeated items benefit from spaced presentation (Cepeda et al., 2006), semantically similar items often show enhanced recall with massed presentation (Borges & Mangler, 1972; Glanzer, 1969; Puff, 1966). Participants will study lists that interleave unrelated items with either twice-presented items or pairs of highly semantically similar items. The design also varies both the spacing between presentations of key items and the proximity of presentations to the beginning or end of lists, enabling investigation of how spacing and serial position effects manifest for different types of item relationships. Applying a retrieved context framework to these data, I will extend

traditional measures of repetition effects to study non-repeating but similar word pairs. Analyses will interrogate whether encoding of similar or repeated items triggers the reinstatement of similar context, whether similarity or repetition can modulate the benefit of spaced over massed study for memory performance, and whether these factors similarly interact with serial position to structure response organization. Results will support formal comparison between models that variously treat semantic and repetition effects as engaging distinct or overlapping processes, informing the development of an integrated account of how item and contextual relationships shape memory search in free recall.

This work is planned as part of a broader trajectory of research aimed at developing a more integrated understanding of how relationships between study events in a sequence even beyond basic feature overlap shape the structure of memory. While this thesis focuses on free recall for word lists, parallel and future work will explore how these principles extend to other domains of memory – particularly narrative comprehension where relationships between expressions are fundamental to constructing a coherent mental representation of story content (McNamara & Magliano, 2009). By testing and refining the CMR model's integrated account of repetition and similarity effects in its typical domain of research, this research will provide a foundation for understanding how relationships between study events shape memory across diverse conditions and tasks.

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An instance-based retrieved-context theory of memory search in free recall

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Retrieved-context theory (RCT) is a prominent framework used to understand the organization of responses in the free recall task (Kahana et al., 2022). Models based on RCT typically use a connectionist approach, where the relationships between items and their contexts are represented by adjustable weights in a network, influencing how activation flows within it (Howard & Kahana, 2002; L. J. Lohnas et al., 2015; Polyn et al., 2009). In contrast, instance models capture each event as a distinct memory trace within an expandable memory matrix, showing versatility across various memory domains like category learning and semantic memory (Cox & Criss, 2020; Hintzman, 1984, 1988; Jamieson et al., 2018; Shiffrin & Steyvers, 1997; Turner, 2019). Here we bridge these methodologies by introducing an instance-based version of the Context Maintenance and Retrieval (CMR) model, which constructs individual memory traces for each item and its contextual associations, driving recall by activating these traces based on their context (Polyn et al., 2009). Our initial simulations confirm that both the instance-based and traditional connectionist versions of CMR are functionally equivalent and equally explain benchmark phenomena in free recall tasks. For my thesis, I propose to further adapt the model to explain semantic contiguity effects and illustrate how RCT can be integrated with other instance-based models. We discuss the implications of these findings for understanding the dynamics of memory search in free recall tasks and the potential for integrating different modeling traditions in cognitive science (Newell, 1973).

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Introduction

Computational models of the free recall task attempt to define a set of cognitive representations and processes that support human memory search. Retrieved-context models have been especially effective at explaining many aspects of free recall task performance (Howard & Kahana, 2002; Morton & Polyn, 2016; Polyn et al., 2009). According to these models, as we learn new items, we also maintain an internal representation of temporal context. This representation keeps updating to include a recency-weighted summary of past experiences. As we study each item, its details get linked in memory with the current state of temporal context. Later, current temporal context guides memory search by prioritizing items that are most associated with the state of temporal context at that time. After recalling an item, the context updates based on that item's time-related details. This mechanism allows the model to predict durable patterns in response sequences, such as the tendency to recall items in the order they were learned or to remember items from the beginning and end of a list better (Kahana, 1996; Murdock Jr, 1962).

Nearly all retrieved-context models are implemented as connectionist networks (Howard & Kahana, 2002; L. J. Lohnas et al., 2015; Polyn et al., 2009). Associations between items and contexts are stored in a set of adjustable weights that control the flow of activation in the network. The weight matrix has a fixed size which doesn't change with learning events. Instead, scalar values are adjusted to reflect the strength of association between item and contextual features. Retrieval in these models involves activating input features that then propagate through the network, activating output features based on how strongly connected they are to input features (Hebb, 1949).

A major alternative to this approach is the multi-trace or instance-based framework. Instance models posit

that each experience is encoded as a distinct memory trace, stored in separate rows of a memory matrix that grows with each new experience (Hintzman, 1984). At retrieval, traces are activated based on their similarity to a probe. Traces are then aggregated in a weighted mean such that the most similar traces are reflected the most strongly in the response. Models based on this framework are influential in a broad range of cognitive domains such as category learning (Nosofsky & Zaki, 2002; Stanton et al., 2002; Turner, 2019) and other episodic and semantic memory tasks (Cox & Criss, 2020; Hintzman, 1988; Jamieson et al., 2018; Shiffrin & Steyvers, 1997).

Research comparing multi-trace and composite memory theories has provided diverse explanations for how we perform memory tasks, such as recognition, semantic memory, and category learning. Researchers in these areas frequently attribute differences in model performance to the distinct ways in which these approaches access associations (Jamieson et al., 2018; e.g., Nosofsky & Zaki, 2002). Whereas instance-based models can selectively retrieve records of specific experiences from memory, composite models like connectionist networks aggregate over memory traces during learning and are less able to selectively activate individual traces. Yet, other studies have shown that these theories share many similarities across these areas (Anderson, 1995; Kelly et al., 2017; Ramsauer et al., 2020; e.g., Turner, 2019). Despite these exchanges, comparatively little work has been done to directly compare the implications of these approaches for the dynamics of memory search in free recall. This lack of direct comparison creates a gap in our understanding of how these theories interact. This gap hampers effective comparisons between models and hinders the development of integrative computational models of memory that incorporate elements from different modeling traditions.

To bridge this gap, we adapted the connectionist Context Maintenance and Retrieval (CMR) model into an instance-based format (Polyn et al., 2009). This adaptation allows us to merge the connectionist and instance modeling traditions within a single framework, helping us to clearly see how each approach affects memory search dynamics in free recall tasks. Whereas the original CMR model (ConnectionistCMR) uses a pair of linear associative memories to encode associations between items and contexts, our instance-based CMR model (InstanceCMR) creates a separate memory trace for each experience, combining the features of the current state of temporal context with the features of the item studied.

In our initial simulations, we demonstrate the functional equivalence between Instance CMR and Connectionist CMR in their ability to explain benchmark phenomena in free recall tasks. We analyzed the memory structures' latent connectivity patterns to understand how the models equivalently encode and retrieve item-context associations to account for these phenomena. We also fit and formally compared both models to three free recall datasets that varied in list length and patterns of item repetition. Our findings showed that both models are equally effective at explaining well-known phenomena in free recall tasks. Next, to show the flexibility and portability of the instance-based framework, we we propose to extend Instance CMR to account for semantic contiguity effects equivalently to a pre-existing semantic extension of the CMR model (Morton & Polyn, 2016). Our analyses of the modeling problem and the results of our simulations clarify how Instance CMR can be combined to interact with other instance-based models and how it can be adapted to account for the implementational details of CMR using a single instance-based memory that concatenates temporal context, semantic, and item features. In our discussion, we examine the conditions under which Instance CMR and Connectionist CMR might behave differently. In particular, we focused on the unique strength of multi-trace models in selectively retrieving specific memory instances while ignoring others. While simulations confirm that these unique properties can influence model behavior, our results highlight the ambiguity of the distinction between instance and composite memory models in the context of free recall tasks. Together, these outcomes suggest that the core principles of retrieved-context theory, as used in CMR, are highly adaptable and can be implemented across various cognitive modeling frameworks.

Connectionist and Instance Memory Models

In retrieved-context models of memory search, memory representations encode associations between items and their contexts to support the retrieval of these associations using retrieval cues. In this section, we explore and compare two main approaches for implementing these principles: connectionist models and instance models. For our exploration of instance theory, we focus on the MINERVA-2 architecture, a well-established example of instance-based modeling (Hintzman, 1984, 1986, 1988). In MINERVA-2, each

learning experience is captured as a discrete memory trace, which is a vector of feature values. All these vectors are the same length and are organized into a stack, making it easy to access and retrieve them individually:

$$M = \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \vdots \\ \mathbf{m}_n \end{bmatrix} \tag{0.1}$$

The model's memory is probed by a cue vector *cue*, which is compared to each memory trace m using a dot product similarity function, S. Compared to the cosine similarity (another popular similarity mechanism for instance models), dot product is a useful similarity measure here because it does not normalize the vectors, allowing differences in learning rates to be directly reflected in trace magnitudes, simplifying model structure. This approach measures the product of the cue and memory trace vectors, providing a straightforward measure of their similarity:

$$S(cue, m_i) = cue \cdot m_i \tag{0.2}$$

An important parameter, termed choice selectivity, influences how the activation A_i from each trace diminishes based on its similarity, elevating the dot product to the power τ:

$$A_i(cue, m_i) = S(cue, m_i)^\tau \tag{0.3}$$

At a neutral value (1), activation is a linear function of similarity. When choice selectivity is especially low (for example, 0), the model's response is broad, activating many traces almost equally. Conversely, a high choice selectivity value sharpens the model's response, significantly activating only the traces that are

most similar to the cue. The ability to precisely target relevant traces in response to a probe is the key feature of instance models.

The model's output, often visualized as an "echo," represents the sum of all memory traces weighted by their activation:

$$E = \sum_i A(cue, m_i) \tag{0.4}$$

This metaphor likens the model's output to a sound wave bouncing back to its source, reflecting the selective recall process of the memory system.

The Context Maintenance and Retrieval (CMR) model uses a type of memory called a linear associative memory, which follows a Hebbian learning rule (Polyn et al., 2009). Unlike instance models that store each memory as a separate trace, basic connectionist models like this one capture associations between items and context features in a weight matrix. This matrix, where n represents item features and m context features, looks like this:

$$W = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1m} \\ w_{21} & w_{22} & \cdots & w_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1} & w_{n2} & \cdots & w_{nm} \end{bmatrix} \tag{0.5}$$

During learning, the model updates these weights using the Hebbian rule, which changes the weights based on the interaction between input and output vectors. Each update is calculated by multiplying a learning rate η with the input feature x_i and the output feature y_j:

$$\Delta w_{ij} = \eta x_i y_j \tag{0.6}$$

In connectionist models, learning involves integrating all the associations directly into the weight matrix. The model's response to a cue is a vector of activations, calculated as the dot product of the cue vector and the weight matrix, adjusted by a scaling parameter τ:

$$A = (cue \cdot W)^{\tau} \tag{0.7}$$

This scaling parameter τ modifies how much each feature influences the overall response, allowing the model to emphasize certain features more than others based on their relevance to the cue. However, this mechanism adjusts the activation of features collectively, rather than activating individual memory traces as instance models do. Instance models uniquely retrieve specific memory traces during the recall process, based on their similarity to the cue and the selectivity parameter τ.

This difference highlights two fundamental distinctions between these modeling approaches. First, instance models organize and retrieve memory traces individually at the time of recall, whereas connectionist models integrate information during the learning phase, storing it in the weight matrix. Second, instance models can precisely target specific memories for retrieval via trace-based activation scaling. In contrast, connectionist linear associator networks used in CMR generate a response by combining the influences of all features learned, using the weight matrix, and cannot isolate individual memory traces under typical conditions. Because the weight matrix aggregates over memory traces during learning in this way, linear associators are normally classified as composite memories memory models – or prototype models when contrasted with instance models (Anderson, 1995; Jamieson et al., 2018; Nosofsky & Zaki, 2002).

The Prototype Model Distinction

This distinction has important caveats. The defining feature of an instance model is that it can retrieve records of specific experiences (instances) without interference from other experiences. This definition is behavior-based rather than architecture-based, and many different kinds of models can achieve realize

it, even including connectionist models that implement separable memory traces (Anderson, 1995). For example, while simplified connectionist models lack the flexibility of instance models to separately activate individual memory traces, more complex connectionist models can be designed to achieve this functionality (Hintzman, 1988; Kelly et al., 2017) using hidden layer(s) to differentiate between memory traces.

Even simple connectionist models' will depend on the nature of the input-output pairs encoded by the network. If the inputs are highly differentiated—having few or no overlapping features—then the network will operate like an instance model. Instead, whether a linear associator network is a prototype model or an instance model depends on the nature of the input-output pairs that are encoded by the network. If the inputs are highly separated such that they share few or no overlapping features, then the network will behave like an instance model. This separation ensures the model can retrieve output patterns corresponding to a specific training instance without activating interfering instances. Conversely, even when features across inputs substantially overlap, a linear associator may not behave like a prototype model, especially if the number of input-output pairs encoded is relatively small. To act like a prototype model, it's not enough that outputs corresponding to specific training instances are inaccessible. The memory must also exhibit a prototype effect, where a prototype – or, average – over training instances organizes retrieval outcomes instead of individual instances. This distinction underscores how input and output characteristics shape the operational mode of connectionist networks.

Anderson (1995) provides a framework for understanding how prototype- and instance-based retrieval can occur in connectionist models. In this framework, individual inputs linked to the same output g$ are analyzed as a sum of a prototypical component p (the average vector of all inputs) and a unique component d_i – a deviation or 'noise' vector that shows how each input differs from the average.

Here's the mathematical representation:

$$f_i = p + d_i \tag{0.8}$$

The overall connectivity matrix W for n inputs that share the same output is then calculated as:

$$W = \sum_i g f_i^T \tag{0.9}$$

$$W = g \sum_i (p^T + d_i^T) \tag{0.10}$$

$$W = ngp^T + g \sum_i d_i^T \tag{0.11}$$

This analysis reveals that the behavior of the network primarily hinges on the total size of the noise terms $\sum_i d_i$. If the noise is minimal, indicating high similarity among the inputs, the connectivity matrix simplifies to:

$$W = ngp^T \tag{0.12}$$

This represents the matrix of a prototype model, where the output relies on the collective average of the input features. In effect, this model's behavior is akin to having encoded just the prototype pattern p, treating it as the representative of all inputs. Conversely, if the noise component is significant, reflecting high dissimilarity among the inputs, then this noise dominates the connectivity matrix, and the model functions like an instance model. Here, the output depends on how similar an input is to each training input's distinctive features:

$$W = g \sum_i d_i^T \tag{0.13}$$

This analysis underscores that the behavior of a linear associator model is influenced as much by the specifics of the input-output pairs it encodes as by its structural design. Thus, comparisons between connectionist models and those explicitly based on instance theory must carefully consider these elements.

Theoretical Competition Between Instance and Prototype Memory Models

While connectionist models can be designed to exhibit instance-like behavior, their default behavior is to collapse information across multiple experiences into a single best-fitting representation that best represents the item's associations. Models that behave in this way are often referred to as prototype models. In the study of memory, the debate between instance and prototype models has been a recurring theme across various fields, such as recognition, semantic memory, and category learning (Hintzman, 1988; Jamieson et al., 2018; Nosofsky & Zaki, 2002). The key issue in this research is whether memory tasks are better supported by dynamically reactivating and combining specific experiences, or by using generalized representations that summarize common features across multiple experiences. This section reviews examples of how how instance and prototype models can differently predict human behavior in memory tasks.

A clear illustration of these differences is seen in studies of category learning. In typical experiments, participants are presented with examples (exemplars) that belong to one of two categories. Each exemplar is characterized by several binary features (such as having a short or long body, a closed green eye or an open red eye, etc.), with some features strongly linked to category membership and others less so. Participants learn these categories by observing exemplars and then try to classify new ones.

Both instance and prototype models predict how participants might categorize new exemplars based on features typical of a specific category. However, the models diverge when dealing with exemplars that have ambiguous features regarding category membership (Stanton et al., 2002). Prototype models usually assign such novel exemplars to categories based on their similarity to a composite representation of each category. In contrast, instance models configured to retrieve specific instances from memory tend to categorize new exemplars based on the closest matching examples seen during training. Across this and other comparisons (Homa et al., 1981; e.g., Nosofsky & Zaki, 2002), the consistent theme is that models tend to make predictions that are more consistent with the average of experience over a label class, while instance models tend to make predictions that are more consistent with the most similar and relevant training instances.

Instance and prototype models also differ in how they handle word-sense disambiguation, a process where the meaning of a word is clarified based on context (Jamieson et al., 2018). These models use orthographic (textual) cues to trigger semantic (meaning-based) memories. A challenge arises with homonyms—words with the same spelling but different meanings depending on the context. Prototype models have been criticized for typically representing homonyms with a single, overarching semantic representation, despite their distinct meanings (Griffiths et al., 2007).

In a study by Jamieson et al. (2018), the effectiveness of traditional prototype models was compared to a novel instance model of semantic memory, known as ITS, in a task of word-sense disambiguation. While prototype models generally performed well in distinguishing between common meanings of homonyms, ITS excelled at recognizing less common meanings through a dynamic activation of stored language instances. This model's strength was particularly evident when orthographic cues were linked with distinct contextual clusters, with one cluster being much more common than others. Under such conditions, prototype models often skew towards the dominant meanings, neglecting or losing access to lesser-known senses. This discrepancy leads to a mismatch with observed human behavior, where people can often discern less frequent meanings based on nuanced cues.

This pattern underscores a consistent theme across memory studies: instance models are superior when it comes to pinpointing specific memories for retrieval, due to their flexible targeting of individual memory instances. Despite these insights, the implications of these differences for free recall—recalling items from memory without cues—have not been thoroughly investigated. Understanding how these models influence free recall could bridge the gap between modeling approaches and lead to integrated memory models that leverage the strengths of both instance and prototype systems. This study aims to address this gap by comparing instance and prototype models within a unified framework, assessing their impact on the dynamics of memory search in free recall.

Modeling Memory Search in Free Recall

Memory can be thought of as a system that links retrieval cues with information based on past experiences. Memory search involves using this system repeatedly to find information that wasn't initially recalled. Information retrieved using previous cues informs the construction of successive retrieval cues, supporting the search for further unrecalled information. Because this search process involves ongoing interactions with the memory system, the fundamental ideas we have about how memory works significantly affect how the search process unfolds. Here we review the basic principles of retrieved-context theory and the Context Maintenance and Retrieval model, and then describe how these principles can be implemented in both instance and connectionist memory models.

Research into how people recall items has helped us identify basic constraints on theories of the representations and mechanisms that underlie memory search. Three important patterns, in particular, have been emphasized in studies of free recall (Kahana, 2020). One such pattern is the serial position effect, which shows a U-shaped relationship between the position of an item on a study list—its serial position—and its likelihood of being recalled (Murdock Jr, 1962). Researchers have noted that items presented at the beginning of a list and those at the end tend to be recalled more frequently. The enhanced recall of the first few items is known as the primacy effect, while the often stronger recall of the last few items is called the recency effect. Primacy and recency effects demonstrate that the temporal structure of the list affects the memorability of the items within it.

This temporal structure can also be seen in the organization of responses throughout the response sequence, not just for initial and terminal items or recall positions. Free recall task data exhibits a lag-contiguity effect where items studied at nearby serial positions tend to be recalled near one another at the retrieval phase of an experiment (Kahana, 1996). To analyze this phenomenon, researchers calculate the conditional probability of recalling an item based on the distance, or lag, between its position and the position of the last item recalled – the lag-CRP. Studies consistently show that participants are more likely to transition between items that are near each other in the list, demonstrating a preference for recalling items in a temporally contiguous sequence. Additionally, there is a forward bias in recall, where participants are

more likely to recall items that follow, rather than precede, the last item recalled.

Retrieved context models have successfully explained many aspects of performance in free recall tasks (Howard & Kahana, 2002; Morton & Polyn, 2016; Polyn et al., 2009). These models propose that as items are encoded, a representation of the temporal context is also formed and continually updated to reflect a recency-weighted summary of past experiences. Each item's features are linked in memory to the features of the current temporal context. This context then guides memory search by acting as a retrieval cue that prioritizes items most associated with the current context. To account for the recency effect, these models suggest that the temporal context at the end of the study phase closely resembles the context at the beginning of the retrieval phase, enhancing the recall of the most recently studied items. When an item is recalled, it can also trigger the retrieval of the temporal context active when it was first learned, updating the current context and making it more likely that the next item recalled will be one that was studied nearby—this is the basis for the lag-CRP effect. On the other hand, primacy effects are primarily attributed to the greater attention and effort devoted to the first items during the study phase, which makes them more likely to initiate the recall process.

The present work focuses on a specific implementation of the retrieved context theory, the Context Maintenance and Retrieval model (Morton & Polyn, 2016; CMR, Polyn et al., 2009). Briefly, CMR employs a pair of linear associator memories to encode associations between items and states of temporal context. Temporal context here is a vector feature representing a recency-weighted history of items studied, where more recent items have a heavier influence than older ones. The model utilizes an item-to-context memory to link item cues with features of the temporal context, and a context-to-item memory to associate contextual cues with corresponding items. These mechanisms facilitate context reinstatement and context-driven recall. Beyond the standard phenomena previously discussed, CMR has been applied successfully in diverse areas such as financial decision-making (Wachter & Kahana, 2019), emotion regulation (Talmi et al., 2019), and even in studying neural signal dynamics within the medial temporal lobe (Kragel et al., 2015). The model's development has also embraced integrations with theories of semantic knowledge (Morton & Polyn, 2016) and age-related cognitive changes (Healey & Kahana, 2016).

In the forthcoming sections, we introduce an instance-based specification of CMR. This version encodes associations by forming memory traces for each experience that merge features of the item studied with the current state of temporal context. Initially, like its connectionist counterpart, this model version uses dual memory representations. However, we later expand it to a singular memory representation that incorporates temporal context, semantic, and item features, enhancing its capacity to handle semantic contiguity effects along with the standard phenomena.

By exploring an instance-based variation of CMR, we aim to discern whether the model's effectiveness stems from the connectionist approach or the broader principles of retrieved context theory. Additionally, we seek to demonstrate how instance-based models might enhance the explanatory scope of retrieved context theory. Showing that these principles can be adapted across different memory architectures sets the stage for future integrations of retrieved context theory with other successful memory models. The following sections will provide a more detailed comparison of connectionist and instance-based approaches within the context of free recall, further illuminating the framework of retrieved context theory.

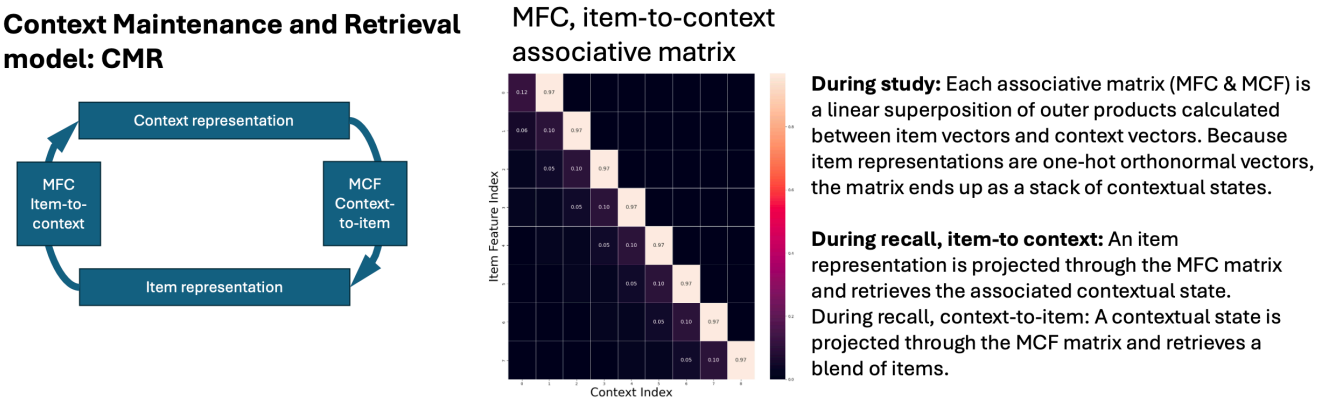
Model Structure

We base our implementations of Connectionist CMR and Instance CMR on the likelihood-based specification provided by Morton & Polyn (2016) and Kragel et al. (2015). The original connectionist version of the Context Maintenance and Retrieval model (ConnectionistCMR) uses this concept by employing two types of memory networks, M^{FC} and M^{CF}. These networks are linear associative networks where the connection weights symbolize the forward and backward relationships between the two layers. Essentially, these weights measure how strongly specific items and contexts are linked based on how often they occur together. In contrast, our initial specification of a multi-trace version of CMR (InstanceCMR) also uses M^{FC} and M^{CF} to represent the relationships between items and contexts. However, rather than blending these associations into a cumulative pattern of weights, each memory in Instance CMR keeps an individual trace for every experience. This means it concatenates the states of F and C at the time of encoding, preserving a distinct record for each event. To fully explain how each model is structured, we first outline the memory systems used by each model to encode and retrieve these item-context associations. We then explore how both models manage these memories to support the process of searching for memories and performing tasks like free recall, where items must be remembered without prompts. This comparison shows not just how the models store information but also how they use it to reconstruct past events during memory tasks.

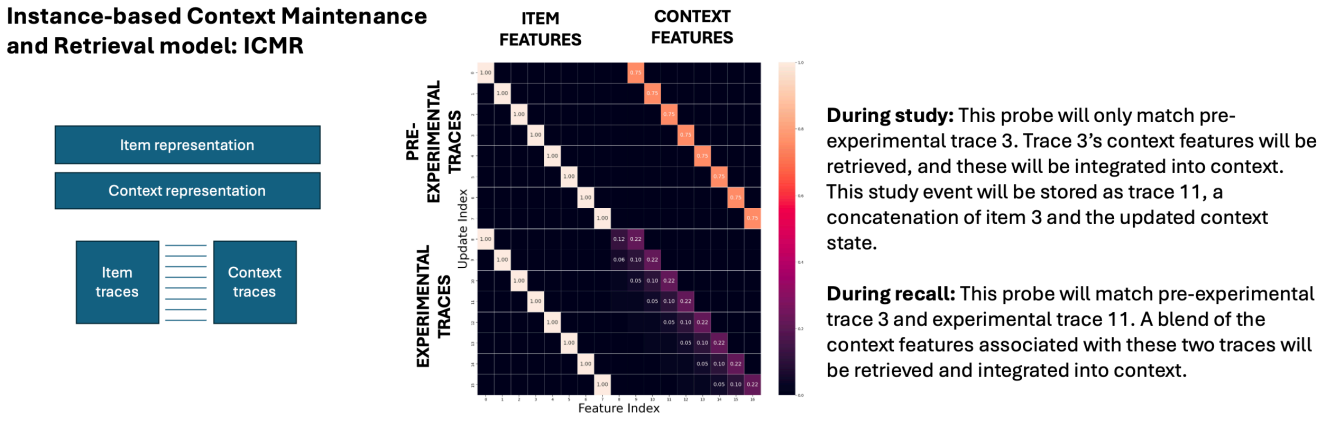
Symbol	Name	Description
C	temporal context	A recency-weighted average of encoded items
F	item features	Current pattern of item feature unit activations

Symbol	Name	Description
M^{FC}	item-to-context memory	encoded feature-to-context associations
M^{CF}	context-to-feature memory	encoded context-to-feature associations
β_{enc}	encoding drift rate	Rate of context drift during item encoding
β_{start}	start drift rate	Amount of start-list context retrieved at start of recall
β_{rec}	recall drift rate	Rate of context drift during recall
α	shared support	Amount of support items initially have for one another
δ	item support	Initial pre-experimental contextual self-associations
γ	learning rate	Amount of experimental context retrieved by a recalled item
ϕ_s	primacy scale	Scaling of primacy gradient on trace activations
ϕ_d	primacy decay	Rate of decay of primacy gradient
θ_s	stop probability scale	Scaling of the stop probability over output position
θ_r	stop probability growth	Rate of increase in stop probability over output position
τ_c	choice sensitivity	Exponential weighting of item supports during the recall competition
τ_t	trace sensitivity	Exponential weighting of trace activations during probes of M^{CF}

Table 1: Parameters and structures specifying CMR.



(a) Connectionist CMR



(b) Instance CMR

Figure 1: **Aggregation of information across memory traces happens at different time points in the two versions.** The connectionist model aggregates over traces during encoding, because there is a fixed-size set of associative weights. The instance model creates separate traces for each event, so aggregation happens during retrieval.

Connectionist Memory Representations

In Connectionist CMR, the memories M^{FC} and M^{CF} are are structured as two weight matrices, where each element is linked to the features of items and contexts respectively. For example, M_{ij}^{FC} represents the association between the ith feature of an item and the jth context in feature-to-context memory Figure 1a.

Before any experiment begins, we set up M^{FC} to summarize existing associations between relevant item features and potential contextual states. This initial configuration is guided by:

$$M_{pre(ij)}^{FC} = \begin{cases} 1-\gamma, \text{ if } i = j \\ 0, \text{ if } i \neq j \end{cases} \tag{0.1}$$

This setup connects each item feature unit on F to a unique context unit on C using a parameter γ, which dictates how much pre-experimental associations influence the model's behavior.

Similarly, the associations from context to feature, managed by M^{CF}, are initially set as follows:

$$M_{pre(ij)}^{CF} = \begin{cases} \delta, \text{ if } i = j \\ \alpha, \text{ if } i \neq j \end{cases} \tag{0.2}$$

Here, δ adjusts the impact of pre-experimental context-to-feature associations compared to those acquired during the experiment. The α parameter establishes a uniform semantic support across items, supporting each other in the recall competition.

The weights between any two units i and j in M^{FC} or M^{CF} are updated based on the Hebbian learning rule, which relies on the activity levels in the item and context layers. Specifically, in M^{FC}, the learning rate of these updates is governed by the γ parameter again:

$$\Delta M_{ij}^{FC} = \gamma \cdot F_i \cdot C_j \tag{0.3}$$

This formula adjusts the strength of the connection based on the simultaneous activity of item features F_i and context C_j, with the γ parameter controlling the learning rate.

In M^{CF}, a function ϕ enforces a primacy effect, scaling the amount of learning based on the serial position of the studied item according to:

$$\phi_i = \phi_s e^{-\phi_d(i-1)} + 1 \tag{0.4}$$

where ϕ_s and ϕ_d are parameters that control the strength and decay of the primacy effect, respectively.

The weight between units i and j in M^{CF} is updated according to:

$$\Delta M_{ij}^{CF} = \phi_i C_i F_j \tag{0.5}$$

To retrieve contextual associations for a given item, the model computes the dot product of the item's feature vector f with M^{FC}.

$$\hat{C} = M^{FC} f \tag{0.6}$$

Similarly, to retrieve item associations for a given state of context, the model computes the dot product of the context's feature vector c with M^{CF}.

$$\hat{F} = M^{CF} c \tag{0.7}$$

These details describe how the connectionist CMR model encodes and retrieves associations between items and contexts in memory and enforces a primacy effect during learning to support memory search in the free recall task.

Instance-Based Memory Representations

In contrast to the traditional connectionist implementation of CMR, our new instance-based specification of CMR (Instance CMR) represents M^{FC} and M^{CF} as instance-based memories. That is, they store a distinct trace for every experience, keeping each event's unique details separate Figure 1b. Initially, like the traditional model, Instance CMR uses two different memory matrices to capture the associations between items and contexts (M^{FC} and M^{CF}). This differentiation helps to account for differences in how M^{FC} and M^{CF} represent pre-experimental associations between item and context features. Later on, we will explore how merging these into a single matrix that also incorporates semantic features can streamline this process.

In the current specification, M^{FC} and M^{CF} are represented as two multi-trace memories. Each trace is represented as a vector that concatenates feature vectors representing the states of F and C. Each trace combines feature vectors from the item layer (F) and the context layer (C), forming a complete snapshot of the encoding event. These vectors are arranged in matrices where each row corresponds to a different experience, and each column aligns with a particular feature or context unit.

To ensure a clear comparison between the traditional composite and the new multi-trace architectures, we initialize M^{FC} and M^{CF} in Instance CMR to mirror the pre-experimental associations found in Connectionist CMR. For M^{FC}, this means setting up the memory with a trace for each feature in F that connects to a specific context unit in C. Initially, all features in these traces are set to 0, except for the ones directly linked, which are set to 1 and $1 - \gamma$ respectively.

Similarly, for M^{CF}, each context unit in C starts connected primarily to one feature in F but maintains a weaker link to all other features. Here, context units in the traces are set to 0 except for the primary connection, which is marked as 1. Item features in these traces are mostly set to α, except for the directly associated feature, which is initialized at δ. This setup provides a structured foundation from which both memories can evolve as they encode new experiences and interactions during the experimental phase.

In the Instance CMR model, instead of updating a single weight between units for each new experience,

we add a new trace that captures the state of both the item features F and the context features C at the time of encoding. Differences in trace strengths ("learning rates" in connectionist memories) are taken into account at retrieval rather than encoding. So in M^{FC} and M^{CF} matrices, each experience simply contributes a new trace that concatenates the item and context feature vectors:

$$M^{FC}_{exp_i} = [f_i, c_i] \tag{0.8}$$

$$M^{CF}_{exp_i} = [c_i, f_i] \tag{0.9}$$

Retrieving associations from these multi-trace memories involves a couple of steps. First, the probe vectors for M^{FC} consist only of F features, formatted as $[f_{probe}, 0]$, and for M^{CF}, they consist only of context features, similarly formatted. These vectors are concatenated with a zero vector to align with the dimensionality of the traces in M^{FC} and M^{CF} . The retrieval process then calculates a weighted sum across both pre-experimental and experimental traces. The weight of each trace in this sum is determined by the dot product between the probe vector and the trace, as well as the trace's learning strength, which corresponds to the learning rates in the connectionist models.

In the M^{FC} matrix, the parameter γ scales the association between the item features and the contextual features within these new traces; it's set to 1 for pre-experimental traces:

$$[\hat{C}...] = \sum_{i=1}^{n} \gamma M^{FC}_i [f, 0] \tag{0.10}$$

Similarly, in the M^{CF} matrix, the parameter ϕ_i scales the association of contextual features with the item features for each new experience, where i is the index of an encoded trace and ϕ_i is held to 1 for pre-experimental traces:

$$[\hat{F}...] = \sum_{i=1}^{n} (\phi_i M_i^{CF}[c, 0]]^{\tau_t} \tag{0.11}$$

For experimental traces, the equation defining ϕ_i is the same as described above for Connectionist CMR. However, the parameter τ_t is unique to the instance model. In M^{CF} (but not in M^{FC}), the parameter τ_t is used to exponentiate the dot product between the probe vector and each trace. This method enhances the difference in activation between traces, allowing the most relevant or best-supported traces to stand out more prominently during the retrieval process. This feature, characteristic of instance models, accentuates the selectivity of memory retrieval at the trace level. Since connectionist memories do not support trace-level activation modulation in the same way, when enforcing equivalence to Connectionist CMR, τ_t is set to 1 to effectively disable this mechanism and ensure that the model's behavior is consistent with the connectionist specification.

Memory Dynamics During the Free Recall Task

Remaining aspects of the model are shared between Connectionist and Instance CMR.

Encoding Phase

When an item i is presented for study, its feature representation f_i activates within the item layer F, and its contextual associations are retrieved from M^{FC}, regardless of whether it's the composite or multi-trace version. This retrieval modifies the current state of the context layer C. The input to the context from this retrieval is defined by:

$$c_i^{IN} = \hat{C} \tag{0.12}$$

Here, $\hat{C}$ represents the context vector retrieved from M^{FC}. This vector is then normalized to maintain a consistent scale, specifically so its length equals 1.

The context state is updated based on this input:

$$c_i = \rho_i c_{i-1} + \beta_{enc} c_i^{IN} \tag{0.13}$$

In this formula, β_{enc} controls how much the context changes–or drifts–with each new item presented, reflecting the influence of the new contextual input. The parameter ρ is used to ensure the length of c_i remains normalized at 1. It is calculated as follows:

$$\rho_i = \sqrt{1 + \beta^2 \left[\left(c_{i-1} \cdot c_i^{IN} \right)^2 - 1 \right]} - \beta \left(c_{i-1} \cdot c_i^{IN} \right) \tag{0.14}$$

This calculation adjusts ρ_i to balance the contribution of the previous context state c_{i-1} and the new input c_i^{IN}, ensuring the overall context vector doesn't grow too large or become too small.

At this point, the model captures and solidifies the association between the current states of the context and item features in both M^{FC} and M^{CF}. This dynamic ensures that each new experience slightly alters the context, which in turn influences how subsequent items are encoded and later retrieved.

Retrieval Phase

To account for the primacy effect in the model, we posit that between the encoding and retrieval phases, the content of the context layer C drifts back towards its state prior to the experiment. We model this drift and set the initial state of context at the start of retrieval as follows, with ρ calculated as previously specified:

$$c_{start} = \rho_{N+1} c_N + \beta_{start} c_0 \tag{0.15}$$

During each recall attempt, this current state of context serves as the cue to M^{CF}, attempting to retrieve a studied item and producing an activation vector A on the item layer F:

$$A = \hat{F} \tag{0.16}$$

To ensure all items have a chance of being recalled, each is given a baseline activation of 10^{-7}. The likelihood of ending the recall session, or the probability of stopping the recall without retrieving any more items, depends on the output position j and is given by:

$$P(stop, j) = \theta_s e^{j\theta_r} \tag{0.17}$$

Here, θ_s and θ_r govern the initial stopping probability and the rate at which this probability increases, respectively, which is modeled as an exponential function. If recall continues, the probability $P(i)$ of recalling a specific item i primarily depends on its relative activation strength:

$$P(i) = (1 - P(stop))\frac{A_i^{\tau_c}}{\sum_k^N A_k^{\tau_c}} \tag{0.18}$$

The parameter τ_c enhances the contrast in activation strengths between items: a high τ_c value amplifies differences, making the most activated item even more likely to be recalled, whereas a lower τ_c value diminishes these differences, equalizing recall probabilities across items with varying activations. This mechanism can be implemented for either instance-based or connectionist memories, even on top of pre-existing trace-based activation scaling.

If an item is recalled, it is reactivated on F, and its contextual associations are retrieved to update the context. This new context input is:

$$c_i^{IN} = \hat{C} \tag{0.19}$$

Context is then updated based on this input using β_{rec} (used during retrieval rather than encoding) to adjust for the current phase of the task, setting the stage for the next recall attempt. This iterative process continues until no further items are recalled, effectively concluding the recall session.

Simulation 1: Functional Equivalence Between Connectionist and Instance CMR

We have established two versions of the Context Maintenance and Retrieval model: Connectionist CMR and Instance CMR that both instantiate the principles of retrieved context theory to explain memory search in free recall tasks. But are the models truly equivalent, or are there subtle differences in their behavior that could influence their performance in different scenarios? While we hold the trace-based activation scaling parameter τ_t constant at 1, the answer to this question is yes. The reason why is that the two models use the same feature and contextual representations, and perform the same mathematical operations upon them to encode and retrieve associations: The pre-scaled activation of an output feature in either model is determined by the dot product similarity between probe vectors and output features' associated input features, weighted by each trace's respective learning rate. Here we present simulation analyses of the functional equivalence between Connectionist CMR and Instance CMR, demonstrating how both models encode and retrieve the same associations despite their structural differences. Establishing the equivalence of Instance CMR and Connectionist CMR under these conditions allows us to explore how trace-based activation scaling might enhance the model's performance in free recall tasks.

To clarify the roles of memory structures in the simulation of CMR and underscore how these models encode and retrieve the same associations despite their structural differences, we provide a functional analysis of the connectivity structure of M^{FC} and M^{CF} in both models that measures associations between item and context features irrespective of how these associations are represented within a memory architecture. To set up this analysis, we use parameters fitted to the subset of the PEERS dataset as reported by (Healey

& Kahana, 2014) as described in the next section, but with τ_t set to 1. Then we simulate encoding of an arbitrary study list of size 8 with each model to construct the memory structures M^{FC} and M^{CF} that would organize retrieval in a free recall task.

To compare the connectivity matrices of Connectionist CMR and Instance CMR, we individually activate each input unit and measure the resulting activation in the output layer for each possible input unit, activating one unit at a time. The resulting vector represents the connectivity between the activated input unit and all output units. Stacking these vectors together, we obtain the full latent connectivity matrix for each memory structure. For connectionist M^{FC} and M^{CF}, this extracted structure is identical to the weight matrices used in the model. But for instance-based M^{FC} and M^{CF}, this structure is effectively a conversion of the multi-trace memory into a composite memory, where each row summarizes associations between an input unit and all output units across all study events.

Along with producing latent M^{FC} and M^{CF} connectivity matrices, we also analyze their combined influence on the dynamics of memory search in free recall tasks by extracting a latent M^{FF} matrix that passes F input vectors through M^{FC} and then passes the resulting context vectors through M^{CF} to measure the strength of associations between item features across all study events. This matrix reflects the interplay between M^{FC} and M^{CF} in guiding memory search – how recalling an item will bias the retrieval of subsequent items based on the context reinstated by the recalled item via M^{FC} and then used as a retrieval cue via M^{CF}. In Figure 1, we illustrate that across Connectionist CMR and Instance CMR, these three latent connectivity structures are identical, despite the differences in how they store and retrieve associations.

To further illustrate how the memory structures of Connectionist CMR and Instance CMR support the temporal organization of memory search, we analyze extracted latent connectivity matrices from both models in a lag-connectivity analysis. In latent M^{FC} representations, this analysis measures the connection strength between each considered item's F representations with each other item's pre-experimentally associated context units C as a function of the other items' serial lag from the considered item. The connection weight between F_i and C_j for all items i and j is calculated for each lag $j - i$ and plotted as a function of this lag. For example, if item i is considered, the lag-connectivity analysis measures the strength of the

connection between F_i and C_j as a function of the lag $j - i$. A converse analysis is performed for M^{CF}, measuring the connection strength between each considered context unit C_i with each other item's pre-experimentally associated feature units F as a function of the other items' serial lag from the considered item.

Finally, to measure the combined influence of M^{FC} and M^{CF} on memory search, we perform a lag-connectivity analysis on the latent M^{FF} matrix, measuring the connection strength between each considered item's F representations with each other item's F representations as a function of the other items' serial lag from the considered item. This analysis, visualized in Figure 2, provides a visual representation of how each of M^{FC} and M^{CF} supports the lag-contiguity effect in free recall tasks, with M^{FC} primarily supporting transitions to backward neighbors of the last recalled item and M^{CF} supporting transitions to forward neighbors. The analysis applied to M^{FF} demonstrates how the combined influence of M^{FC} and M^{CF} instantiates a bidirectional but forward-biased lag-connectivity structure that supports the asymetrical lag-contiguity effect in free recall tasks. These analyses together demonstrate that the latent connectivity structures of Connectionist CMR and Instance CMR are functionally equivalent, supporting the same dynamics of memory search in free recall tasks.

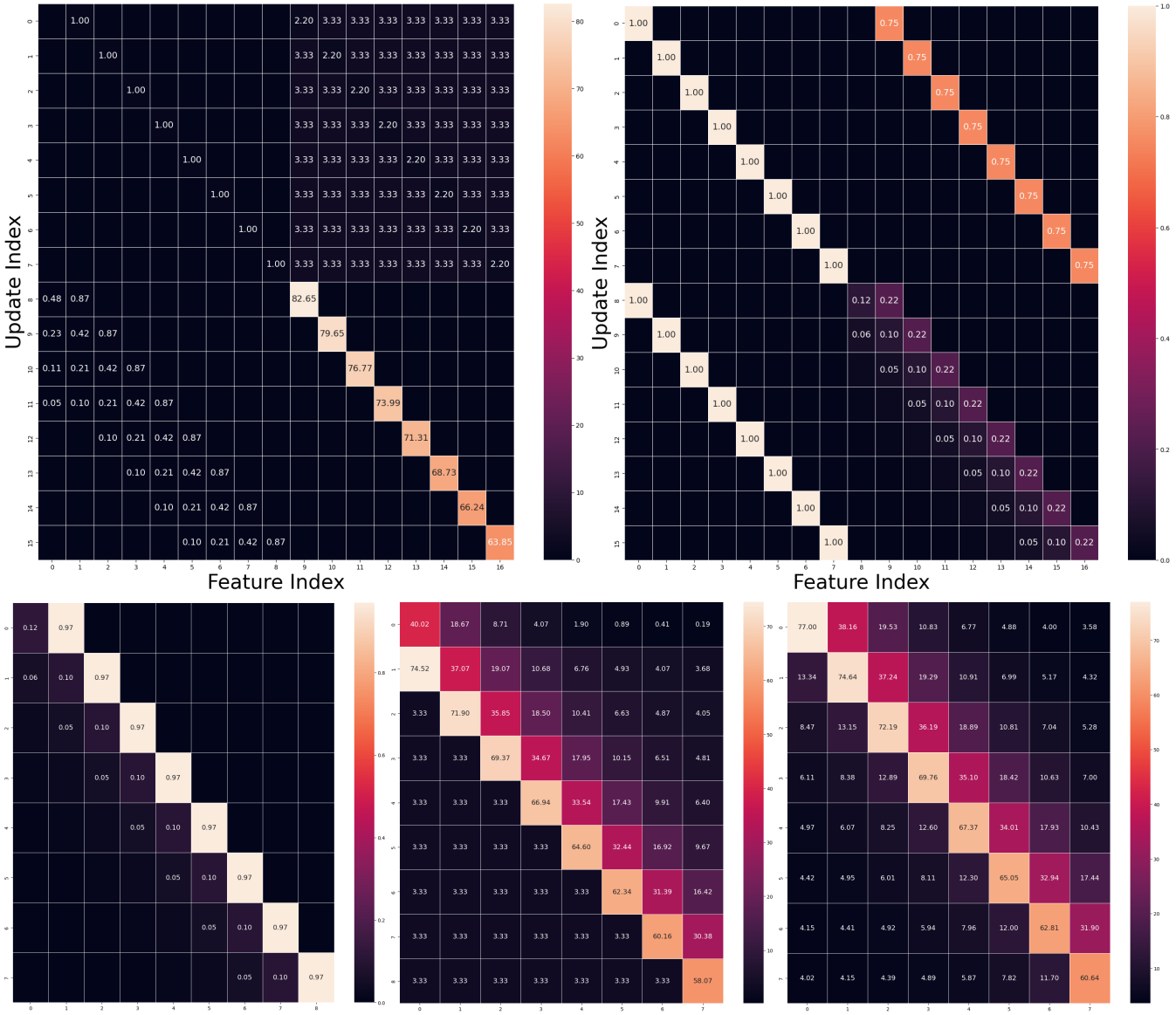


Figure 1: **Comparison of latent connectivity matrices between Connectionist CMR and Instance CMR**. Top Row: Instance CMR's M^{FC} (left), and M^{CF} (right) matrices. Each row represents a memory trace associating item and context features. Bottom Row: Extracted latent connectivity matrices from either CMR variant's M^{FC} (left), and M^{CF} (middle) memories as well as the combined influence of M^{FC} and M^{CF} on memory search in the M^{FF} matrix (right). These are identical between the Instance and Connectionist models and also identical to Connectionist CMR's M^{FC} and M^{CF} weight matrices.

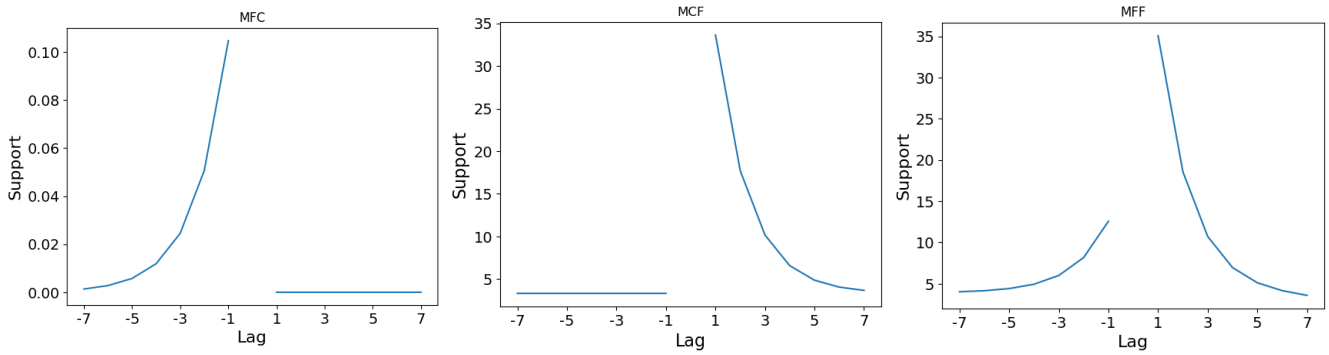


Figure 2: **Lag-connectivity analysis of latent connectivity matrices** from M^{FC} (left), M^{CF} (middle), and M^{FF} (right) in both models. These analyses show how each memory structure supports the lag-contiguity effect in free recall tasks.

Simulation 2: Performance of Connectionist and Instance CMR on Free Recall Datasets

As shown, in our specification, Connectionist and Instance CMR models are functionally equivalent when the trace-based activation scaling parameter τ_t is set to 1. This setup ensures that with identical parameter configurations, both models should yield the same results in free recall tasks. This equivalence is maintained by using consistent association encodings between item and context features, identical learning rates, primacy scaling parameters across experiences, and by disabling the trace-based activation scaling in the instance model. To demonstrate how both models equivalently handle the dynamics of memory search in free recall, we simulate their performance on a selection of free recall datasets previously used to assess the Connectionist CMR, as well as exploring how Instance CMR performs when τ_t is varied during model fitting. This will help determine the potential benefits of trace-based activation scaling in these tasks.

Datasets

For our initial comparisons, we utilize a segment of the PEERS dataset as reported by (Healey & Kahana, 2014). In this subset, each of 126 participants, aged 17 to 30, encountered 112 trials where they studied lists of 16 unique words, followed by immediate recall. The words in each trial were unique and selected for low similarity, randomly drawn from the Toronto Word Pool (Friendly et al., 1982), which includes high-frequency nouns, adjectives, and verbs.

One significant aspect of the CMR model is its ability to demonstrate the relative scale-invariance of serial position effects, regardless of list length. As documented by (Murdock Jr, 1962), variations in list length did not affect the primacy or the shape of the recency effects observed, though they did influence the overall probability of recalling initially encoded items and the list-list asymptote. To assess whether our multi-trace and composite implementations of CMR can replicate these findings, we utilize a subset of the original behavioral data from (Murdock Jr, 1962) where 15 subjects completed 240 trials with list lengths varying between 20, 30, and 40 words.

Additionally, past research (L. Lohnas & Kahana, 2014) has highlighted CMR's capacity to model various repetition effects observed in free recall experiments. N otably, recall sequences often show a spacing effect, where the likelihood of recalling an item increases with the lag between its repeated presentations in a study list. (L. Lohnas & Kahana, 2014) demonstrated that CMR could account for this through two mechanisms: contextual variability and study-phase retrieval. In the study-phase retrieval, when an item appears multiple times, recalling its prior instances and their contexts enriches the associations made with the current presentation. We test our model implementations against the dataset from (L. Lohnas & Kahana, 2014), where 35 subjects recalled 48 lists over four sessions. Each session included unique words, with controlled semantic relatedness below a threshold of .55 according to WAS (Steyvers et al., 2005), and featured four types of list structures:

1. **Control lists**: Each item appeared only once.
2. **Pure massed lists**: Items appeared twice consecutively (e.g., 1, 1, 2, 2).
3. **Pure spaced lists**: Items appeared twice with variable spacings from 1 to 8 positions.
4. **Mixed lists**: Contained once-presented, massed, and spaced items, with each spacing amount distributed equally.

Through these simulations, we aim to further understand how different memory model architectures influence recall dynamics and whether certain features, like trace-based activation scaling, enhance the model's predictive accuracy and applicability across varied recall scenarios.

Likelihood-based model optimization and comparison

To determine how accurately each model captures the sequence of items recalled in our experiments, we employ a likelihood-based model comparison technique, as suggested by Kragel et al. (2015). This method evaluates model variants based on their precision in predicting the exact order of recalled items. According to this method, repeated items and intrusions (responses naming items not presented in the list) are excluded from participants' recall sequences.

Here's how it works: For each trial, after simulating the encoding of each item as presented in the study list, the model begins by predicting the likelihood of the first item recalled by the participant. Subsequently, the model simulates the retrieval of this item, updates its state, and uses this updated context to predict the next recall event—this could either be the retrieval of another item or the termination of the recall session. This process continues until no further items are recalled. The probability assigned by the model to each recall event, conditional on prior events in the sequence, is recorded, log-transformed, and summed to derive the log-likelihood for the entire sequence. Across multiple trials in a dataset, these sequence log-likelihoods are summed to calculate the total log-likelihood for the dataset under a given model and parameter set. Higher log-likelihoods indicate a model's superior capability to account for the observed data.

Optimizing the parameters for each model to maximize the likelihood of the observed data is achieved using the differential evolution technique (Storn & Price, 1997), implemented in the scipy Python library. This method involves maintaining a population of potential parameter configurations. In each iteration, the algorithm mutates each member of this population by combining them with other members in a stochastic manner. If the new configuration offers an improvement, it replaces the older one; otherwise, it is discarded. This iterative process helps in progressively refining the parameters until they converge on a configuration that maximizes the observed dataset's log-likelihood. We conduct optimization at the individual subject level and compare models by examining the distribution of log-likelihood values across subjects, which helps in identifying the most effective model configuration for explaining recall dynamics.

Summary Statistic Simulation

In our evaluations of model performance, we utilize a set of summary statistics to characterize recall sequences. To calculate these statistics for the models, we first simulate recall sequences based on the specified model mechanics. For each unique study list in our dataset, we simulate 1000 trials. Each trial involves simulating the encoding of each item into memory followed by the stochastic simulation of free recall, which continues according to the calculated probabilities for each recall attempt until the session ends. The summary statistics for each model are then computed across all these simulations.

Our analysis primarily focuses on four well-documented regularities in free recall tasks that significantly influence model evaluation:

1. **Serial Position Effect**: This effect is crucial for understanding how the order of items in a study list affects their likelihood of recall. We calculate and visualize the retrieval rate for each item based on its position in the study list. The rates for items presented early in the list help quantify the primacy effect, while the retrieval rates for later items illustrate the recency effect.
2. **Probability of First Recall**: This statistic measures the rate at which items are retrieved first across recall sequences. This metric helps identify the items that are most likely to initiate the recall process.
3. **Lag-Contiguity Effect**: This phenomenon, where items close together in the list order tend to be recalled in close sequence, is another focal point of our analysis. To assess this, we employ lag-based conditional response probability (lag-CRP) analyses. The "lag" here refers to the distance between two items in the study list. We calculate the probability of a participant or model making a transition to recall an item at a specific lag, given that such a transition is possible. A high degree of temporal contiguity is indicated by frequent transitions to nearby items and infrequent transitions to items further apart.
4. **Spacing Effect**: The spacing effect describes how the intervals between item presentations within a study list influence recall probability. We plot the probability of recalling an item based on its lag

from the last presentation (or whether it is a novel item). This analysis helps us understand how the model's memory dynamics are influenced by the spacing between repeated items.

We plot simulated data against observed data to evaluate how well the models capture these regularities across different datasets.

Results

Healey & Kahana (2014)

	Connectionist CMR	Instance CMR	Trace Scaling CMR	Multi Scaling CMR
fitness	590.11 +/- 16.87	589.14 +/- 16.92	589.04 +/- 16.87	588.32 +/- 16.97
encoding drift rate	0.79 +/- 0.02	0.80 +/- 0.02	0.75 +/- 0.03	0.71 +/- 0.04
start drift rate	0.29 +/- 0.04	0.27 +/- 0.04	0.23 +/- 0.04	0.26 +/- 0.04
recall drift rate	0.85 +/- 0.02	0.85 +/- 0.02	0.89 +/- 0.02	0.88 +/- 0.02
shared support	22.73 +/- 5.26	23.83 +/- 5.51	0.14 +/- 0.07	25.02 +/- 5.08
item support	22.68 +/- 4.65	26.20 +/- 5.16	22.82 +/- 5.19	28.25 +/- 5.30
learning rate	0.49 +/- 0.06	0.49 +/- 0.06	0.33 +/- 0.04	0.51 +/- 0.06

	Connectionist CMR	Instance CMR	Trace Scaling CMR	Multi Scaling CMR
primacy scale	25.59 +/-	30.50 +/-	48.46 +/-	10.24 +/-
	5.55	5.78	5.80	3.53
primacy decay	10.53 +/-	14.09 +/-	10.78 +/-	15.66 +/-
	4.42	4.96	3.75	4.41
stop probability scale	0.01 +/-	0.01 +/-	0.01 +/-	0.01 +/-
	0.00	0.00	0.00	0.00
stop probability growth	0.40 +/-	0.40 +/-	0.40 +/-	0.40 +/-
	0.02	0.02	0.02	0.02
choice sensitivity	38.70 +/-	31.31 +/-	1.00 +/-	48.98 +/-
	6.24	5.55	0.00	6.09
mcf trace sensitivity	1.00 +/-	1.00 +/-	2.87 +/-	2.02 +/-
	0.00	0.00	0.26	0.28

Table 1: Confidence intervals of parameters fit to data from Healey & Kahana (2014), computed across subjects. Column 1: Connectionist CMR follows the specification in Morton & Polyn (2016). Column 2: Instance CMR with τ_t set to 1. Column 3: Instance CMR with τ_t optimized during fitting but τ_c set to 1. Column 4: Instance CMR with both τ_t and τ_c optimized during fitting.

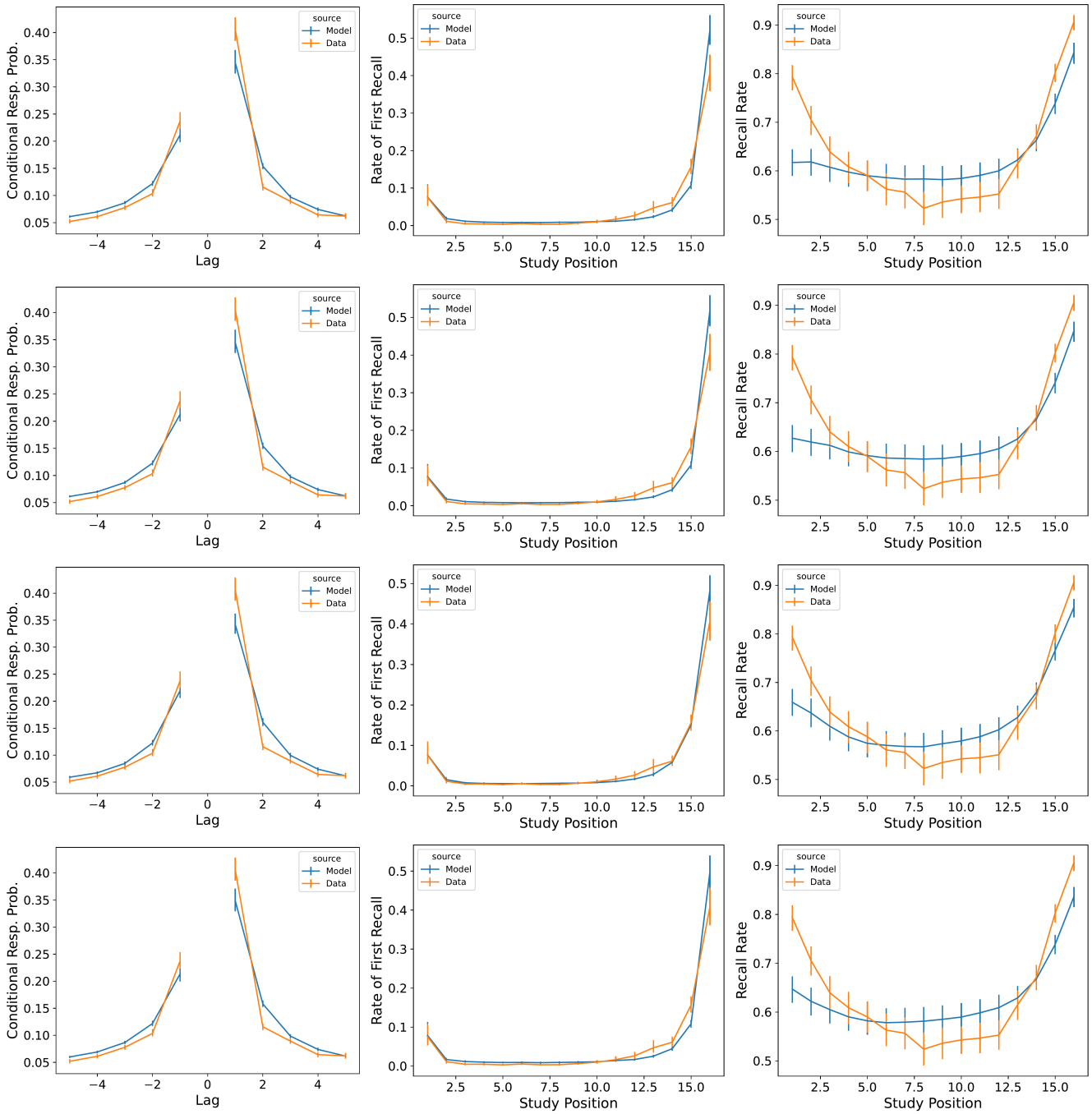


Figure 1: Summary statistic fits to Healey & Kahana (2014). Top: Connectionist CMR. Second Row: Instance CMR, τ_t set to 1. Third Row: Trace Scaling CMR – Instance CMR, τ_c set to 1 and τ_t optimized during fitting. Fourth Row: Multi Scaling CMR – Instance CMR, both τ_t and τ_c optimized during fitting. Left: conditional response probability as a function of lag. Middle: probability of starting recall by serial position. Right: recall probability by serial position.

Murdock (1962)

	Connectionist CMR	Instance CMR	Trace Scaling CMR	Multi Scaling CMR
fitness	5260.73 +/- 290.76	5283.63 +/- 303.93	5287.44 +/- 302.87	5267.38 +/- 304.15
encoding drift rate	0.47 +/- 0.09	0.51 +/- 0.09	0.58 +/- 0.13	0.61 +/- 0.13
start drift rate	0.21 +/- 0.13	0.20 +/- 0.13	0.12 +/- 0.07	0.27 +/- 0.19
recall drift rate	0.87 +/- 0.05	0.84 +/- 0.06	0.90 +/- 0.02	0.93 +/- 0.03
shared support	3.37 +/- 4.11	6.00 +/- 7.87	0.03 +/- 0.01	7.12 +/- 5.62
item support	17.00 +/- 13.71	27.35 +/- 20.16	45.09 +/- 16.45	25.82 +/- 21.40
learning rate	0.14 +/- 0.07	0.11 +/- 0.05	0.17 +/- 0.09	0.13 +/- 0.13
primacy scale	15.93 +/- 14.21	21.36 +/- 17.07	54.97 +/- 13.58	10.24 +/- 8.70
primacy decay	25.26 +/- 21.17	14.94 +/- 14.91	26.49 +/- 19.13	19.80 +/- 17.12
stop probability scale	0.02 +/- 0.00	0.02 +/- 0.00	0.02 +/- 0.00	0.02 +/- 0.00
stop probability growth	0.28 +/- 0.02	0.29 +/- 0.02	0.28 +/- 0.02	0.28 +/- 0.02

	Connectionist CMR	Instance CMR	Trace Scaling CMR	Multi Scaling CMR
choice sensitivity	30.78 +/- 19.05	38.38 +/- 22.18	1.00 +/- 0.00	37.44 +/- 17.25
mcf trace sensitivity	1.00 +/- 0.00	1.00 +/- 0.00	3.24 +/- 1.13	1.13 +/- 0.84

Table 2: Confidence intervals of parameters fit to data from Murdock Jr (1962), computed across subjects. Column 1: Connectionist CMR follows the specification in Morton & Polyn (2016). Column 2: Instance CMR with trace activation scaling turned off. Column 3: Instance CMR with only trace activation scaling and no item activation scaling. Fourth model Column 4: Instance CMR with both trace and item activation scaling parameters freed for fitting.

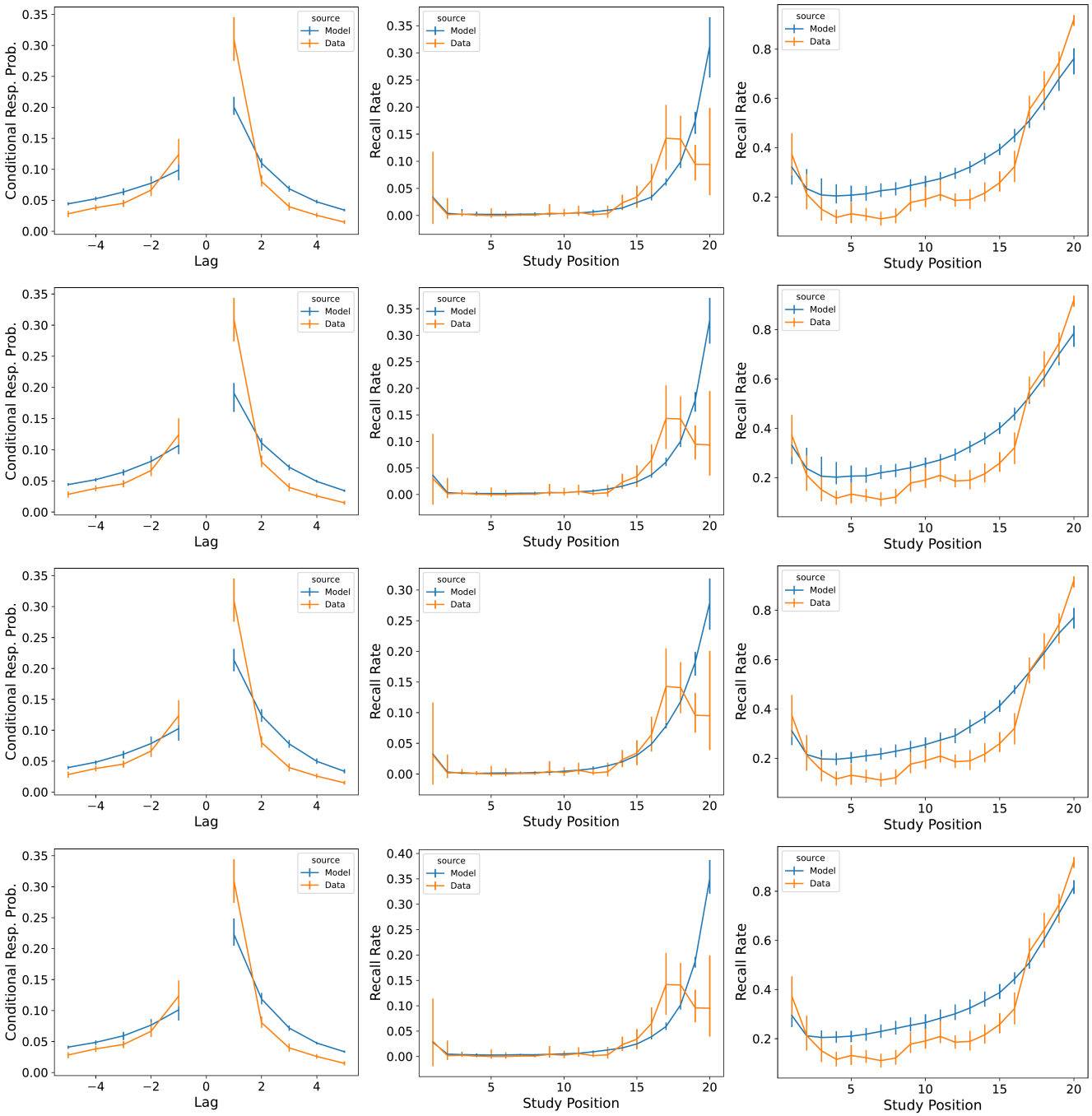


Figure 2: Summary statistic fits to Murdock Jr (1962) where list length = 20. Top: Connectionist CMR. Second Row: Instance CMR, τ_t set to 1. Third Row: Trace Scaling CMR – Instance CMR, τ_c set to 1 and τ_t optimized during fitting. Fourth Row: Multi Scaling CMR – Instance CMR, both τ_t and τ_c optimized during fitting. Left: conditional response probability as a function of lag. Middle: probability of starting recall with each serial position. Right: recall probability as a function of serial position.

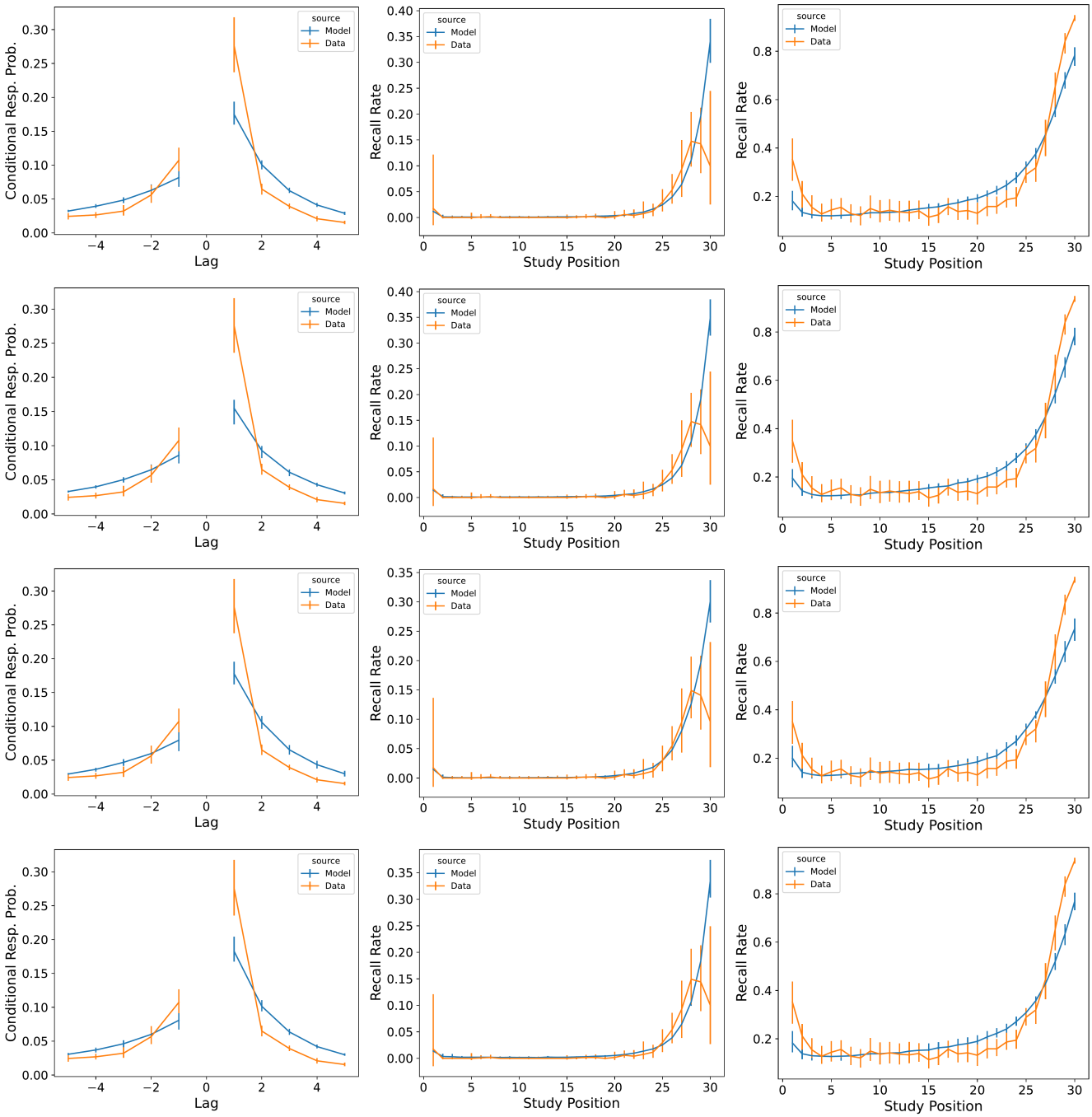


Figure 3: Summary statistic fits to Murdock Jr (1962) where list length = 30. Top: Connectionist CMR. Second Row: Instance CMR, τ_t set to 1. Third Row: Trace Scaling CMR – Instance CMR, τ_c set to 1 and τ_t optimized during fitting. Fourth Row: Multi Scaling CMR – Instance CMR, both τ_t and τ_c freed for fitting. Left: conditional response probability as a function of lag. Middle: probability of starting recall with each serial position. Right: recall probability as a function of serial position.

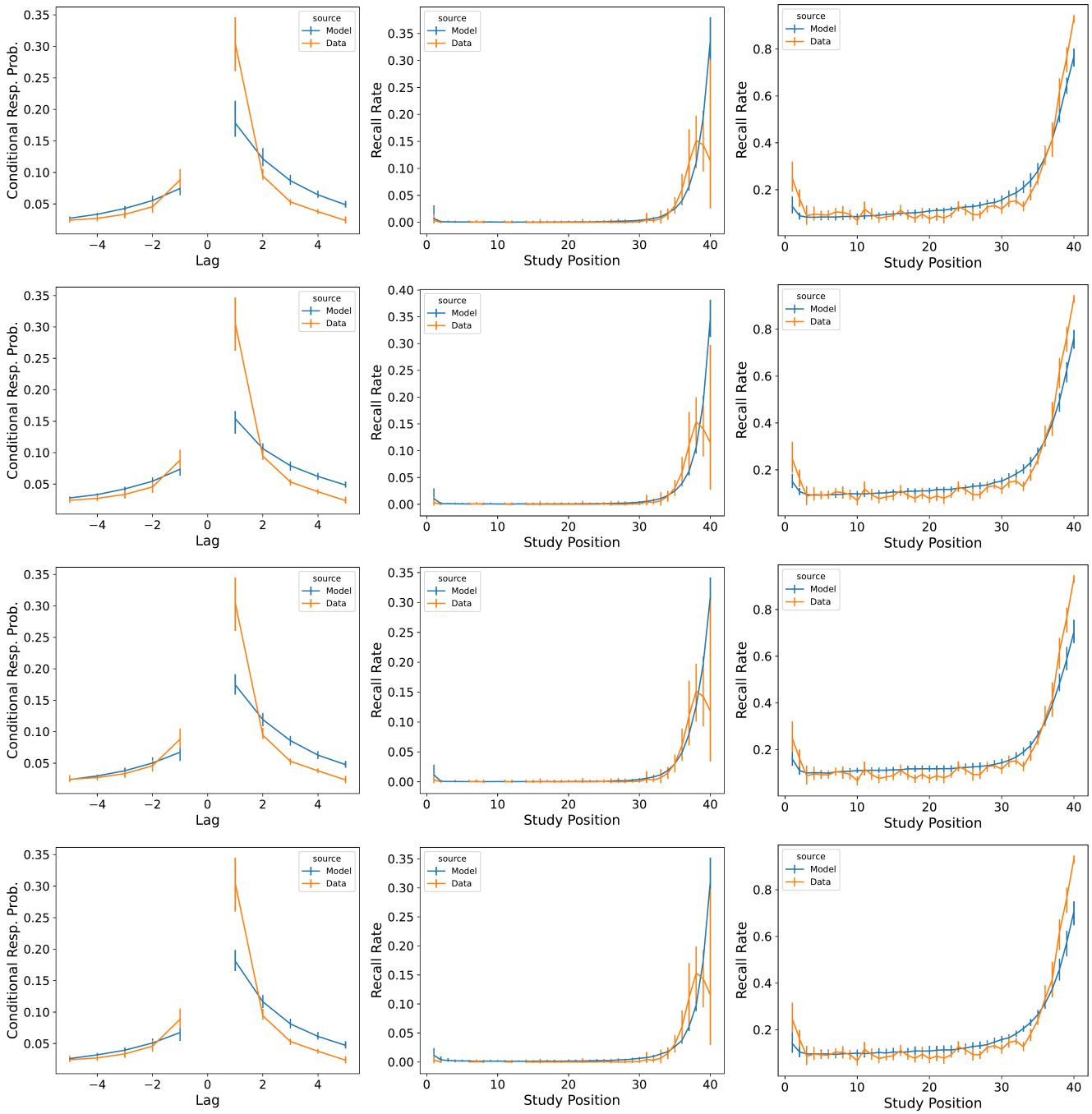


Figure 4: Summary statistic fits to Murdock Jr (1962) where list length = 40. Top: Connectionist CMR. Second Row: Instance CMR, τ_t set to 1. Third Row: Trace Scaling CMR – Instance CMR, τ_c set to 1 and τ_t optimized during fitting. Fourth Row: Multi Scaling CMR – Instance CMR, both τ_t and τ_c freed for fitting. Left: conditional response probability as a function of lag. Middle: probability of starting recall with each serial position. Right: recall probability as a function of serial position.

Lohnas & Kahana (2014)

	Connectionist CMR	Instance CMR	Trace Scaling CMR	Multi Scaling CMR
fitness	1670.26 +/- 147.36	1672.85 +/- 147.45	1667.82 +/- 146.36	1669.93 +/- 148.38
encoding drift rate	0.75 +/- 0.03	0.74 +/- 0.03	0.55 +/- 0.06	0.56 +/- 0.06
start drift rate	0.58 +/- 0.11	0.59 +/- 0.10	0.56 +/- 0.08	0.61 +/- 0.07
recall drift rate	0.95 +/- 0.02	0.95 +/- 0.01	0.90 +/- 0.02	0.90 +/- 0.02
shared support	4.80 +/- 2.43	3.79 +/- 2.57	1.24 +/- 0.60	16.06 +/- 8.39
item support	7.11 +/- 3.13	4.92 +/- 2.80	8.03 +/- 4.62	17.83 +/- 7.59
learning rate	0.53 +/- 0.08	0.52 +/- 0.07	0.65 +/- 0.08	0.68 +/- 0.06
primacy scale	11.89 +/- 7.78	9.45 +/- 7.12	14.54 +/- 5.55	7.43 +/- 4.31
primacy decay	15.62 +/- 8.73	21.84 +/- 11.44	8.52 +/- 7.05	10.90 +/- 7.90
stop probability scale	0.02 +/- 0.01	0.02 +/- 0.01	0.02 +/- 0.01	0.02 +/- 0.01
stop probability growth	0.20 +/- 0.02	0.19 +/- 0.02	0.19 +/- 0.02	0.19 +/- 0.02

	Connectionist CMR	Instance CMR	Trace Scaling CMR	Multi Scaling CMR
choice sensitivity	21.39 +/- 11.06	15.15 +/- 9.67	1.00 +/- 0.00	19.32 +/- 10.90
mcf trace sensitivity	1.00 +/- 0.00	1.00 +/- 0.00	3.44 +/- 0.42	2.76 +/- 0.43

Table 3: Confidence intervals of parameters fit to data from L. Lohnas & Kahana (2014), computed across subjects. Column 1: Connectionist CMR follows the specification in Morton & Polyn (2016). Second Row: Instance CMR, with τ_t set to 1. Third Row: Trace Scaling CMR – Instance CMR with τ_t set to 1 and τ_c optimized during fitting. Fourth model Column 4: Instance CMR with both trace and item activation scaling parameters freed for fitting.

Table 1, Table 2, and Table 3 present the confidence intervals of the parameters fit to the data across subjects for each model variant. The tables also present the mean and confidence intervals of the log-likelihood values for each model variant. Figure 1, Figure 2, and Figure 3 illustrate the summary statistic fits for each model variant. We additionally performed paired t-tests to compare the log-likelihood values of the models across subjects, using a significance level of 0.05 and Connectionist CMR as the reference model. Alternative models Instance-CMR, Trace Scaling CMR, and Multi Scaling CMR were compared to Connectionist CMR to determine if subject-level log-likelihoods were significantly lower than the reference model (lower scores indicate better fit), with outcome p-values ~0.0176, ~0.0232, and ~0.0008, respectively. Performing the same tests for the data from Murdock Jr (1962) yielded p-values 0.9125, 0.9414, and 0.6252 for the three alternative models, respectively. For the data from L. Lohnas & Kahana (2014), the p-values were 0.8638, 0.1015, and 0.4374 for the three alternative models, respectively. These results suggest that the Multi Scaling CMR model provides the best fit to the data across all three datasets, with the Instance CMR model performing similarly to the Connectionist CMR model. However, even the best-fitting model, Multi Scaling CMR, only marginally outperforms Connectionist CMR in terms of log-likelihood values and capture of summary statistics across all datasets.

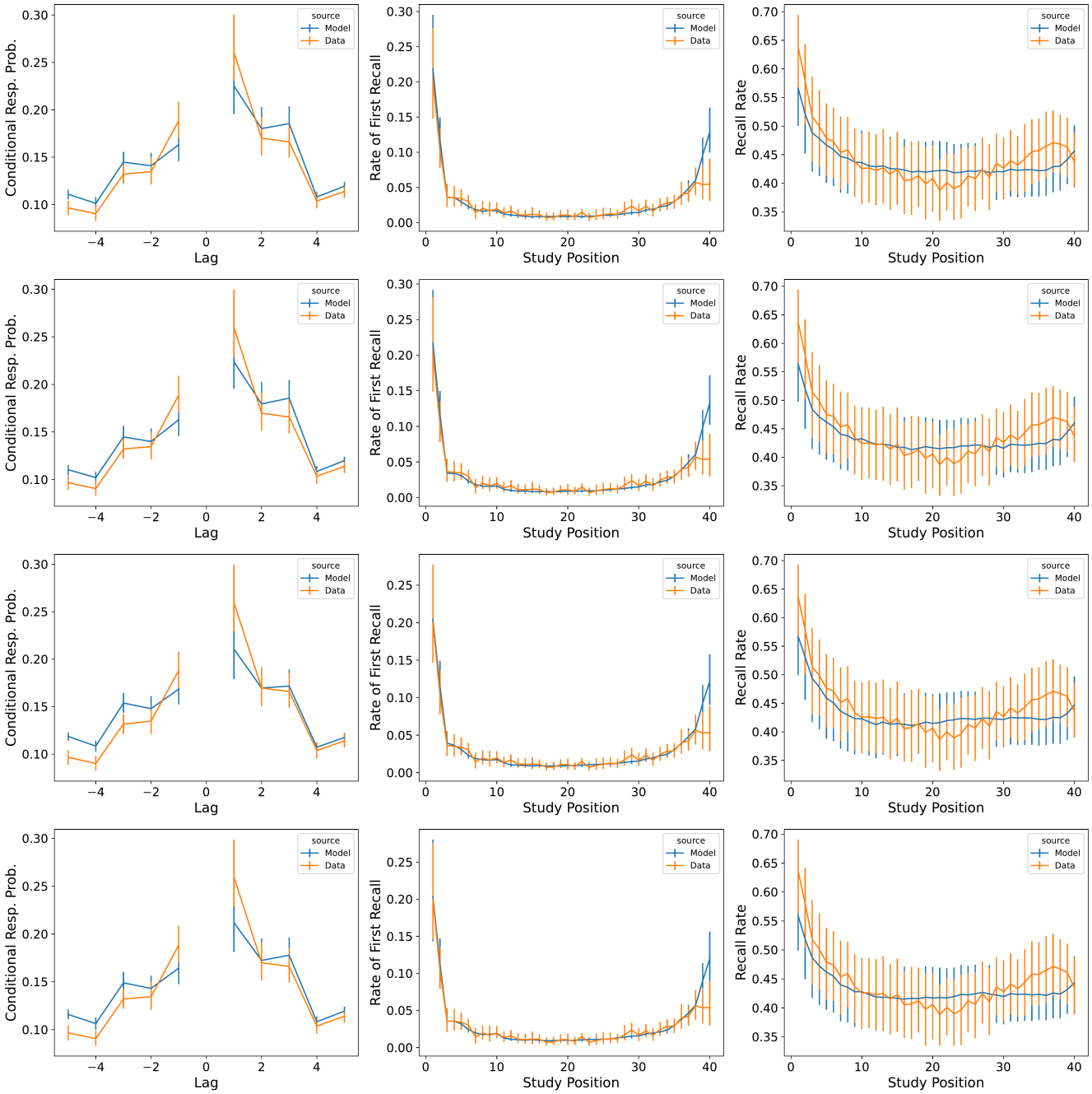


Figure 5: Summary statistic fits to L. Lohnas & Kahana (2014). Top: Connectionist CMR. Second Row: Instance CMR, τ_t set to 1. Third Row: Trace Scaling CMR – Instance CMR, τ_c set to 1 and τ_t optimized during fitting. Fourth Row: Multi Scaling CMR – Instance CMR, both τ_t and τ_c freed for fitting. Left: conditional response probability as a function of lag. Middle: probability of starting recall with each serial position. Right: recall probability as a function of serial position.

These several figures help illustrate a single point: the functional equivalence between Connectionist and Instance CMR models. When the trace-based activation scaling parameter τ_t is set to 1, both models yield the same results in free recall tasks, with differences in results only attributable to the stochastic nature of the simulations and fitting process. Even when τ_t is varied during model fitting, either instead of or in addition to the item activation scaling parameter τ_c, model performance is only marginally different. The challenges that the original Connectionist CMR faces in capturing the dynamics of memory search in free recall tasks are shared by Instance CMR, and are not clearly resolved by the latter model. For example, both models overrate the strength of the lag-contiguity effect in the dataset from Murdock Jr (1962), and lack the ability to capture the plateau-like shape of the probability of first recall curve in data both from Murdock Jr (1962) and L. Lohnas & Kahana (2014). These results suggest that the instance-based specification of CMR does not provide a clear advantage over the connectionist specification in terms of capturing the dynamics of memory search in free recall tasks – at least not without additional modifications to the model or novel task conditions.

Simulation 3: Accounting for Semantic Organization in a Single Instance Memory

We have argued that an instance-based specification of CMR can facilitate the integration of retrieved context theory with other models that used an instance-based memory framework. Above we demonstrated how Instance CMR can replicate the dynamics of memory search in free recall tasks, showing that it can be functionally equivalent to Connectionist CMR when trace-based activation scaling is disabled. But a few issues obscure how tenets of retrieved context theory as implemented in CMR can be integrated with other instance-based models to provide a more comprehensive account of memory performance across different tasks. For another set of simulations tackling these concerns, we explore the integration of semantic content into the Context Maintenance and Retrieval (CMR) model using an instance-based framework.

These simulations focus on three main goals. First, we seek to present an instance-based alternative to previous approaches for accommodating semantic organization in CMR. The connectionist specification of CMR presented already uses a simple mechanism to provide semantic support for items in the absence of episodic associations. A parameter α provides a fixed level of pre-experimental support in M^{CF} for all items, which can be adjusted to simulate the strength of pre-experimental semantic associations. More recent work has attempted to provide a more detailed account of semantic support in CMR by introducing a separate memory stack for semantic associations M^{sem}, which is probed in conjunction with M^{CF} during free recall to enable semantic alongside temporal organization of responses. Critically, connections within M^{sem} are configured using similarity operations performed on word embedding vectors trained on a large corpus of text, allowing the model to capture the semantic relationships between items in a more

detailed and data-driven manner than the fixed α parameter. Here we show that a single instance-based memory can be used to capture semantic relationships between items in a similar manner. Our approach directly encodes word embedding vectors alongside item and context representations within memory traces, sidestepping the need for a separate memory stack to represent semantic associations.

Second, addressing this issue also allows for a specification of Instance CMR that uses a single memory structure to store both item-to-context and context-to-item associations without sacrificing the model's ability to capture the dynamics of memory search in free recall tasks. Using a single memory should make it easier to integrate CMR with other instance-based models via a shared memory structure, potentially allowing for a more comprehensive account of memory performance across different tasks. The reason Instance CMR and Connectionist CMR use separate M^{FC} and M^{CF} memories can be traced back to the same assumption above that pre-experimental semantic associations are uniform across items in M^{CF}. By contrast, in pre-experimental M^{FC}, each item feature unit is uniquely connected to a single context unit, with 0 connections to all other context units. These assumptions are incompatible within a unitary memory structure that only stores item features and context features. However, we demonstrate here that CMR can be implemented not only in terms of two instance-based M^{FC} and M^{CF} instance-based memories, but also in terms of a single instance-based memory which stores both item and context representations and can be contacted in either direction (F->C or C->F) without disrupting equivalence with the original CMR model. We show that cleanly distinguishing semantic content from episodic content within the same memory structure and concatenating these representations into a single trace allows Instance CMR to capture the dynamics of memory search in free recall tasks while also providing a more detailed account of semantic organization in memory.

Finally, this demonstration will also help clarify how CMR can be readily integrated with other cognitive models implemented using an instance-based framework and how components will interact during operations like probe-based retrieval. To this end, we interpret word embedding models as themselves basic instance-based models of semantic memory, and show how they can be integrated with CMR in a single memory stack to account for semantic organization in free recall. We use this example as an opportunity to provide a straightforward mathematical analysis of the capacity of instance-based models to be com-

bined with other instance-based models using this pattern of concatenation of memory trace features. We show that the concatenation of feature configurations within memory traces and the use of dot product operations to retrieve associations between these features and a probe vector are equivalent to a summation of dot product operation outcomes across separate memory traces. The mathematical equivalance of this approach to pre-existing approaches for integrating semantic content into CMR will further clarify the relationship between instance and connectionist memory models and provide a foundation for future integrations of CMR with other cognitive models.

Model Extension

First we outline the extensions of Connectionist CMR and Instance CMR that allow them to capture semantic organization in free recall tasks using either a uniform semantic support across items or a more detailed semantic representation based on word embedding models. To represent word embeddings, we use the GloVe embeddings (Pennington et al., 2014) trained on the Common Crawl corpus (Common Crawl Foundation, 2014) with 300 dimensions. These embedding vectors have the property that the dot similarity between two embedding vectors reflects the semantic similarity between the words they represent. the strength of semantic associations relative to temporal associations during recall.

Semantic Extension of Connectionist CMR

We start with the item-cue semantic extension of Connectionist CMR, which uses a separate memory stack M^{sem} to encode semantic associations between items in the study list, and drives item competition during recall using a linear combination of the context-to-item memory M^{CF} and the item-cue semantic memory M^{sem} (Morton & Polyn, 2016). Next we'll leverage this construction supercedes the original implementation of semantic support in Connectionist CMR by shifting representation of semantic associations from a fixed parameter α in M^{CF} to a separate memory representation M^{sem}. If M^{sem} is configured to provide a

uniform amount of support for all items, the model is functionally equivalent to the original Connectionist CMR model.

Following Morton & Polyn (2016), this extension adds a separate memory stack M^{sem} to the model, which encodes a scaled semantic similarity matrix between items in the study list. The diagonal of this matrix (representing item self-similarity) is set to 0, and the off-diagonal elements are set to the cosine similarity between the word embedding vectors of the items. The model uses these embeddings to calculate the semantic similarity between items in the study list and encodes these similarities in M^{sem}. The model purely updates the the activation function used to drive item competition during recall. For the initial recall, item supports are calculated normally, by projecting the context vector through the context-to-item memory M^{CF}. For subsequent recalls, the model additionally projects the feature representation of the last recalled item through M^{sem} to retrieve semantic associations between items in the study list, allowing it to capture the semantic organization of responses:

$$A = sM^{sem}f_{last} + M^{CF}c \tag{0.1}$$

A scaling parameter s is used to adjust the strength of semantic associations relative to temporal associations during recall. This replaces the fixed parameter α in M^{CF} (now fixed to a value of 0) with a more detailed representation of semantic associations between items in the study list. If M^{sem} is configured to provide a uniform amount of support for all items, the model is functionally equivalent to the original Connectionist CMR model.

Semantic Extension of Instance CMR: A Single Memory Structure

The connectionist M^{sem} could be readily added the same way to Instance CMR (determining activations with a sum of M^{sem} and M^{CF} supports), but for a purely instance-based implementation we'll take a different approach here that allows us to capture semantic associations between items in the study list using a single memory structure. First, rather than using M^{FC} and M^{CF} to represent item-to-context

and context-to-item associations, we'll use a single memory structure M that stores both item and context representations. In our semantic extension of Instance CMR, each trace additionally encodes an embedding vector corresponding to the semantic representation of a word in the study list. The resulting instance-based memory for ICMR would have 3 components for each trace – a semantic component W, an orthonormal F representation component, and a temporal contextual C component. To represent pre-experimental memory associations, we initialize M with a trace for each item in the study list, where each item feature unit is connected to a unique context unit by setting the corresponding elements of the f_i and c_i vectors to 1. In addition, each trace includes the word embedding vector for the item i, designated w_i, which is concatenated with the item and context representations. For experimental traces, f_i and c_i are updated as in the original Instance CMR model, and w_i is set to the embedding vector for the encoded item i:

$$M_i = [f_i, c_i, w_i]$$

The model retrieves associations similarly to the original Instance CMR model, simply including the semantic component in the activation function used to drive item competition during recall. For retrieving contextual features $\hat{C}$ to reinstate into the context vector, the model uses a probe vector consisting only of F features, formatted as $[f, 0, 0]$. During activation, the parameter γ is used to adjust the strength of pre-experimental item-to-context associations relative to experimental associations during recall. γ_i is configured to $1 - \gamma$ for pre-experimental traces and γ for experimental traces. Activated context features in the retrieved echo are used to update the current state of context:

$$[...\hat{C}...] = \sum_{i=1}^{n} \gamma_i M_i[f, 0, 0]$$

For the initial recall, item supports are calculated normally, by projecting the context vector through the context-to-item memory M^{CF}, using a probe vector formatted as $[0, 0, c]$. Then for further recalls, the probe contains both the current context vector and the semantic representation of the last recalled item, formatted as $[0, w_{last}, c]$. During activation, the parameter ϕ_i is used to adjust the strength of context-to-

feature associations according to the primacy function specified in Equation 0.4. At the same time, the parameter s is used to adjust the strength of semantic associations relative to temporal associations during recall. These values modulate the contribution of context and semantic trace components to the activation function used to drive item competition during recall:

$$[...\hat{F}...] = \sum_{i=1}^{n} M_i[0, \phi_i c, sw_{last}]$$

Otherwise, model dynamics are identical to the original Instance CMR model, including the calculation of recall probabilities for each item and the updating of memory traces and contextual states during encoding and recall.

How Trace Components Interact: Equivalence with Connectionist Semantic CMR

A major theme of our presentation of Instance CMR is that it shows retrieved context theory is portable: that by concatenating CMR-like representations (context states and orthonormal item representations) to instance-based memories of other models, CMR's dynamics can be added to those models. Here we use our semantic extensions of Connectionist and Instance CMR to clarify how the components of such extended memory traces interact in a combined model. Our analysis shows that, just like our original specifications of Connectionist and Instance CMR, the two semantic models are equivalent when the trace-based activation scaling parameter τ_t is held to a neutral value.

Our results depends on the observation that the dot product of concatenated vectors is the sum of the dot products of the corresponding sub-vectors. Let us consider two vectors w and v concatenated to create the new vector $z = [w; v]$. Because of the linear nature of the dot product, when we then compare z to a probe a vector $x = [x_1; x_2]$ with the same dimensionality as z, the dot product has this relation:

$$z \cdot x = (w \cdot x_1) + (v \cdot x_2)$$

Similarly, in our Semantic Instance CMR, semantic similarity to the last recalled item modulates retrieval supports *additively* with the context-to-feature retrieval supports, just as it does in the Semantic Connectionist CMR using probes of M^{CF} and M^{sem}.

And this principle applies beyond the semantic extension of Instance CMR. In general, concatenating InstanceCMR to other instance-based memory models will instantiate this same linear relationship described above, with the model of semantic memory replaced with the details of the other model. If memory traces in a composite model are a concatenation of other-model features and ICMR-derived features, while memory probes are themselves a concatenation of other-model features and context-features, the resulting echo will be the same if you separately calculated and then linearly combined other-model support and context-driven support in ICMR. This observations helps us more clearly conceptualize how the portability of ICMR really works at a mathematical, as opposed to procedural, level. In our proposed research finalizing this result, we will provide empirical support for this claim by showing that the semantic extension of Instance CMR is functionally equivalent to the semantic extension of Connectionist CMR when the trace-based activation scaling parameter τ_t is set to 1 and the semantic scaling parameter s is allowed to vary during fitting. We also show that, when the semantic components of the traces are configured to provide uniform support for all items according to a free parameter α the model is functionally equivalent to our original specifications of Connectionist and Instance CMR.

Comparison Approach

We will compare the semantic extension of Connectionist CMR to the semantic extension of Instance CMR re-using the subset of the PEERS dataset described in Healey & Kahana (2014). We will fit the models to the data using the same differential evolution optimization technique as before, and compare the log-likelihood values of the models across subjects using paired t-tests. We will also compare the summary

statistics of the models to the observed data to evaluate how well they capture the dynamics of memory search in free recall tasks. Extending previous comparisons, we will use GloVe embeddings to represent semantic content in the models, and compare the semantic extension of Connectionist CMR to the semantic extension of Instance CMR to determine which model provides the best fit to the data. Our analysis of simulations of summary statistics will include a semantic CRP analysis that defines distance bins based on the distribution of cosine similarities between word embedding vectors for items in the experiment word pool and measures the probability of transitioning to recall an item at each distance bin given that such a transition is possible. This analysis measures the semantic contiguity effect, the tendency for items close together in semantic space to be recalled in close sequence, and will help us evaluate how well the models capture the semantic organization of responses in the data.

Discussion

In light of a collection of results across task domains distinguishing between the performance of prototype- and instance-based models of memory-based behavior, we searched for evidence of a similar distinction with respect to the free recall task paradigm. To do this, we specified and compared the original connectionist implementation of an established model of task performance based in retrieved context theory (Connectionist CMR) against a parallel instance-based variant (Instance CMR) across diverse experimental conditions, including a dataset featuring variable study list lengths across trials and a dataset featuring item repetitions within trials and variable item repetition spacing between trials. While our results identify some marginal differences in model performance across datasets, they do not provide clear evidence of a performance distinction between instance- and instance-based models in the free recall task paradigm. The model variants accounted for human performance on the free recall task with similar effectiveness in each dataset considered. In addition, the results of our semantic extension of Connectionist and Instance CMR models will further underscore the portability of retrieved context theory across model architectures. We show that the concatenation of memory traces in a single memory structure is equivalent to a linear combination of the separate retrieval of traces from two memory structures, identify the mathematical basis for this equivalence, and provide a foundation for future integrations of CMR with other cognitive models.

Altogether these results demonstrate that the success of the theoretical commitments made by the Context Maintenance and Retrieval (CMR) model and models like it are likely not especially dependent on the architectures in which they are implemented. Instead, the main insights of retrieved context theories (at least as formalized by CMR) are highly portable, and can likely be esconced within any reasonable

model architecture where memory search via temporal contextual representations might prove valuable. Establishing the portability of these successful theoretical principles across modeling approaches helps advance the historical effort in cognitive science to develop "a general account of memory and its processes in a working computational system to produce a common explanation of behavior rather than a set of lab-specific and domain-specific theories for different behaviors" (Jamieson et al., 2018; Newell, 1973).

An instance-based specification of CMR does not provide a clear advantage over the connectionist specification in terms of capturing the dynamics of memory search in free recall tasks – at least not without additional modifications to the model or novel task conditions. But a few features unique to the instance-based specification of CMR may still prove valuable in future research. Throughout this work, we emphasized one of these: the instance-based specification of CMR is more easily integrated with other instance-based models of memory, potentially supporting lines of research that seek to combine the strengths of different models to provide a more comprehensive account of memory performance across different tasks.

Another feature of instance-based memory models that may prove valuable in future research is their interpretability, the degree to which the model's parameters and internal representations can be understood and reasoned about (Gao & Guan, 2023). Because instance models do not compress information about the history of an item into a single vector, they can be more easily interpreted than connectionist models. Each trace is represented separately and its weighting in the activation function is also calculated separately, making the contribution of each trace to the model's output transparent in its internal structure. These features may make instance-based models favorable for research that seeks to diagnose and address the limitations of existing models, or to develop new models that can capture the dynamics of memory search in free recall tasks more effectively.

The most emphasized advantage of instance models in the literature though is their ability to capture the dynamics of memory search in free recall tasks more effectively than connectionist models via its use of trace-based activation scaling in domains such as category learning and semantic memory (Jamieson et al., 2018; Nosofsky & Zaki, 2002). Why does this advantage not appear when comparing Connectionist and Instance CMR in the free recall task?

One key reason is that Connectionist CMR has some features that allow it to exhibit instance-memory-like behavior in some aspects of its dynamics. First, the item representations on its F layer are orthonormal – each item's representation is unique and does not overlap with any other item's representation. This means that even without the trace-based activation scaling unique to instance models, the model can still selectively activate the representation of a single item in the F layer without activating the representations of other items, making Instance M^{FC} and Connectionist M^{FC} equivalent for all values of τ_t. Similarities between τ_t in the instance-based model and τ_c in the connectionist model also contribute to the similarity in model performance. Despite some fundamental differences, both activation scaling mechanisms help to focus the model's attention on the most relevant traces in memory, perhaps enough to in a broad range of cases obscure the differences in model performance that might be expected from the different architectures.

A more fundamental issue for comparison between Instance and Connectionist CMR is that the free recall task and design of CMR might not instantiate a clear instance vs prototype model distinction that has set up previous successes of instance models in other domains. In these other domains, experiments were configured to support clear contrasts between prototype-based decision-making where the central tendency of a class drives responses, and instance-based decision-making where the the similarity of a probe to a specific study instance drives responses. The free recall task is designed to test the ability of a model to retrieve items from memory in a specific order, and the dynamics of memory search in this task may not be as sensitive to the differences between instance and prototype models as other tasks. However, future variations of the task that introduce category structure as a major factor in recall, or that require participants to make decisions based on the category membership, might provide a clearer test of the distinction between instance and prototype models in the domain of free recall. These results highlight that the advantage of instance- over prototype- models in accounting for memory performance is highly contingent on the task and conditions under which models are tested.

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Same item, different traces: Dissecting the structure of memory for repeated experience

Jordan Gunn Sean Polyn

To investigate the structure of memory for repeated experiences, this study examines how the lag-contiguity effect – the tendency for items positioned close together in a study list to be retrieved successively during free recall (Kahana, 1996) – manifests for items with multiple study positions. Participants will study lists containing both once- and twice-presented items, with the proximity of each presentation of a repeated item to the beginning or end of the study lists varied systematically to probe the influence of serial position effects (Murdock Jr, 1962) on memory trace accessibility. Using a modified lag-conditional response probability (lag-CRP) analysis, we will compare transition probabilities from repeated items to items studied near either their first or second presentation. Preliminary analyses across two archival datasets suggest that, even after accounting for serial position effects, the temporal contiguity effect is more pronounced relative to the first presentation of a repeated item than to the second. Initial results also indicate that the influential Context Maintenance and Retrieval (CMR) model (Polyn et al., 2009) may not accurately predict these differences, pointing to potential limitations in its current framework. Further data collection, follow-up analyses, and proposed modeling aim to clarify these findings, enhancing our understanding of memory search and refining theoretical models of memory for repeated experiences.

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Introduction

What is the structure of memory for repeated experience? It's a basic law of memory that repeated study of a piece of information usually improves our memory for it (Kahana et al., n.d.). Yet, from the pioneering work of Ebbinghaus (Ebbinghaus, 1885) onwards, researchers have understood that the mechanisms behind this improvement are complex—not simply a matter of strengthening the original memory trace (Delaney et al., 2010). A particularly well-documented factor behind the benefits of repetition is the spacing between study events. Studies across various conditions and materials consistently show that spaced repetitions – where intervals between restudy are longer – tend to enhance memory more significantly than when repetitions are grouped together (Cepeda et al., 2006). Several theories attempt to explain this spacing effect, focusing on factors like effort, variability in encoding, rehearsal strategies, and retrieval dynamics. However, no single explanation has emerged as definitively superior (Maddox, 2016). Moreover, repetition doesn't only enhance the memorability of information; it also affects how memories are organized and retrieved in sequences, such as in free recall tasks. This highlights the need for more nuanced theories that can explain how repetitions influence the structural organization of memory retrieval.

A promising framework for understanding the structure of memory for repeated experience is the Context Maintenance and Retrieval (CMR) model (Polyn et al., 2009), a leading computational implementation of the retrieved context theory (RCT) of free recall. RCT posits that each item studied is linked in memory with a continuously updated, recency-weighted context representation, which later guides the retrieval process during free recall as the primary retrieval cue. The CMR model has successfully explained various repetition effects and

recall phenomena, including the spacing effect (Lohnas, 2024; Lohnas & Kahana, 2014a), as well as broader patterns in free recall such as the serial position effect and lag-contiguity effect (Polyn et al., 2009). By modeling the iterative processes of memory search, CMR offers a comprehensive view of how prior experiences influence ongoing encoding and retrieval across different experimental conditions.

While CMR represents a substantial advance in computational modeling of key phenomena, existing evaluations of the CMR model as an account of repetition effects have so far been limited in scope. While the model has been successful in qualitative simulation of key repetition effects, it has yet to receive formal evaluation for its ability to predict entire responses sequences in free recall for lists with repeated items. Existing analyses have focused primarily on the model's predictions regarding study-phase retrieval and the spacing effect, potentially overlooking other critical aspects of repetition in free recall. These gaps raise questions about whether CMR provides a complete account of repetition effects or requires further refinement.

The present work aims to address these gaps in our understanding of repetition effects in memory by combining more robust model evaluation with a new experiment and analyses that more finely dissect the distinct consequences of each presentation of a repeated item for the temporal organization of memory search. Participants in this study will be presented with lists containing items shown either once or twice. For items appearing twice, we'll manipulate their presentation positions to probe their distinct memory contributions. Some items will appear first near the start and again around the middle of the list. Others will appear first near the middle and again near the end. This design leverages the well-established finding that items at the beginning and end of a list are typically remembered better (Murdock Jr, 1962). By varying the positions of repeated items, we aim to isolate and examine the unique memory contributions of each occurrence by manipulating the strength of their associated memory traces.

To interpret the results of this manipulation, I propose novel analyses that identify the independent effects of each study event of a repeated item on free recall task performance. The key analysis will focus on the temporal contiguity effect—the tendency for sequentially proximate

items in a study list to be recalled in close sequence (Kahana, 1996). Using a modified version of the lag-conditional response probability (Lag-CRP) analysis, we will measure transition rates from recalls of twice-presented items to neighbors of either their initial or subsequent presentation positions, reflecting the reinstatement of context associated with each presentation of a repeated item. This analysis approach allows us to quantify how the contextual features associated with each presentation of a repeated item individually shape the memory search process. And to extend prior evaluations of the CMR model, I propose to apply the likelihood-based model comparison (Kragel et al., 2015), generalization criterion (Busemeyer & Wang, 2000) and related techniques to evaluate the model's ability to account for these results and examine alternatives that address its limitations. Developing and evaluating computational models designed to account for these effects will help refine our understanding of the structure of memory for repeated experience.

Theoretical Framework

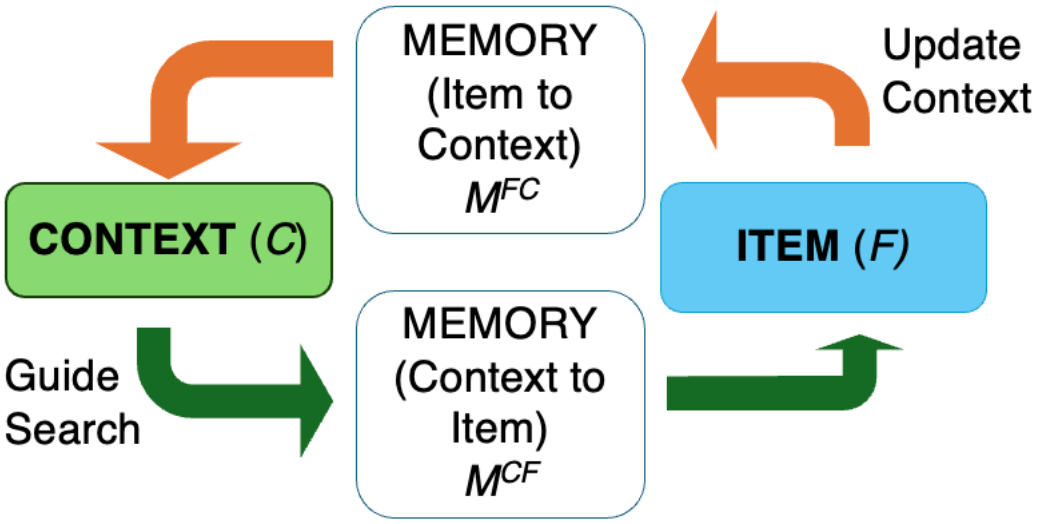


Figure 1: **The Context Maintenance and Retrieval (CMR) model**. The model posits that each item studied is associated with a continuously updated, recency-weighted context representation. During free recall, the current state of the context vector acts as the primary retrieval cue, guiding memory search through the sequence of studied items. The model's mechanisms of contextual evolution and reinstatement enable it to account for key repetition effects and recall phenomena in free recall tasks.

The present research emphasizes the free recall task because it offers a unique blend of experimental control and behavioral richness that has made it a powerful tool for studying the structure of memory (Howard & Kahana, 2002). In the task, participants study a sequence of items and then are asked to recall as many of the items as they can, in any order they choose. This ordering has provided insights into the underlying processes and representations

that guide memory search, enabling accounts of repetition and other effects that go beyond item memorability measures to explain the sequential organization of recall. And the straightforward study-test structure of the task has allowed researchers to precisely manipulate which items are studied and when, achieving fine-grained control over the memories encoded and their relationship to one another. Notably, spacing effects – the most evident among repetition phenomena – are especially pronounced in free recall, highlighting its utility for investigating how repetition influences memory (Cepeda et al., 2006).

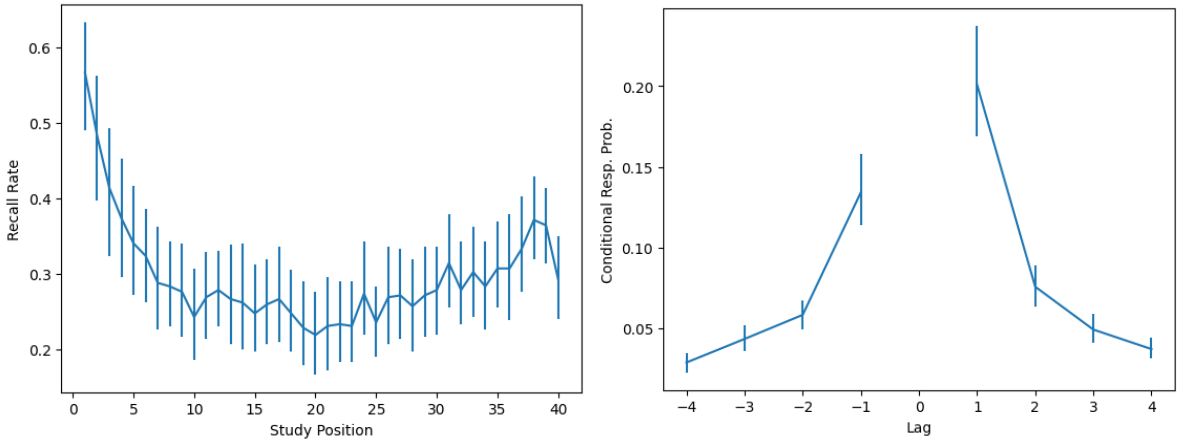
Theoretical interpretations of free recall conceive it as a dynamic search through memory, where each retrieval act updates the cue for the next, essentially guiding navigation through a mental timeline of events (Tulving et al., 1972). A converging body of work suggests that the Context Maintenance and Retrieval model (CMR; Polyn et al., 2009) can simultaneously account for basic features of memory search and key repetition effects in free recall (Lohnas, 2024; Lohnas & Kahana, 2014a). Rooted in the retrieved context theory (RCT) of free recall, CMR elaborates on how interactions between item-context associations and context-driven retrieval processes manifest repetition effects and other recall phenomena.

Central to the CMR model is the concept of temporal context, envisioned as a vector of features that evolves over time, reflecting a recency-weighted average of past experiences. When an item is studied, it becomes associated with the current state of the temporal context vector. This linkage is bidirectional: the context not only cues the item but also updates itself with the item's associated contextual features in a process of context reinstatement. For items appearing the first time in a list, the reinstated context derives from pre-experimental experiences—contexts tied to earlier occurrences of the item outside the ongoing trial. For items appearing a second time, the reinstated context blends pre-experimental contexts with those linked to the item's first appearance. This reinstated context partially modifies the existing context vector, maintaining some influence from earlier encoding events. This residual effect facilitates the associative linkage between temporally adjacent items, enabling the retrieval of one item to cue the next ones close in the sequence.

During free recall, the current state of the temporal context vector acts as the retrieval cue.

Items strongly associated with the current context are more likely to be recalled. Following the retrieval of an item, its associated context is reinstated, mirroring the encoding phase. This reinstatement updates the temporal context vector, configuring subsequent retrieval attempts. This iterative process of context-driven retrieval and updating continues until no more items are recalled, producing an interplay of memory encoding, retrieval, and updating that shapes how we remember sequences of events.

The Temporal Organization of Free Recall



(a) Serial Position Curve

(b) Lag-CRP

Figure 2: **The temporal organization of free recall**. Example analysis outcomes using control lists from Lohnas & Kahana (2014a) (no item repetitions). The U-shaped serial position curve reflects the combined influence of the primacy and recency effects on recall performance. The lag-CRP analysis identifies an assymetric lag-contiguity effect, where participants are more likely to transition to items studied near the serial position of the last recalled item, especially in forward transitions.

The emphasis here on the retrieved context account of memory search stems from the theory's ability to account for key features of the temporal organization of free recall. The Context Maintenance and Retrieval (CMR) model provides a formal generative account of the temporal organization of the course of memory search in the free recall task, explaining not just differences in recall rates between items, but also predicting the structure of recall sequences

themselves. CMR addresses three particularly robust benchmark phenomena in the free recall task: the primacy effect, the recency effect, and the lag-contiguity effect.

Primacy and recency effects in free recall compose the well-established serial position effect that has been used to argue for the importance of temporal organization in memory search (Murdock Jr, 1962). The primacy effect refers to the enhanced recall of items at the beginning of a list, attributed to increased attention or rehearsal at the start of learning. The recency effect, conversely, pertains to the improved recall of the most recently studied items, driven by their fresher mental representations. The two are most commonly measured by the serial position curve, which plots the probability of recall for each item in a study list as a function of its serial position in the list. The U-shaped curve that typically results from this analysis reflects the combined influence of the primacy and recency effects on recall performance (Figure 2a). Analysis also extends to how often recall initiates with each item, providing insights into how the position affects the likelihood of an item starting the recall sequence. Across both analyses, the boost in recall performance for items from the beginning of the list is called the primacy effect. The boost in recall for items from the end of the list is called the recency effect: more recently studied items are more likely to be recalled than items studied earlier.

The CMR model explains the recency effect by the similarity between the current contextual state during recall and the context associated with recently studied items. In scenarios where there's a delay filled with distractions between the study phase and recall (delayed free recall), the recency effect tends to diminish over time as the contextual link degrades, allowing primacy effects to emerge more strongly. To explain the primacy effect, CMR posits two main mechanisms. The first is superior encoding: early items in a study list are hypothesized to be encoded more strongly due to heightened attention or active rehearsal at the beginning of the learning session (Sederberg et al., 2008). This leads to robust associations between these items and their corresponding contextual states, enhancing their retrievability. The second is context reinstatement: at the start of the recall phase, there is an intentional partial reinstatement of the context from the beginning of the list, priming the recall of items in those early study positions (Polyn et al., 2009). This strategic retrieval of initial contextual states

facilitates the recall of early items, and has been shown to even facilitate performance on the free recall task overall (Zhang et al., 2022). The influence of these mechanisms on recall performance varies depending on several factors, including the length of the study list and any intervening delays between study and recall. These dynamics are robustly modeled by CMR, which has been shown to align well with empirical data under a variety of conditions.

The lag-contiguity effect refers to the tendency during free recall to transition to items that were studied near the serial position of the last recalled item. It is arguably the most pivotal empirical constraint for accounts of the temporal dynamics of memory search in the free recall task (Howard & Kahana, 2002; Kahana, 2020; Polyn et al., 2009). The effect is measured using the lag-conditional response probability (lag-CRP). In this analysis, each transition in a recall sequence is classified by the number of intervening items, or "serial lag," between the currently recalled item and the next. The lag-CRP quantifies the likelihood of transitioning to an item at a specific lag, provided that such a transition is possible – that is, the target item was studied and has not yet been recalled (Figure 2b).

This analysis has revealed that participants consistently show a preference for transitioning to temporally adjacent items. But while this clustering is bidirectional, the effect is asymmetric: forward transitions (e.g., from item 5 to item 6) are more common than backward transitions (e.g., from item 2 to item 1). The lag-contiguity effect substantially constrains the ways theories of free recall can explain not just differences in recall rates between items, but also the structure of item-to-item transitions in recall sequences.

The Context Maintenance and Retrieval (CMR) model explains the lag-contiguity effect through its mechanisms of temporal context management and retrieval cueing. In CMR, the context associated with each studied item carries forward traces of earlier contexts, making it more similar to contexts of temporally adjacent items. This similarity decreases with increasing lag between items. Moreover, the structure of these contextual associations is asymmetric. Normally, the context for an item is absent until its point of study but continues to influence the encoding of subsequent items, making forward-going (positive lag) transitions more likely than backward-going (negative lag) ones. Thus, when an item is recalled, the

reinstated context closely aligns with that of following items rather than preceding ones, leading to a predominance of forward transitions and an assymetric lag-contiguity effect.

Repetition Effects in Free Recall

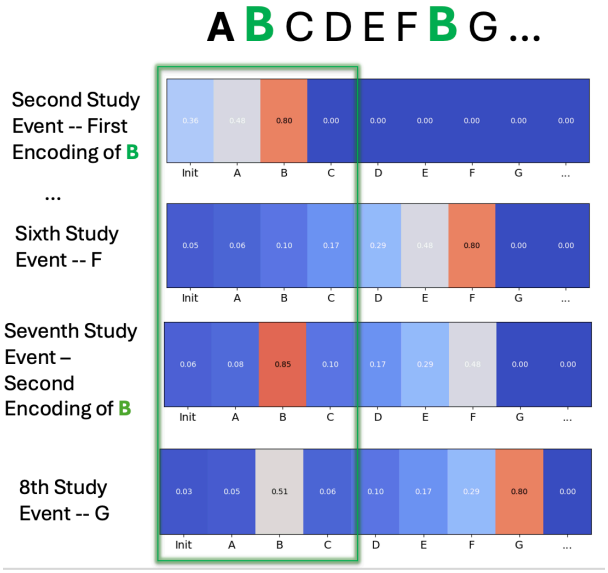


Figure 3: **CMR's contextual states when encoding a study list featuring an item repetition.** CMR implements two hypotheses explaining repetition effects in free recall. According to the *contextual variability hypothesis*, repeating an item in spaced study positions correspondingly associates the item with distinct contextual states, enhancing its accessibility during recall by allowing for more cues to prompt retrieval. According to the *study-phase retrieval hypothesis*, when an item is encoded for the second time, the context associated with its first presentation is reinstated, elaborating on the initial encoding and enhancing the item's accessibility during recall.

From its inception, the Context Maintenance and Retrieval (CMR) model has addressed the role of prior experience in shaping memory search, particularly prior study lists and semantic associations (Polyn et al., 2009). More recently, research has explored how the model uses similar mechanisms to account for within-list interactions between items, particularly those that are repeated during study (Lohnas & Kahana, 2014a). This research has helped to clarify how the model's mechanisms of context reinstatement and study-phase retrieval interact to

produce repetition effects in free recall.

The spacing effect is classically understood as a repetition effect, and has been a central focus of research on memory for repeated experience (Cepeda et al., 2006) However, it applies even to pairs of distinct items, not just repeated items. By measuring the probability of retrieving either of two study items during a free recall trial – an analysis termed the "OR score" – research consistently finds that pairs of spaced items were more likely to be recalled than pairs of massed items (Lohnas et al., 2011). Notably, in a phenomenon called "superadditivity", the mnenomic benefit of spacing tends to be more pronounced for repeated items than for distinct items . The probability of recalling an item repeated in a spaced fashion is higher than the probability of recalling one of two distinct but similarly spaced items, violating statistical independence (Benjamin & Tullis, 2010). This has been interpreted as evidence that additional mechanisms are at play when items are repeated, beyond those that govern the spacing effect for distinct items.

CMR's mechanisms implement two influential hypotheses that have been advanced to explain these features of the spacing effect: encoding variability and study-phase retrieval (Lohnas & Kahana, 2014a). Encoding variability posits that the more distinct the contexts associated with each presentation of an item, the more cues are available to prompt retrieval of the item, increasing the probability of recall (Bower et al., 1972). CMR implements this account because it posits that the state of temporal context evolves with each encoding event in a sequence memory experiment and then associates with the item studied at that time. In addition, the model's context reinstatement mechanisms enact a form of study-phase retrieval, where the context associated with an item's first presentation is reinstated at its second presentation, elaborating on the initial encoding (Greene, 1989; Lohnas & Kahana, 2014a; Maddox, 2016). This mechanism is perhaps particularly relevant for addressing the superadditivity of item repetition, as it suggests that memory traces from items' initial and subsequent presentations interact during encoding to enhance the item's accessibility during recall.

Lohnas & Kahana (2014a)'s major contribution was to simulate and measure key predictions CMR makes about the structure of memory for repeated experience as a consequence of these

mechanisms. Using parameters fitted to a previous dataset, they simulated free recall of study lists following one of four conditions, comparing the model's performance to that of human participants. Within each session, subjects encountered three lists of each of four types. Pure control lists contained all once-presented items and provided baseline recall performance measures for comparison against the other conditions. Pure massed lists contained all twice-presented items, while pure spaced lists contained items presented twice at lags 1-8. Finally, mixed lists contained a mix of once-presented, massed, and spaced items, allowing for examination of repetition effects in the free recall task.

The results of the study were consistent with CMR's predictions. Both the data and CMR reproduced a monotonic increase in recall probability with increasing spacing between repeated items, a hallmark of the spacing effect. Similarly, OR score analyses revealed across applicable conditions that pairs of spaced items were more likely to be recalled than pairs of massed items. And while superadditivity was not directly examined, the spacing effect was more pronounced for repeated items than for distinct items in both data and model simulations, consistent with the phenomenon.

Lohnas & Kahana (2014a) also examined CMR's assertion that studying an item for a second time retrieves and reinstates into context the contextual features associated with its first presentation (Figure 4). Lohnas & Kahana (2014a) identified a potential consequence of this mechanism: since the context from an item's first presentation is reinstated at its second presentation, the contextual states associated with items studied successively to each presentation should be particularly similar, compared to if the study events were unrelated. This increased similarity should in turn lead to heightened rates of transitions from neighbors of the first presentation of a repeated item to neighbors of the second presentation of the item – even if the repeated item itself is not recalled. To test this prediction, in what we will call a "neighbor contiguity analysis", Lohnas & Kahana (2014a) measured the probability of transitioning from a +1 or +2 neighbor of the first presentation of a repeated item to a +1 or +2 neighbor of the second presentation of the item and vice versa across repeated items in experimental data. They used data from control lists (no item repetitions) to establish a baseline rate of

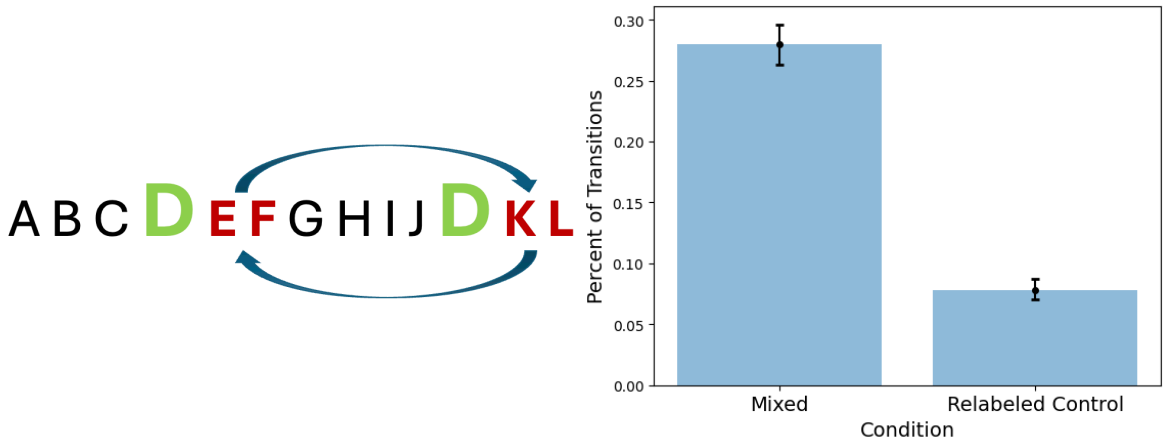
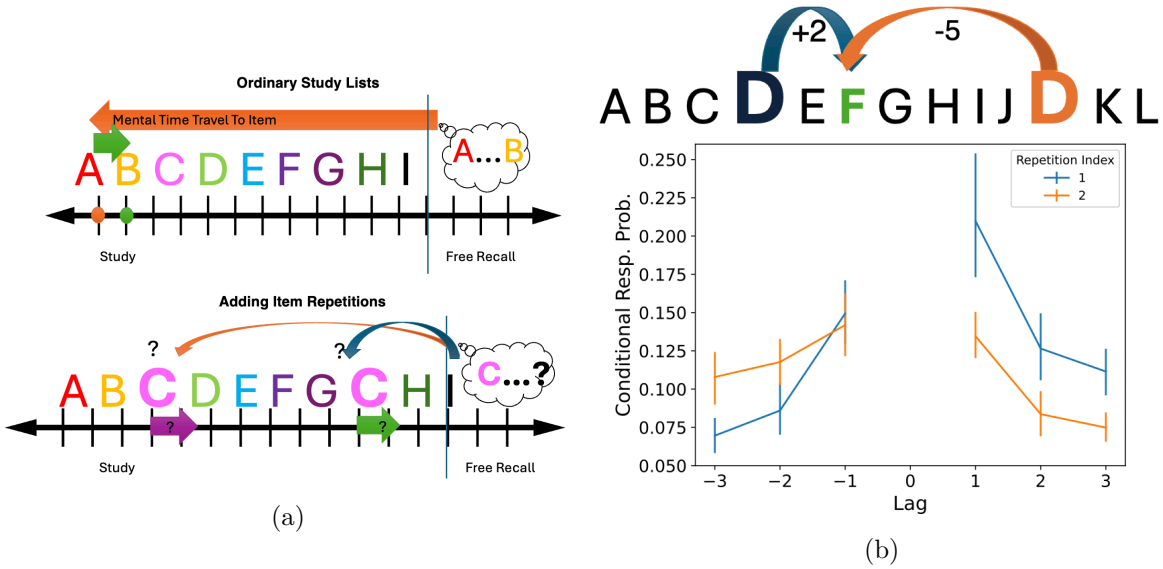


Figure 4: **The repetition neighbor contiguity analysis**. CMR predicts that the context associated with the first presentation of a repeated item is reinstated at the second presentation, associating neighbors of the two presentations. To test this prediction, Lohnas & Kahana (2014a) measured the probability of transitioning from a +1 or +2 neighbor of the first presentation of a repeated item to a +1 or +2 neighbor of the second presentation of the item and vice versa. To control for serial position effects, they used data from control lists (no item repetitions) with study positions matched to shufflings of mixed lists to establish a baseline rate of these transitions.

these transitions, matching study positions used for the analyses to control for serial position effects. Model simulations and participant data were both consistent with this prediction, finding a robust boost in transition rates between repetition neighbors relative to control lists. These results together substantially burnished CMR and retrieved-context theory as accounts of repetition effects in free recall.

Motivation for New Experiment

The lag-CRP analysis played a key role in the advancement of models of free recall by providing a detailed quantitative measure of the temporal structure of memory for word lists that substantially constrained potential generative accounts of memory search in free recall (Howard & Kahana, 2002; Kahana, 2020; Polyn et al., 2009). Recent work in the study of repetition has similarly shifted emphasis from characterizing changes in recall likelihood to



(a) (b)

Figure 5: **The repetition Lag-CRP analysis**. **A.** The lag-contiguity effect has informed a conceptualization of memory search in the free recall task as a traversal over a mental timeline of events, with transitions between items reflecting the temporal proximity of their study positions. However, this notion is disrupted when considering transitions from items with multiple study positions, which have multiple positions on a hypothetical mental timeline. **B.** To investigate the structure of memory for repeated experience, we propose a novel analysis that extends the lag-CRP to repeated items, measuring the probability of transitioning to items at each lag relative to the first and second presentations of repeated items. A comparison of these transition probabilities identifies a differential repetition contiguity effect, where transitions to neighbors of the first presentation of a repeated item are more likely than to neighbors of the second presentation. Figure generated using data from mixed list condition of Lohnas & Kahana (2014a).

understanding how item repetition affects the course of memory search in free recall tasks (Lohnas & Kahana, 2014a). The proposed research will build on this work by applying the lag-CRP analysis to repeated items in free recall to measure the probability of transitioning to items at each lag relative to the first and second presentations of repeated items. This repetition lag-CRP analysis will consider transitions from recalls of repeated items and measure the probability of transitioning to items studied near either presentation of the repeated item. This and related analyses will provide a detailed characterization of the structure of memory for repeated experience in free recall and help evaluate the ability of the Context Maintenance and Retrieval (CMR) model to account for repetition effects in memory.

Preliminary analyses of trials with item repetitions in archival datasets (Kahana & Howard, 2005; Lohnas & Kahana, 2014a) implement the proposed Repetition lag-CRP analysis. Applying these analyses to the archival datasets, we consistently observe a **differential repetition contiguity effect** (Figure 5). Even after accounting for serial position effects, the temporal contiguity effect is more pronounced relative to the first presentation of a repeated item than to the second. Transition rates to the neighbors of the first presentation of a repeated item are higher than to the neighbors of the second presentation, suggesting that the context associated with the first presentation has a stronger influence on memory search. Critically, fitting the CMR model to these datasets and simulating the repetition lag-CRP analysis reveals that the model does not accurately predict these differences, pointing to potential limitations in its current framework.

The central manipulation of the proposed experiment is to vary the initial position of the first presentation of repeated items to test the robustness of the repetition contiguity effect with respect to the primacy and recency effects. The data will prove pivotal for testing the hypothesis that relative differences in the strength of memory traces for repeated items are primarily responsible the repetition contiguity effect, supporting competition between the CMR model and alternative accounts of repetition effects in memory. Other departures from considered archival datasets – such as the choice of immediate free recall over delayed free recall and more controlled allocation of repeater study positions – will help clarify the extent to which

results from the archival datasets can be generalized to new experimental conditions.

The proposed research addresses the limitations of prior evaluations of the CMR model by applying a likelihood-based model comparison technique to assess and quantify the model's ability to predict entire response sequences in free recall for lists with repeated items. This form of model evaluation is more complete than simulation of qualitative patterns as the assessment necessarily penalizes for any systematic discrepancies between the model's predictions and the observed data, even patterns that are not explicitly targeted by the model. We will also apply the generalization criterion method to evaluate the model's ability to account for data from conditions that were not included in the model fitting process. This work will represent the first formal application of the generalization criterion method to evaluate CMR and advance our understanding of the model's robustness and generalizability. Finally, evaluation of alternative models that address the limitations of CMR's current framework will provide a more complete assessment of the mechanisms underlying repetition effects in free recall.

Proposed modeling will focus on adjudicating between three broad hypotheses about the causes of the differential repetition contiguity effect and CMR's difficulty in accounting for it. A deficient-processing hypothesis suggests that learning events for the initial presentation of repeated items are reliably superior to those for the second presentation, especially under conditions of massed repetition. This hypothesis posits that the strength of memory traces for repeated items is the primary driver of the repetition contiguity effect, and that a modification of CMR that modifies learning for the second presentation of repeated items could account for the effect. Compound cueing theory suggests that recalls of repeated items tend to be preceded by recalls of items that were studied near the first presentation of the repeated item, biasing the recall transition toward neighbors of the first presentation of the repeated item. This account points to a potential deficiency in CMR's ability to account for the data's substantial lag-contiguity effects and suggests that the model may be underpredicting the consistency of compound cueing effects across responses that would create a differential repetition lag-contiguity effect. Thirdly, a study-phase retrieval hypothesis posits that CMR's account of reinstatement processes enacted upon restudy of repeated items should be modified

to better account for the differential repetition contiguity effect. By enabling reinstatement and relearning of forward-neighbor context states relative to the first presentation of repeated items, a revised mechanism might drive a differential repetition contiguity effect in the model in a way that aligns with the data. Adjudicating between these variants will involve specifying reasonable implementations, identifying confirmatory or discrepant evidence in the data, and formally comparing the models.

Proposed Experiment

The proposed experiment manipulates the strength of memory traces for repeated items by varying the study position of the first presentation of repeated items. When the position of the first presentation of a repeated item occurs in the middle of a study list, the differential repetition contiguity effect may be ameliorated or even reversed due to interaction with the recency effect: transitions to neighbors of the second presentation of a repeated item may be more strongly cued. By contrast, when the initial position of the first presentation of a repeated item occurs at the beginning of a study list, the differential repetition contiguity effect may be strengthened due to interaction with the primacy effect: transitions to neighbors of the first presentation of a repeated item may be more strongly cued. The overall experiment will further investigate the reproducibility of the deficient repetition contiguity effect and other results in new data. This manipulation will allow us to examine the robustness of the repetition contiguity effect and its interaction with other factors that influence the structure of memory for repeated experience and are addressed for by the CMR model. Such results combined with additional organizational analyses and more robust model comparison techniques will enable full assessment of new potential constraints for models of memory for repeated experience

Methods

The proposed design is a partial replication and extension of the Lohnas & Kahana (2014a) study. Key deviations from the original design aim to examine the reproducibility of key organizational repetition effects under new conditions and to manipulate the strengths of memory

traces associated with repeated items by varying the initial position of the first presentation of repeated items. The original design included pure massed and pure spaced conditions, which are not examined here. New manipulations of the mixed list type will be introduced to examine the interaction of the repetition contiguity effect with the primacy and recency effects.

The experiment will involve approximately 35 subjects who will engage in free recall tasks across 24 lists within a 90-minute session. Word stimuli will be selected from the PEERS word pool, which consists of 1,638 words derived from the University of South Florida free association norms. To control for semantic relatedness, we will ensure that the GloVe embedding similarity value between any two items does not exceed 0.3. Within a session, no word will occur in more than one list. Words will be presented for 2 seconds each, with a 800ms interstimulus interval where a fixation cross will be displayed. After the study phase, participants will perform immediate free recall of the list, with 60 seconds to recall as many items as possible in any order. Participants will be encouraged to continue recalling until the time limit is reached.

The study will utilize a within-subjects design, allowing the examination of individual performance across different conditions under time constraints. To comprise lists, we define two types of items: singletons (items presented only once) and repeaters (items presented twice). Sessions will contain control lists (containing only singletons) and mixed lists (containing both singletons and repeaters). Word lists across conditions will be divided into 32 positions, with 20 singletons and 6 repeaters in the mixed lists. The spacing between repeaters will range from 1 to 6, with equal probability of each lag within a session. We further classify repeaters into primacy repeaters (initial presentation in one of the first five positions of the list), mid-list repeaters (initial presentation in one of the middle ten positions of the list), and recency repeaters (second presentation in one of the last five positions of the list). Mixed lists will contain 2 primacy repeaters, 2 mid-list repeaters, and 2 recency repeaters, with applicable positions randomized within each list.

Archival Datasets

The proposed experiment will be conducted in conjunction with analyses of two archival datasets that have been previously collected and analyzed.

Lohnas & Kahana (2014a)

Across 4 sessions, 35 subjects performed delayed free recall of 48 lists. Subjects encountered four different types of lists:

1. Control lists that contained all once-presented items;
2. pure massed lists containing all twice-presented items;
3. pure spaced lists consisting of items presented twice at lags 1-8, where lag is defined as the number of intervening items between a repeated item's presentations;
4. mixed lists consisting of once presented, massed and spaced items.

Within each session, subjects encountered three lists of each of these four types.

In each list there were 40 presentation positions, such that in the control lists each position was occupied by a singleton item, and in the pure massed and pure spaced lists, 20 unique repeater words occupied the 40 positions. In the mixed lists 28 singletons and six repeater words occupied the 40 positions. In the pure spaced lists, spacings of repeated items were chosen so that each of the lags 1-8 occurred with equal probability. In the mixed lists, massed repetitions (lag=0) and spaced repetitions (lags 1-8) were chosen such that each of the 9 lags of 0-8 were used exactly twice within each session. The order of presentation for the different list types was randomized within each session.

Kahana & Howard (2005)

Sixty-six students studied and attempted free recall of 15 different lists of high-frequency nouns drawn from the Toronto Noun Pool (Friendly et al., 1982). The lists consisted of 30 words, each

repeated three times for a total of 90 presentations per list. List presentation was auditory, and the subjects made their responses vocally into a headset microphone. The words were presented at a rate of 1.5 sec. After list presentation, the subjects were given a distractor task involving simple arithmetic problems of the form A B C ?. The subjects had to correctly answer 15 problems in a row before they could proceed to the recall phase.

There were three list types: massed, spaced short, and spaced long. In the massed lists, each word was repeated three times successively. In the spaced-short lists, the presentation order was randomized, subject to the constraint that the lag between repetitions was at least 2 and no more than 6. For the spaced-long lists, presentation order was randomized, subject to the constraint that inter-repetition lags were at least 6 and not more than 20.

Proposed Analyses

The goal of this project is to characterize the structure of memory for repeated experience and evaluate the ability of the CMR model and plausiable variants to simultaneously account for both key repetition effects and basic benchmark phenomena in free recall across new and archival datasets. Unless specified otherwise, repetition effects analyses will be conducted separately for each of the three types of repeated items: primacy repeaters, mid-list repeaters, and recency repeaters to enable examination of the impact of serial position on repetition effects. Piloting of the task will be used to ensure we are seeing substantial primacy and recency effects.

Recall Performance

A serial position analysis will characterize recall performance by list position. Probability of Nth recall analyses will measure the probability of recalling an item at a specified recall position as a function of its serial position in the list. Primacy and recency effects can also be reflected in this version analysis that focuses on the probability of initiating recall with an

item as a function of its serial position. With respect to repetition effects, we will measure the probability of recall for repeated items as a function of the number of intervening items between their presentations – the spacing effect. This analysis will be conducted for each of the three types of repeated items: primacy repeaters, mid-list repeaters, and recency repeaters. If the data for any variation of these analyses are insufficient to estimate recall rates for specific lags, I will aggregate lags into broader intervals (e.g., 1-2, 3-4, 5-6) to ensure each lag bin contains a sufficient number of observations for reliable estimation. A default hypothesis for the interaction between serial position and repetition effects is that the spacing effect will be more pronounced for primacy and recency repeaters than for mid-list repeaters. Because the strength of the primacy and recency effects tends to vary across experiments, it is unclear a priori whether a differential spacing effect will be observed between primacy and recency repeaters.

Lag-Contiguity Effect Across Singletons and Repeaters

The lag-CRP is the probability of transitioning to an item at a given lag, conditional on such a transition being possible (i.e., the item of the given lag was studied and has not yet been recalled). It involves tabulating for each transition from one item to another in a recall sequence the number of items that intervened between the two items in the study list – its serial "lag" as a numerator value, and the total number of transitions that could have been made to items at that lag as a denominator value. This analysis has revealed that participants are more likely to transition to items that were studied close together in time. The lag-CRP analysis will be applied to the new experimental data and the archival datasets to measure the lag-contiguity effect.

We implement a novel technique to account for item repetitions in the calculation of the basic lag-CRP. Repeaters add a level of complexity to the lag-CRP analysis because they have multiple study positions and thus are considered as having multiple lags during tabulation of possible and observed lag transitions. Previous implementations of the lag-CRP analysis for

mixed lists (Lohnas & Kahana, 2014a) applied a random selection of one of the repeated item's study positions to represent the item in the analysis, repeating this selection over multiple runs of the analysis and then averaging the results. Our implementation is to tabulate the lag-CRP based on the item (rather than study position) actually or possibly recalled, counting lags relative to each of the item's study positions during simulation simultaneously, and removing lags simultaneously when the item is removed as a possible recall transition. This deterministic approach is more computationally efficient and provides a more accurate representation of the lag-CRP for mixed lists.

Repetition Lag-CRP Analysis

I propose a Repetition lag-CRP analysis that separately measures lag-contiguity effects relative to the first and second presentations of a repeated item. For each recall of a repeated item, lags are defined relative to each of its study positions. For example, if a repeated item was studied at positions 5 and 15, a +1 lag transition from the first presentation would be to an item studied at position 6, and a +1 lag transition from the second presentation would be to an item studied at position 16. The repetition lag-CRP separately measures the probability of transitioning to an item at each position-centered lag, conditional on such a transition being possible (i.e., the item of the given lag was studied and has not yet been recalled). This analysis complements the neighbor contiguity analysis from Lohnas & Kahana (2014a), which specifically examined transitions from neighbors of repeated items rather than the repeated items themselves. The neighbor contiguity analysis informs us how contextual states after studying the same item are related to one another, while the repetition lag-CRP analysis informs us how contextual states associated with repeated items are reinstated and influence memory search.

Whereas the initial analysis focus on forward-going transitions from repeated items to their neighbors, we will also consider relationships between recalls of repeated items and the items that were recalled immediately before them. These results will help interpret potential causes of

other observed repetition effects as well as any gaps in model performance. The specification of the analysis involves shifting the repetition lag-CRP analysis to consider transitions TO recalls of repeated items, rather than transitions FROM recalls of repeated items. Lags between the recalled item and the item that was recalled immediately before it are tabulated, and the lag-CRP is calculated as before, one centered on the first presentation of repeated items and one centered on the second presentation of repeated items. The outcome of the analysis will measure whether recalls of repeated items tend to proceed from items that were studied near the first or second presentation of the repeated item. A parallel modification of the lag rank analysis will be applied to summarize the results of this analysis and support statistical comparisons.

Repetition Neighbor Contiguity Analysis

The repetition neighbor contiguity analysis introduced by Lohnas & Kahana (2014a) will be applied to the new experimental data and the dataset from Lohnas & Kahana (2014a). This analysis measures the probability of transitioning from a +1 or +2 neighbor of the first presentation of a repeated item to a +1/+2 neighbor of the second presentation of the item and vice versa across repeated items in experimental data. Critically, this analysis does not directly examine recalls of repeated items, instead testing a prediction about how item repetition shapes the evolution of context during the study phases of list learning experiments to influence transitions during free recall. Reproducing this result in a new dataset would help confirm that this result is robust and generalizable across different experimental conditions and subject pools, while also providing an additional qualitative constraint for model development.

Accounting for Serial Position in Repetition Effects Analyses

While many of our analysis specifically measure the impact of serial position on repetition effects, we will also apply a control analysis to account for serial position effects in the repetition effects analyses. Serial position effects have historically been considered a confound in

analyses of repetition effects in free recall (Lohnas & Kahana, 2014a). For example, if the first presentation of a repeated item is at the beginning of the list while the second presentation is in the middle, observed differences in position-relative repetition lag-CRPs could reflect the primacy effect rather than more specific repetition effects. A proposed technique for controlling for serial position effects is to match the study positions used for the repetition effects analyses to those used for the control analyses (Lohnas & Kahana, 2014a). For 100 iterations, matchings between control list trials and study positions from experimental mixed list trials can be repeatedly sampled with replacement. Repetition effect analyses are then applied to the control list data, assuming that the serial position of the repeated items in the control list is the same as in the experimental data. The outcomes of these analyses can provide baseline expectations of how performance on the repetition effects analyses should differ from chance, given the serial position of the repeated items in the study list.

Statistical Testing

For statistical comparisons, it's useful to summarize measurements of differences between first and second presentation lag-CRPs for each repeated item with a single scalar value. To summarize the effect and control for serial position effects, we measure +1 lag conditional transition rates relative to each repetition index. We perform the repetition lag-contiguity analysis, measuring the +1 lag-conditional transition rates relative to the first and second presentations of repeated items. Where applicable, we also do the same for the control version of the analysis, applied to the control list data using the same study positions as the mixed list data and shuffling assignment 100 times. We then subtract control analysis +1 transition rates from the repetition analysis +1 transition rates to obtained control-corrected +1 transition rates relative to each repetition index. Finally, we perform a subject-level one-tailed paired t-test of scores relative to the first presentation position versus relative to the second presentation position to determine if the difference in transition rates is statistically significant.

Proposed Model Evaluation

To more rigorously evaluate the ability of the Context Maintenance and Retrieval (CMR) model and considered variants to account for repetition effects in free recall, we will employ a combination of quantitative model comparison techniques, generative tests, and the generalization criterion method. First, we will use a likelihood-based model comparison technique introduced by Kragel et al. (2015) to assess how well each model variant can predict the specific sequences of recalls made by participants. This technique compares models based on the log-likelihood they assign to the observed recall sequences, with higher log-likelihoods indicating better model performance. Next, we will test the generative performance of the models by fitting them to data and then simulating new data from the fitted models. By applying our analyses to the simulated data and comparing the results to the observed data, we can directly assess each model's ability to reproduce the key effects of interest. Finally, we will apply the generalization criterion method (Busemeyer & Wang, 2000) to evaluate the models' ability to account for data from conditions that were not included in the fitting process. Specifically, we will fit the models to data from trials without repeated items and then test their ability to predict data from trials with repeated items, using the previously estimated parameters. This will allow us to assess the generalizability of the models' mechanisms and identify any limitations in their ability to account for repetition effects.

Sequence-Based Model Evaluation

According to the likelihood-based evaluation technique, repeated items and intrusions (responses naming items not presented in the list) are excluded from participants' recall sequences. Given an arbitrary parameter configuration and sequences of recalls to predict, a model simulates encoding of each item presented in the corresponding study list in its respective order. Then, beginning with the first item the participant recalled in the trial, the probability assigned by the model to the recall event is recorded. Next, the model simulates retrieval of that item, and given its updated state is used to similarly predict the next event in the recall sequence -

either retrieval of another item, or termination of recall - and so on until retrieval terminates. The probability that the model assigns to each event in the recall sequence conditional on previous trial events are thus all recorded. These recorded probabilities are then log-transformed and summed to obtain the log-likelihood of the entire sequence. Across an entire dataset containing multiple trials, sequence log-likelihoods can be summed to obtain a log-likelihood of the entire dataset given the model and its parameters. Higher log-likelihoods assigned to datasets by a model correspond to better effectiveness accounting for those datasets.

Generative Performance

While quantitative model comparison helps measure relative evidence in favor of one model over another, tests of the ability of candidate models to produce specific effects from data can provide more direct evidence for or against a model's account of memory (Palminteri et al., 2017). Failure to reproduce an effect that is well-established in the literature can effectively falsify a model, requiring new or modified models to account for the data. To evaluate the generative performance of the CMR model and its variants, we will first fit model variants to applicable data using the likelihood-based evaluation and fitting techniques reviewed above. Next, we will simulate data from the model using the fitted parameters and apply proposed analyses to the simulated data to measure the model's ability to predict the outcomes of these analyses. The mean squared error between the model's predictions and the observed data will be used to quantify the model's generative performance.

Generalization Criterion

The generalization criterion method (Busemeyer & Wang, 2000) is a formal method for evaluating the ability of a model to account for data from conditions that were not included in the model fitting process. The method is based on the idea that a model's ability to account for data from a condition not included in the fitting process is a measure of its ability to generalize to new conditions. If a model fits well to one condition but poorly generalizes to a

new condition under this method, it suggests that the model's account of the first condition is not generalizable to the second condition. That is, there are details of the first condition that are not captured by the model and that are necessary to account for the data from the second condition. In the context of the present work, the generalization criterion method can be used to evaluate the ability of the CMR model to account for repetition effects in free recall using mechanisms originally fitted to account for free recall of lists with no repeated items.

The generalization criterion employs samples from two different research designs:

1. During the calibration stage, competing models are fit to data from one design.
2. Next, the previously estimated parameters from the first design are used again to compare predictive accuracy for the second design.
3. If (A) model parameters were estimated with sufficient precision in the first stage and (B) model performance is distinguishable in the second stage, you have grounds for selecting one model over the other.

In general, the application of the generalization criterion to models of free recall has been limited despite a range of relevant research designs and data that could be used to apply the method. But in a new experiment featuring trials with repeated items and trials with no repeated items, the generalization criterion can be used to identify the limitations of the CMR model's account of repetition effects in free recall that may otherwise be obscured by the model's flexibility in fitting to the data and identify advantages of alternative accounts. CMR applies the same mechanisms to account for free recall of lists with either repeated or nonrepeated items. Indeed, since it used parameters fitted to a separate dataset that featured only once-presented items, the initial demonstration of CMR's ability to account for repetition effects in free recall (Lohnas & Kahana, 2014a) was itself a test of the model's ability to generalize to lists with repeated items. However, differences in task design and subject pool properties between the two datasets limited the interpretability of this test since other factors besides the use of repeated items could have contributed to differences in performance between the two datasets. Therefore CMR performance on a more formal test of generalization to a

new condition can help clarify its ability to account for repetition effects in free recall.

Expected Outcomes and Preliminary Research

Our experiment seeks to characterize the structure of memory for repeated experience in free recall through a strategy of dissection. First, the proposed repetition lag-CRP analysis measures the distinct influence of each presentation of a repeated item on the temporal organization of memory search by assessing the probability of transitioning from recalls of repeated items to neighbors of each presentation. Second, the proposed experimental manipulation of the presentation positions of repeated items allows control over the relative contribution of initial and secondary study events to memory for repeated items. This manipulation provides an additional window into the separate and combined influences of how each memory trace for repeated items separately contributes to repetition effects. Together, this approach will help identify factors underlying the differential repetition contiguity effect, clarify its interaction with primacy and recency effects, and provide a more complete evaluation of the CMR model's account of repetition effects in free recall.

Here we discuss more concretely the empirical and theoretical outcomes that we expect from the proposed experiment. In particular, we expect to reproduce key outcomes of our preliminary analyses of archival datasets, observing a differential repetition contiguity effect in new experimental data that is not accounted for by the CMR model while otherwise replicating the model's ability to account for well-known repetition and benchmark organizational effects in free recall. We expect that placing the initial presentation of repeated items closer to the beginning of the list will strengthen the differential repetition contiguity effect by boosting the influence of context associated with their first presentation in memory search, with opposite effects expected when repeated items' second presentation is placed near end of the list. De-

pending on the outcomes of these and related analyses, we will be able to adjudicate between competing hypotheses about the causes of the differential repetition contiguity effect and the CMR model's ability to account for it, and identify potential modifications to the model that could better account for the data.

Replication of Preliminary Analyses of Archival Data

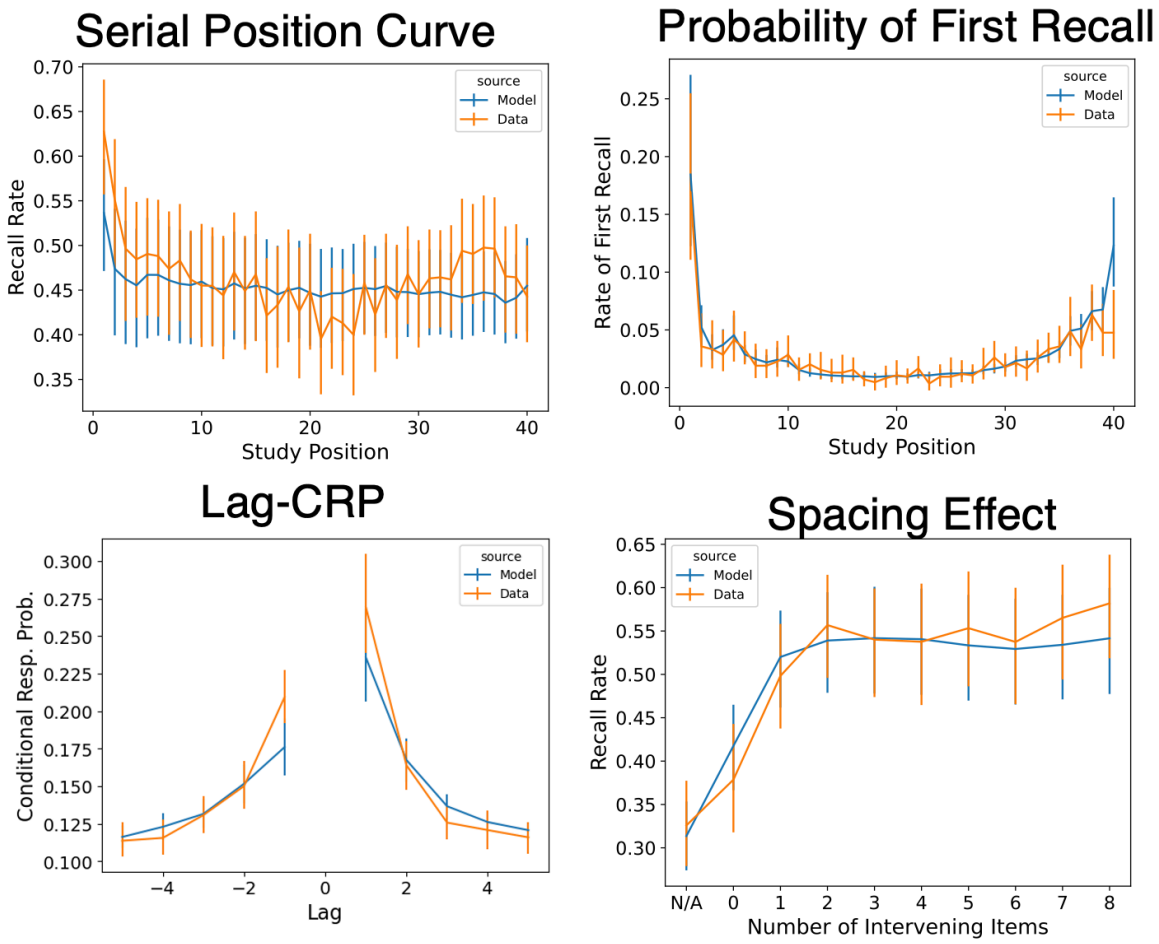


Figure 1: **Simulation of the CMR model fitted to the Lohnas & Kahana (2014a) dataset**. The CMR model was fit to the mixed list condition of the Lohnas & Kahana (2014a) dataset using the likelihood-based fitting technique. The model broadly accounted for benchmark temporal organizational phenomena (primacy effect, recency effect, lag-contiguity effect) as well as the spacing effect.

Previous analyses of the Lohnas & Kahana (2014a) and Kahana & Howard (2005) archival datasets substantially inform expectations and provide motivating questions for the proposed experiment. Following the methodology described above, we fit the CMR model to each dataset and simulated both benchmark and repetition effects analyses. Matching with results from Lohnas & Kahana (2014a), CMR successfully accounted for the spacing effect in the Lohnas & Kahana (2014a) dataset, with the probability of recall increasing with increasing spacing between repeated items. The model also reproduced outcomes of the repetition neighbor contiguity analysis, showing a robust boost in transition rates between neighbors of repeated items relative to control lists, consistent with the hypothesis that encoding the second presentation of an item retrieves and reinstates the context of the first presentation, influencing the context of subsequent study events. The model also broadly accounted for outcomes of the lag-CRP and serial position effect analysis, showing a preference for transitions to temporally adjacent items and a U-shaped serial position curve reflecting the primacy and recency effects (Figure 1).

However, the model could not account for outcomes of the repetition lag-CRP analysis, which revealed a differential repetition contiguity effect in both the Lohnas & Kahana (2014a) and Kahana & Howard (2005) datasets. In both datasets, participants reliably showed a preference for transitioning to items studied near the first presentation of a repeated item relative to the second. To measure the proportion of transitions expected in the absence of repeated items, using just the Lohnas & Kahana (2014a) dataset we match the serial positions used in the mixed lists to corresponding items in the control lists, shuffling assignment 100 times. The effect disappeared under this control analysis. Applying our proposed statistical testing to the control-corrected +1 transition rates relative to each repetition index, we find that the difference in transition rates is statistically significant (t=3.704; p=0004), suggesting that the differential repetition contiguity effect is meaningful and not due to serial position effects. Applying the same statistical test to the Kahana & Howard (2005) dataset without controlling for serial position effects (because control list data is not available), we find that the difference in transition rates is also statistically significant (t=2.909; p=0.0025), suggesting that the

differential repetition contiguity effect is robust across datasets.

When we apply the same analyses and statistical tests to data simulated from the CMR model after fitting to the Lohnas & Kahana (2014a) dataset, we find that the model does not predict the differential repetition contiguity effect beyond that accounted for by our control analysis for serial position effects. Instead, the model predicts that participants should transition to items studied near the second presentation of a repeated item at roughly the same rate as to items studied near the first presentation – $t=1.07$; $p=0.1456$ for Lohnas & Kahana (2014a), and $t=0.93$; $p=0.1771$ for Kahana & Howard (2005). This suggests that the model may not be able to account for the differential repetition contiguity effect, even after fitting to data that exhibit the effect. These results together suggest that the differential repetition contiguity effect may reflect a more complex interaction between initial and secondary study events for repeated items that is not directly captured by CMR or addressed by other analyses of repetition effects in free recall.

We expect that the proposed experiment will replicate the differential repetition contiguity effect observed in the archival datasets in the new experimental data. We further expect CMR to continue to struggle to account for the differential repetition contiguity effect, even after fitting parameters directly to applicable data. On the other hand, we expect CMR's ability to account for other repetition and benchmark effects to be robust, as it has been in prior evaluations. These outcomes would validate the need for alternative models that integrate other accounts of repetition effects and allow examination of how our proposed experimental manipulations affect the strength of the effect and potentially elucidate its underlying causes.

Figure 2: Simulation of repetition lag-CRP analysis and control for serial position effects from Lohnas & Kahana (2014a) dataset. **Top Row**: Main and control (left and right) repetition lag-CRP analysis results for the Lohnas & Kahana (2014a) dataset. **Middle Row**: Simulated main and control analysis results from the CMR model fitted to the Lohnas & Kahana (2014a) dataset. **Bottom Row**: Observed and simulated (left and right) analysis results for the Kahana & Howard (2005) dataset.

Manipulation of Initial Position of Repeated Items: Testing the Deficient Processing Hypothesis

Our experimental manipulation of the initial position of the first presentation of repeated items directly addresses a particularly intuitive hypothesis about the causes of the differential repetition contiguity effect. According to the deficient processing hypothesis, when items are repeated, the encoding of the second presentation of the item is less detailed than the first, leading to a weaker memory trace and less effective retrieval cues (Greene, 1989; Maddox, 2016). The most detailed specification of this hypothesis is that the strength of memory traces associated with repeated items is inversely proportional to the lag between their presentations, explaining the spacing effect in free recall using mechanisms entirely distinct from those proposed by CMR. A range of mechanisms for this deficiency have been proposed, including reduced allocation of attentional resources, interference from the first presentation, and a lack of novelty in the second presentation (Murdock Jr, 1961). Following this hypothesis, the differential repetition contiguity effect may be driven by differences in the accessibility of memory traces associated with repeated items. The second presentation of a repeated item may be less effectively encoded than the first, making it less influential in the course of memory search when a repeated item is recalled.

Our manipulation of the initial position of the first presentation of repeated items uses the primacy and recency effects to examine this hypothesis by examining how the strength of the repetition contiguity effect changes with the primacy and recency effects. Importantly, though we manipulate the influence of primacy and recency effects on the encoding of repeated items, our hypothesis is not primacy and recency effects explain the differential repetition contiguity effect. Instead, we seek to explore whether differences in the strength of memory traces associated with repeated items can influence the effect, identifying a mechanism relevant to its structure. A deficient processing account of differential repetition contiguity would predict that the effect will be most pronounced when the initial position of the first presentation of a repeated item occurs at the beginning of a study list due to interaction with the primacy

effect. It would further predict that the effect is weakest when the second presentation of a repeated item occurs near the end of a study list.

If our experiment observes these patterns, these results might motivate an extension of retrieved context theory and the CMR model that incorporates the deficient processing account of repetition effects in memory. Since deficient encoding would only be engaged when items are repeated, such a mechanism could account for the differential repetition contiguity effect in mixed list trials without affecting simulations of control list data and allow for a more complete account of repetition effects in free recall. Speculating about the possibility that CMR may require further refinement to account for repetition effects in memory, Lohnas & Kahana (2014a) suggested a way it could integrate the deficient-processing theory of the spacing effect into its existing framework. Its proposal in the literature highlights its intuitive appeal and makes it deserving of further investigation. The authors propose adding deficient processing by having the amount of context retrieved for a repeated item decreases as a function of its lag/spacing.

In a more detailed proposed, Collins et al. (2020) presented MINERVA-DE, an extension of the classic MINERVA 2 model (Hintzman, 1984), that integrates a deficient-processing mechanism into the instance-based model to account for spacing and repetition effects in recognition memory. MINERVA-DE conceptualizes deficient-processing as an attentional artifact, driven by "temporary, familiarity-mediated shifts in exogenous attention that bias attention toward unfamiliar, not-repeated words and away from familiar, repeated words". In MINERVA-DE, a short-term memory store called primary memory is maintained with a long-term secondary memory. When a word is attended, its familiarity is computed relative to the items in primary memory. The more familiar a studied item is to existing items in primary memory, the less well it is encoded in secondary memory. This process is called discrepancy encoding, because encoding is strongest for items that are discrepant with information already stored in primary memory. CMR already has an attention process mediating learning rates; Sederberg et al. (2008) proposes one to enforce the primacy effect, scaling the amount of learning based on the serial position of the studied item according to:

$$\Delta M_{exp}^{CF} = \phi_i f_i c_i^{'}$$

I tried adding another mechanism that similarly scales the amount of learning based on the "familiarity" of the encoded item, calculated on the basis of (cosine) similarity to the current state of C. A familiarity_scale parameter ζ_s would control the amount of this scaling, similar to ϕ_s:

$$\zeta_i = (1 - S(f_i, c_i))^{\zeta_s}$$

Configured this way, when ζ_s is 0, the effect of familiarity on learning will be nil, but as it grows, that effect will increase.

A range of options exist for including such a mechanism in the CMR model. During encoding of the second presentation of an item in a list, this mechanism could be applied selectively to variable scale learning of feature-to-context associations, learning of context-to-feature associations, the amount of context retrieved during recall, the amount of context associated with the item reinstated during encoding, or a combination of these. Furthermore, a simpler version of the mechanism could, instead of scaling processing deficiency as a function of lag, apply a fixed reduction in the encoding quality of successive presentations of an item. By evaluating a range of candidate models that integrate deficient processing into the CMR model, we can identify the most effective and parsimonious account of repetition effects in free recall and provide a more detailed and specific characterization of the structure of memory for repeated experience.

While this approach to account for the differential repetition contiguity effect and other repetition effects in free recall has intuitive appeal, there are reasons to suspect that it may not be supported by our data and in model evaluation. First, diminished encoding of the second presentation of a repeated item potentially fails to address or even directly works against other known repetition effects in memory. In particular, the deficient encoding hypothesis weakens

the basic repetition effect, which predicts that the probability of recall increases with the number of times an item is presented. Furthermore, the deficient encoding hypothesis may not be consistent with the superadditivity of repetition in free recall, where the probability of recall for a repeated item with spaced presentations is greater than the sum of the probabilities of recall for either of two distinct items presented in the same positions (Maddox, 2016). Proposed deficient encoding mechanisms specifically interfere with processing of item repetitions in seeming direct opposition to this phenomenon. Implemented within the CMR model, the tension between the need to account for the differential repetition contiguity effect and the need to account for other repetition effects in memory may require a more complex and nuanced account of repetition effects in memory. This gap in the deficient processing hypothesis is a common challenge in the literature, and our experiment will contribute to the ongoing effort to resolve it (Maddox, 2016).

In addition, my preliminary simulations of model extensions discussed here further suggest this mechanism, despite support in the broader literature (Collins et al., 2020; Greene, 1989), poorly accounts for applicable empirical constraints in free recall for lists with repeated items. In proposed variants manipulating contextual reinstatement for example, deficient encoding of an item presentation in a way that affects the evolution of context during the study phase of a list learning experiment can diminish the ability for an item's recall to cue the recall of nearby items by disrupting the strength of association between study contexts.

On the other hand, proposed variants that ultimately work to manipulate the extent to which the study context associated with an item's first presentation is reinstated upon recall of the item relative to its second presentation may not be a sufficiently strong lever to drive the differential repetition contiguity effect. The reason why is that study contexts associated with the first and second presentations of a repeated item are not especially different from one another. Both study events for a twice-presented item reinstate the item's pre-experimental contextual associations. The second study event for a repeated item additionally reinstates the context of the first study event, but this only serves to enhance the similarity of the contextual states to be associated with the item in the memory traces for each of the item's presentations.

Consequently, manipulations of CMR's simulation to exclusively reinstate the study context of a repeated item's first presentation upon recall of the item have not been successful in supporting a large difference in the probability of transitioning to neighbors of the item's first presentation compared to neighbors of the item's second presentation.

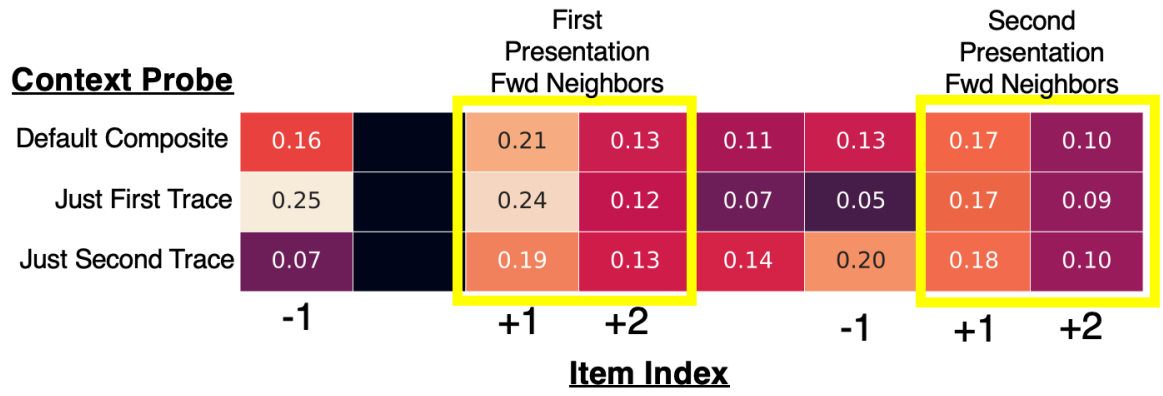
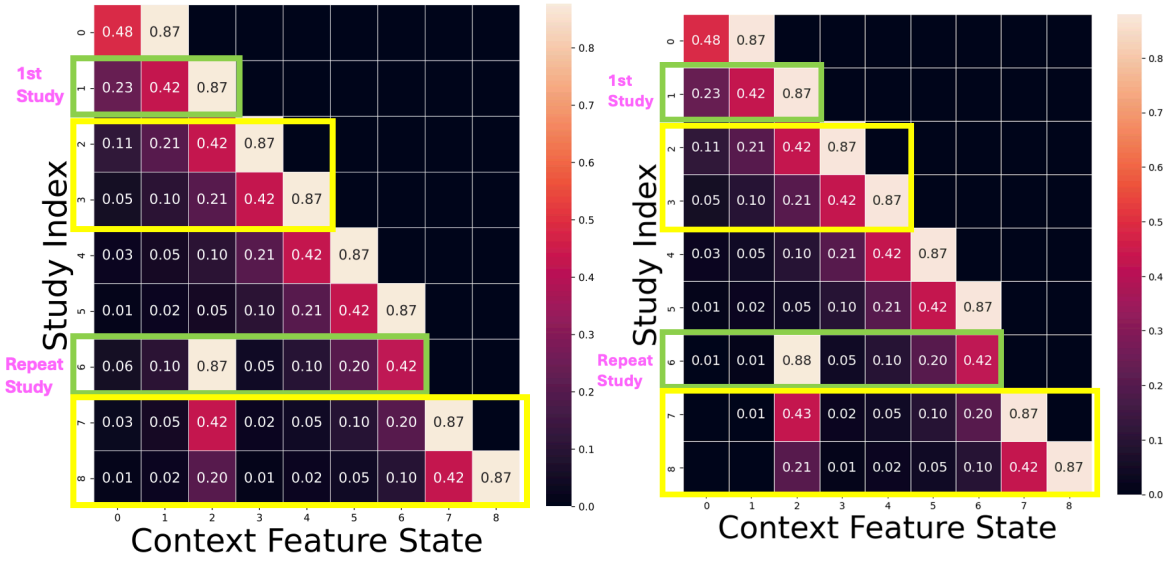


Figure 3: **Predicted transition rate as a function of context cue given arbitrary subject's parameters.** Many proposed implementations of deficient processing and study phase accounts of repetition effects traditionally explain phenomena such as the spacing effect in terms of diminished access at retrieval to the context associated with the second presentation of a repeated item relative to the first. However, driving retrieval with either trace's context does not substantially bias transitions to either presentation's +1 or +2 neighbors compared using a composite of both contexts. This suggests that trace dominance in memory search may not be a strong lever for explaining the differential repetition contiguity effect.

This issue could potentially defeat a broad class of potential mechanisms that emphasize an asymmetry in how study contexts are reinstated upon recall of repeated items to explain the differential repetition contiguity effect. Other mechanisms in this class could include those that selectively strengthen the association between a repeated item and its first presentation context during the second presentation of the item, or retrieval-oriented mechanisms that select individual memory traces for reinstatement based on their similarity to the current context state. In the proposed research, I will develop these simulations to demonstrate why this intuitive approach to account for the differential repetition contiguity effect may be substantilally flawed.



(a) Full CMR Model

(b) Reduced CMR Model

Figure 4: Comparison of contextual states during encoding of a list with a repeated item in the full and reduced CMR models. **A**. When a repeated item is presented for the second time, the full CMR model reinstates the context of the item's first presentation along with its pre-experimental contextual associations. **B**. In the reduced CMR model, only pre-experimental contextual associations are reinstated upon encoding of the second presentation of a repeated item. In either model, since pre-experimental contextual associations are reinstated upon encoding of either presentation, +1/+2 neighbors of either presentation (study indices 2/3 and 7/8) exhibit similar contextual states. This similarity is only marginally affected by reinstatement of first-presentation context during the second presentation of the item in the full model.

The study-phase retrieval hypothesis represents another influential account of repetition effects in free recall that could potentially explain the differential repetition contiguity effect. In its general form, the study-phase retrieval hypothesis posits that studying an item for a second time may trigger retrieval of the first presentation of the item, facilating memory for the item (Maddox, 2016). Initial proposals of the study-phase reinstatement mechanism argued that retrieval of an item's prior presentations were necessary to support the spacing effect in memory tasks (Hintzman, 1974). Other work, including the CMR model, gives the mechanism a complementary role in explaining repetition effects in free recall (Lohnas & Kahana, 2014a;

Maddox, 2016). In particular, by specifying an interaction between traces of the first and second presentations of a repeated item that does not occur for distinct items, the study-phase retrieval is argued to help explain the superadditivity of item repetitions – the phenomenon where spacing effects are more pronounced for repeated items than for distinct items in the same study positions in study lists. A study-phase retrieval mechanism could also explain the differential repetition contiguity effect by suggesting that retrieval of the first presentation of a repeated item upon encoding its second presentation might bias memory for repeated items toward neighbors of the first presentation.

The CMR model's context reinstatement mechanism already implements a form study-phase retrieval, where the context associated with an item's first presentation is reinstated during the second presentation of the item. This mechanism is argued to support both the model's ability to account for the spacing effect, the superadditive advantage of item repetition in free recall, and the repetition neighbor contiguity effect in free recall (Lohnas & Kahana, 2014a). Given these successes, it is perplexing that the model does not predict a differential repetition contiguity effect in free recall, as observed in the Lohnas & Kahana (2014a) and Kahana & Howard (2005) datasets. But a portion of my preliminary research has focused on re-examining how CMR implements the contextual reinstatement retrieval mechanism argued to underlie key repetition effects in free recall (Lohnas & Kahana, 2014a), interpret why it does not contribute to a differential repetition contiguity effect, and consider whether the mechanism deserves revision. Strikingly, this research has called into question whether CMR's implementation of study-phase retrieval contributes at all to its ability to explain any repetition effects in free recall discussed in this proposal.

To investigate the mechanism, we can implement a variant of the CMR model that does not include the study-phase retrieval mechanism and compare its performance to the full model. In the reduced variant, when an item is presented for a second time, rather than reinstate the context of the first presentation of the item along with its pre-experimental contextual associations, only pre-experimental contextual associations are reinstated – the same as when the item is presented for the first time. This variant can then be fitted to the Lohnas & Kahana

(2014a) dataset and evaluated using the likelihood-based model comparison and simulation of applicable repetition and benchmark effect analyses. We can also perform focused analyses of the internal states of the model after simulation of study lists to measure how the model's implementation of study-phase retrieval affects associations between repeated items and their neighbors.

In preliminary simulations of the reduced CMR model, we find that the model accounts for performance on the Lohnas & Kahana (2014a) dataset comparably to the full model. Average log-likelihoods assigned to the dataset by the reduced model are indistinguishable from those assigned by the full model and the model's ability to account for repetition and benchmark effects in free recall is largely unchanged. These outcomes suggest that the study-phase retrieval mechanism does not substantially contribute to the model's ability to account for the data.

The most striking result of this research is that the reduced CMR model successfully accounts for outcomes of the repetition neighbor contiguity analysis initially proposed by Lohnas & Kahana (2014a) specifically to test CMR's implementation of study-phase retrieval. According to the premise of the analysis, reinstatement of first presentation context during the second presentation of a repeated item should enhance the similarity of contextual states associated with neighbors of each presentation of the item, driving a boost in transition rates between neighbors of repeated items relative to control lists. However, reinstatement of pre-experimental context alone upon encoding of either presentation already enforces this pattern of similarity between neighbors of repeated items, with the addition similarity contributed by reinstatement of first-presentation context during the second presentation of the item only marginally affecting the outcome by comparison.

Similarly, CMR's implementation of study phase reinstatement does not clearly improve the model's ability to account for the spacing effect in free recall. While preliminary analyses so far do not evaluate either CMR or its reduced variant's simulation of recall rates for spaced distinct items (using the OR score analysis described above), the model's ability to account for the spacing effect in free recall after parameter fitting is not substantially affected by the presence or absence of the study-phase retrieval mechanism. The effect of excluding the

mechanism on simulation of the differential repetition contiguity effect is also minimal. Indeed, by enhancing the similarity between first and second presentation study contexts associated with repeated items, the study-phase retrieval mechanism may actually disrupt the model's ability to account for the differential repetition contiguity effect by reducing the ability of the model to selectively target neighbors of the first presentation of a repeated item upon recall of the item. The proposed research will seek to confirm these results and provide a more detailed and specific characterization of CMR's mechanisms for encoding and retrieval of repeated items.

The CMR model's current implementation of study-phase retrieval effectively works to associate the context of a repeated item's first presentation with that the item's second presentation, but does not clearly contribute to the basic repetition effect or the kind of dominance of first-presentation context during retrieval that would support a differential repetition contiguity effect. An alternative mechanism could be supposed that targets for directly strengthening initial presentation memory traces during the second presentation of a repeated item. Compared to the deficient processing hypothesis, this mechanism would similarly support the superadditivity of item repetitions in free recall. At the same time, it would sidestep the need to reduce the encoding quality of the second presentation of a repeated item, potentially avoiding conflicts with other repetition effects in memory that focus on its mnemonic advantages over encoding of distinct items. However,depending on its implementation, since contextual states associated with the first and second presentations of a repeated item would still be similar, this mechanism could still struggle to account for the differential repetition contiguity effect for the same reasons that a second-presentation deficient processing mechanism might. A strategy for avoiding this issue could ignore modification of trace accessibility during context reinstatement enacted during free recall (involving modifications to the evolution of context during encoding or to learned item-to-context associations), and instead focus on the strength of associations between repeated items and their neighbors during encoding. New data will help constrain possible mechanisms for study-phase retrieval in the CMR model and identify potential modifications that could better account for differential repetition lag-contiguity in

interaction with other repetition and temporal organization effects in free recall.

Exploring the Compound Cueing Hypothesis

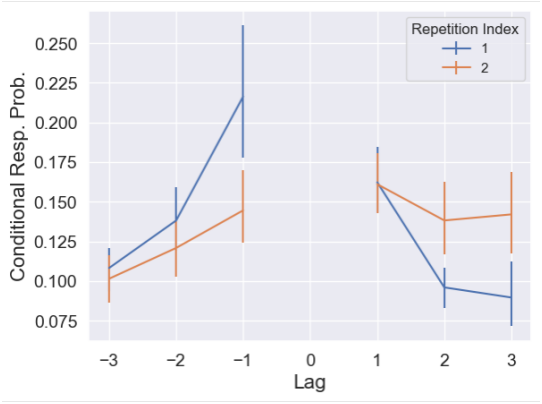


Figure 5: **Reverse repetition lag-CRP analysis results for the mixed list condition of the Lohnas & Kahana (2014a) dataset.** The specification of the analysis involves shifting the repetition lag-CRP analysis to consider transitions TO recalls of repeated items, rather than transitions FROM recalls of repeated items. Lags between the recalled item and the item that was recalled immediately before it are tabulated, and the lag-CRP is calculated as before, one centered on the first presentation of repeated items and one centered on the second presentation of repeated items. Analysis outcomes suggest that recalls of repeated items tend to be preceded by recalls of items studied near the first presentation of the repeated item, potentially supporting the differential repetition contiguity effect in successive transitions from repeated items.

Another attractive account of the differential repetition contiguity effect focuses on the possibility that items recalled before recalls of repeater items bias the following recall transition toward neighbors of the first presentation of the repeated item. CMR posits that temporal context changes gradually, only partially replacing its previous state with context reinstated from the current item. This implies that context reinstated from retrieval of an item can have persistent influence beyond the first successive retrieval event (Lohnas & Kahana, 2014b). If recalls of repeated items tend to be preceded by recalls of items that were studied near the first presentation of the repeated item, this could bias the recall transition toward neighbors

of the first presentation of the repeated item and explain the effect.

A few preliminary results make this account attractive. First, an important result of applying our *reverse* variant of the repetition lag-CRP analysis to archival datasets was the discovery that recalls of repeated items do indeed tend to be preceded by recalls of items that were studied near the first presentation of the repeated item (Figure 5). Second, our initial fits of the CMR model to the Lohnas & Kahana (2014a) dataset suggest that along with failing to predict a differential repetition contiguity effect in our repetition lag-CRP analysis, the model also simulates overly flat lag-contiguity effects (Figure 1) across responses in both control and mixed list conditions when applying our regular lag-CRP analysis. This suggests that the model may be underpredicting the frequency of +1 transitions across the dataset, reducing occurrences of compound cueing effects that would bias transitions toward neighbors of the first presentation of repeated items. This observation potentially explains CMR's failure to predict a differential repetition contiguity effect in terms of its failure to accurately account for the data's substantial lag-contiguity effects. Manually increasing parameters in CMR responsible for the strength of the lag-contiguity effect do indeed also create a substantial differential repetition contiguity effect, but at the cost of disrupting other benchmarks and overall model fit.

These preliminary results indicate that CMR is systematically underpredicting the strength of temporal organization, even in trials without item repetitions. The compound cueing hypothesis provides an account for why such systematic underprediction could cause the CMR model to fail to predict the differential repetition contiguity effect by limiting the extend to which persistent context from prior recalls can bias transitions from repeated items toward neighbors of their first presentation. To investigate this possibility, we will explore strategies to diagnose gaps in preliminary fits of the CMR model to the Lohnas & Kahana (2014a) dataset and develop new fits that better account for the data's substantial lag-contiguity effects. This work will help clarify the role of compound cueing in the differential repetition contiguity effect and allow fuller arbitration between all hypotheses considered here with greater confidence.

Conclusion

This study examines how the lag-contiguity effect – the tendency for items positioned close together in a study list to be retrieved successively during free recall (Kahana, 1996) – manifests for items with multiple study positions. The lag-contiguity effect has informed a conceptualization of memory search in the free recall task as a traversal over a mental timeline of events, with transitions between items reflecting the temporal proximity of their study positions. However, this notion is disrupted when considering transitions from items with multiple study positions, which have multiple positions on a hypothetical mental timeline. Given the central role of the lag-contiguity effect in theories of memory search (Howard & Kahana, 2002; Kahana, 2020; Polyn et al., 2009), understanding how it manifests for recalls of repeated items is crucial for refining our understanding of memory for repeated experience.

In the broader context of my thesis, the proposed research will contribute to understanding of how sequence learning and memory depends on content-based relationships between items in the sequence. Other proposed research examines the same question where content relationships are provided by semantic similarity and narrative structure. These constructs are comparatively complex and abstract next to the simple repetition relationships considered in the current proposed research, and require addressing more complex questions about how applicable relations are perceived in represented in memory. A robust account of repetition effects in free recall could help ground such research in a more fundamental understanding of how memory for sequences is structured and help advance toward a unified account that addresses both simple and complex relationships between items in a sequence.

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Similarity versus repetition: An integrative retrieved-context account of semantic and repetition effects in free recall

Jordan Gunn Sean Polyn

This study examines whether the mechanisms underlying repetition and similarity effects in free recall are fundamentally distinct or reflect a continuum over a single underlying process. Despite the intuitive appeal of the continuum hypothesis, an apparent contradiction arises from the contrasting effects of spacing on memory performance: while repeated items benefit from spaced presentation (Cepeda et al., 2006), semantically similar items often show enhanced recall with massed presentation (Borges & Mangler, 1972; Glanzer, 1969; Puff, 1966). Participants will study lists that interleave unrelated items with either twice-presented items or pairs of highly semantically similar items. The design also varies both the spacing between presentations of key items and the proximity of presentations to the beginning or end of lists, enabling investigation of how spacing and serial position effects manifest for different types of item relationships. Applying a retrieved context framework to these data, I will extend traditional measures of repetition effects to study non-repeating but similar word pairs. Analyses will interrogate whether encoding of similar or repeated items triggers the reinstatement of similar context, whether similarity or repetition can modulate the benefit of spaced over massed study for memory performance, and whether these factors similarly interact with serial position to structure response organization. Results will support formal comparison between models that variously treat semantic and repetition effects as engaging distinct or overlapping processes, informing the development of an integrated account of how item and contextual relationships shape memory search in free recall.

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Introduction

What is the relationship between semantic similarity and repetition effects in free recall? Similarity at its most extreme is identity. This perspective suggests a continuum between the two concepts, with repetition representing the most extreme form of similarity and similarity conversely representing a kind of partial repetition – the repetition of shared features between items. This continuum hypothesis posits that the mechanisms underlying both repetition and similarity effects in memory search are fundamentally the same, differing only in the degree of overlap between item features. However, in the study of memory search, the processing of similar and identical items is often treated as distinct phenomena, each with its own theoretical explanations and empirical predictions.

In free recall tasks, the most prominent effect of similarity is the semantic contiguity effect, where participants are more likely to transition from one studied item to another if the items are semantically similar (Bousfield, 1953; Glanzer, 1969; Howard & Kahana, 2002b; Romney et al., 1993). On the other hand, repetition effects in free recall have primarily been examined through the spacing effect, which highlights the advantage of distributed over massed study for memory performance (Cepeda et al., 2006; Ebbinghaus, 1885). Notably, the benefit for repeated items tends to be superadditive, suggesting that memory mechanisms unique to item repetition facilitate this effect (Benjamin & Tullis, 2010; L. J. Lohnas et al., 2011; Maddox, 2016). Despite the intuitive appeal of the continuum hypothesis, an apparent contradiction arises from the contrasting effects of spacing on memory performance: while repeated items benefit from spaced presentation, semantically similar items often show enhanced recall with

massed presentation (Borges & Mangler, 1972; Glanzer, 1969; Puff, 1966). This contrast challenges the idea that the same mechanisms underlie both repetition and similarity effects.

The Context Maintenance and Retrieval (CMR) model (Polyn et al., 2009), a computational implementation of the retrieved context theory (RCT) of memory search, offers a promising approach to addressing this challenge by providing a structured framework for understanding how both repetition and similarity effects might operate within a single, integrated system. RCT asserts that performance in free recall tasks is mediated by a representation of temporal context that evolves by reinstating and gradually integrating the context associated with items as they are studied. Studied items become associated with the current context, and this context is updated with item-specific information each time an item is studied or recalled. During recall, the current state of context drives memory search, with the likelihood of recalling an item increasing as the similarity between the current context and the context associated with that item grows. Moreover, recalling an item reinstates its associated context, providing a cue for subsequent retrievals that tend to prioritize nearby items in the study list (Kahana, 1996, 2020). These mechanisms primarily address the temporal organization of memory search, especially components of the serial position effect (Murdock Jr, 1962) and the lag-contiguity effect (Howard & Kahana, 2002a; Kahana, 1996). However, the CMR model has also been extended to account for a range of repetition and semantic organizational effects in memory search, including simulations of recall for lists with repeated (L. L. Lohnas & Kahana, 2014) or related items (Morton & Polyn, 2016; Polyn et al., 2009). By providing a unified framework for both types of effects, CMR allows for the possibility that repetition and similarity effects might share common underlying processes.

Although CMR successfully models both repetition and semantic effects in memory search, the model is traditionally applied to these effects separately, with distinct mechanisms employed to explain each. Repetition effects are often modeled through the dynamics of temporal context, where items reinstate contextual states that are largely orthogonal (unrelated) to those associated with other items (L. L. Lohnas & Kahana, 2014). By contrast, variants of CMR that address semantic organizational effects often differ in how to represent pre-existing rela-

tionships between items, with some models positing a secondary layer of semantic context that gradually integrates shared features among related items (Polyn et al., 2009) and others that propose a separate memory matrix for semantic features that is inert until probed alongside episodic contextual associations during retrieval (Morton & Polyn, 2016). Because repetition and semantic effects in free recall are typically studied in isolation, the relationship between the two remains unclear, and the extent to which they can be modeled as points along a continuum is an open question.

To address this gap, the present work seeks to examine this separation by directly testing whether the same contextual dynamics can account for both semantic and repetition effects in free recall. By comparing the impact of repetition and similarity under parallel study conditions, I aim to determine whether these effects can be understood as manifestations of a single underlying process, as posited by the continuum hypothesis. Our proposed experiment will involve participants studying lists that interleave unrelated items with either repeated items or pairs of highly semantically similar items. By varying the spacing between presentations of these key items and manipulating their positions within the list, we will explore how different item relationships affect memory performance and the structure of recall sequences. This design will allow me to directly compare how repetition and semantic similarity influence the temporal organization of memory search, assessing the extent to which they share common underlying mechanisms.

To interpret the results of this experiment, I will combine benchmark analyses with extensions of traditional measures of repetition effects to study non-repeating but similar word pairs. Analyses apply the retrieved context framework to evaluate three key questions: whether encoding of similar or repeated items triggers the reinstatement of similar contextual states in a way that influences recall performance, whether similarity or repetition can modulate the benefit of spaced over massed study for memory performance, and whether these factors similarly interact with serial position to structure response organization. Proposed modeling will help clarify predictions of competing accounts of repetition and semantic effects in memory search, and support formal comparison between accounts in terms of their ability to explain

the observed data across all trial conditions simultaneously. The results of this study will clarify how the retrieved context framework simultaneously addresses repetition effects and semantic organization in free recall and lay groundwork for a more unified account of how item and contextual relationships mutually shape memory search.

Theoretical Framework

The continuum hypothesis posits that the mechanisms underlying repetition and similarity effects in memory search are fundamentally the same, differing only in the degree of similarity between items. To translate this hypothesis into specific, testable predictions, I leverage the Context Maintenance and Retrieval (CMR) model (Polyn et al., 2009), a computational framework designed to interpret the interaction of semantic and temporal contiguity effects in memory search. The CMR model provides a robust account of how these effects shape the organization of recall responses, considering factors such as the serial position of items in the study list, the spacing between repeated items, and the semantic relationships among items. Central to the CMR model is the concept of temporal context, a representation of the evolving state of memory as items are studied and recalled. My investigation of the continuum hypothesis will focus on how temporal context is influenced by repetition and semantic similarity and, in turn, how its dynamics shape the organization of recall responses.

Encoding Variability and Study-Phase Retrieval as Mechanisms of Repetition Effects

While the previous chapter identified some limitations, L. L. Lohnas & Kahana (2014) and my own follow-up analyses evaluated CMR's ability to account for repetition effects in free recall and found it to be broadly successful. The CMR model implements two influential hypotheses about the mechanisms underlying repetition and spacing effects in memory search: encoding variability and study-phase retrieval (L. L. Lohnas & Kahana, 2014).

The encoding variability hypothesis suggests that as the spacing between presentations of an item increases, the context associated with each presentation becomes more distinct, allowing a broader range of contextual cues to prioritize the repeated item during recall (Bower et al., 1972). CMR models this by positing that temporal context evolves over time, gradually integrating the context associated with each item as it is studied. In contrast, CMR's specification of study-phase retrieval posits that when an item is presented for the second time, it reinstates the context from its initial presentation (Greene, 1989; L. L. Lohnas & Kahana, 2014; Maddox, 2016). This process not only reinforces the association between the item and its initial study context but also associates the item with the new contextual information, potentially leading to a more robust and accessible memory trace.

Importantly, while encoding variability can help explain recall performance for unrelated items presented at different intervals (L. J. Lohnas et al., 2011; L. L. Lohnas & Kahana, 2014), study-phase retrieval in CMR is specific to repeated items and may explain the superadditive benefit of repetition for memory performance (Benjamin & Tullis, 2010). This benefit is measured using OR score, which quantifies the probability of recalling either of two once-presented items presented at varying lags in controlled study lists. When compared to recall probabilities for repeated items presented at the same lags, the OR score usually reveals that while spacing reliably benefits recall for unrelated items, it has a more pronounced effect on the recall of repeated items (L. J. Lohnas et al., 2011).

Although the spacing effect dominates much of the literature on repetition effects, L. L. Lohnas & Kahana (2014) identified a key consequence of item repetition for the organization of recall responses. Under retrieved context theory, the contextual features reinstated when an item is studied for the second time are highly similar to those associated with the item's initial presentation. Since context evolves gradually over time, these shared features can persist through study events succeeding presentations of the repeated item. These features of CMR imply that if an item is presented at positions i and j in a study list, the context associated with forward neighbors of the item at positions $i+1$ and $i+2$ should have enhanced similarity to the context associated with items presented at positions $j+1$ and $j+2$. The analysis of

this effect, termed the neighbor contiguity analysis in this proposal, measures the probability of transitioning between +1/+2 neighbors of repeated items in a recall sequence, ignoring recalls of the repeated items themselves. L. L. Lohnas & Kahana (2014) confirmed that CMR predicts a higher probability of transitioning between neighbors of repeated items than between neighbors of unrelated items, and also identified this effect in a free recall dataset. While my preliminary analyses of CMR, reported in the previous chapter, question whether the model's study-phase retrieval mechanisms are necessary to account for this effect, the neighbor contiguity analysis identifies contextual similarity as a key consequence of item repetition and remains a powerful tool for evaluating how similar contexts influence the organization of recall responses.

Semantic Similarity and the Evolution of Context in Memory Search

The continuum hypothesis, which posits that the mechanisms underlying repetition and similarity effects in memory search are fundamentally the same, is not a new idea within the framework of CMR. In fact, it was a driving force behind the development of CMR as a generalization of the Temporal Context Model (TCM) (Howard & Kahana, 2002a; Polyn et al., 2009). CMR extends the principles of TCM by integrating multiple forms of context – temporal, semantic, and otherwise – into a unified framework for understanding how items are organized and retrieved in memory.

According to this framework, the semantic organization of free recall – the tendency for participants to transition between items that are semantically related – arises from how items drive the evolution of temporal context during encoding. Items evolve context by reinstating the context associated with their prior study events, a blend of temporal contextual states across all experiences involving the item. Since semantically similar items historically tend to occur in similar contexts, this implies that two similar items will reinstate similar contextual states during encoding, associating them with one another in memory and favoring their subsequent retrieval together.

To implement this principle, CMR incorporates a secondary layer of contextual features that capture the shared features among related items, such as those within the same semantic category or using the same encoding task (Polyn et al., 2009). While the base layer of temporal context in CMR treats the context associated with each item as largely orthogonal to that of other items, this secondary layer allows for the gradual integration of shared features among related study events. This gradual integration is crucial as it causes the context to become increasingly dominated by category-specific information as similar items are studied in succession, thereby enhancing the likelihood of category-based clustering during recall. This expansion of TCM's context-related mechanisms to encompass various types of context, including semantic and source context, is fundamental to CMR's ability to explain the simultaneous organization of studied material along multiple dimensions.

A key set of observations supporting this framework focuses on the effect of massed versus spaced study of similar items on memory performance. For example, participants exhibit an advantage for recalling items presented in blocks by category compared to when same-category items are randomly distributed throughout a study list (Puff, 1966). EEG measures have also identified category-specific neural responses during the encoding of same-category items in free recall tasks (Morton et al., 2013; Morton & Polyn, 2017). These signals ramped up as same-category items were successively studied but gradually decreased when items from a new category were presented. The strength of this signal predicted whether studied items would be recalled later, as well as whether they would be recalled near other items from the same category. These results are consistent with the gradual integration of category-specific semantic information into context during encoding, facilitating the retrieval of items from the same category. The CMR model explains these findings by positing that the secondary contextual layer increasingly reflects category-specific information as similar items are studied in succession, thereby enhancing category-based clustering during recall.

These mechanisms, supported by the dual-layer structure of context in CMR, have also been leveraged to explain a range of contextual disruption effects in memory search (Polyn et al., 2009; Talmi et al., 2019). Over time, adaptations of CMR have further developed this con-

cept, addressing other sources of similarity that might influence the organization of memory search [e.g., emotionality; Talmi et al. (2019)] and addressing semantic organization in study lists without systematic semantic structure (Morton & Polyn, n.d.). By demonstrating how semantic similarity influences the evolution of context, these findings implement a key component of the continuum hypothesis. However, while CMR's framework suggests that repetition and similarity effects may be manifestations of the same underlying processes, further work is needed to reconcile these findings with the retrieved context account of repetition effects. Notably, the reviewed research implies a benefit for massed over spaced study of semantically similar items, which contrasts with the classic spacing effect for repeated items (Borges & Mangler, 1972; Bousfield, 1953; Glanzer, 1969; Puff, 1966). The present research aims to address this discrepancy by directly comparing the effects of repetition and semantic similarity on memory search within a unified experimental framework, providing a critical test of the continuum hypothesis and clarifying ambiguities in the retrieved context accounts of semantic and repetition effects.

Extending Encoding Variability and Study-Phase Retrieval to Explain Similarity Effects

CMR provides a powerful framework for conceptualizing the continuum hypothesis by explaining both repetition and semantic similarity effects through the mechanisms of context reinstatement and context-cued retrieval. However, integrating repetition and similarity effects under a single account remains an ongoing challenge, with key theoretical and empirical gaps yet to be fully addressed. To advance this integration, I turn to two influential mechanisms in CMR that have traditionally been applied to repetition effects: encoding variability theory and study-phase retrieval. In this section, I explore how these mechanisms might also drive semantic organization in free recall, and I assess the implications of applying them to semantically similar items.

Encoding variability is a well-established mechanism for explaining repetition effects, where

increased spacing between presentations corresponds with greater variability in the context associated with each presentation. This facilitates the recall of repeated items by allowing a broader range of contextual cues to prioritize them during recall (Bower et al., 1972; L. L. Lohnas & Kahana, 2014; Maddox, 2016). Implemented in CMR, this principle successfully explains why spaced repetition generally results in better recall than massed repetition (L. L. Lohnas & Kahana, 2014).

However, a large body of research has shown that, contrary to the spacing effect observed with repetition, semantically similar items often benefit from massed rather than spaced presentation. For instance, participants are more likely to recall items presented in blocks by category compared to when same-category items are distributed throughout a study list (Puff, 1966). Furthermore, while placing a pair of semantically related items in a study list boosts the memorability of both items, the advantage becomes stronger as the items are presented in closer temporal proximity, rather than more distant, contributing to category-based clustering in response sequences (Borges & Mangler, 1972; Glanzer, 1969). This difference in how spacing affects the recall of repeated versus similar items seemingly challenges the continuum hypothesis examined in my proposed and potentially indicates that the mechanisms underlying repetition and similarity effects may be fundamentally distinct. However, CMR has been separately applied to account for both patterns (L. L. Lohnas & Kahana, 2014; Morton & Polyn, n.d.; Polyn et al., 2009), suggesting that the discrepancy may be reconciled within its framework.

To explain the advantage of massed presentation for semantically similar items, CMR incorporates a secondary contextual layer that allows for the gradual integration of shared features among related items (Polyn et al., 2009). As similar items are studied in succession, their associated contexts become increasingly dominated by category-specific information. This increased dominance makes contextual states associated with similar items better cues for other similar items during recall, facilitating semantic clustering. So whereas variability in contextual states is central to accounts of the spacing effect for repeated items, this account of semantic organization emphasizes similarity between contextual states across study events

as a key driver of memory performance. This presents a seeming paradox within CMR, where contextual variability and contextual similarity can each improve the memorability of items in different ways.

A similar tension exists in the retrieved context account of repetition effects, where the model's dual mechanisms implicated in repetition effects – encoding variability and study-phase retrieval – support memory for repeated items in opposing ways. CMR's implementation of the study-phase retrieval account of repetition effects suggests that the context associated with an item's initial presentation is reinstated when the item is studied for the second time (L. L. Lohnas & Kahana, 2014). Instead of supporting contextual variability, this mechanism drives enhanced similarity between contextual states associated with each presentation of a repeated item. As a result forward neighbors of repeated items are more likely to be recalled together than neighbors of unrelated items, reflecting the shared context reinstated by the repeated item (L. L. Lohnas & Kahana, 2014). The mechanism is also proposed to help explain the superadditive benefit of repetition for memory performance, as measured by comparison to the OR score analysis of recall rates for either of two once-presented items (L. J. Lohnas et al., 2011; Ross & Landauer, 1978). In this way, the retrieved context account of repetition effects emphasizes sources of contextual similarity and of contextual variability as key drivers of memory performance, despite their opposing consequences for contextual dynamics.

An analysis of the conditions under which these mechanisms apply seems to help resolve the tension between the roles of contextual variability and contextual similarity in memory search. In CMR, similarity between states of context associated with two study events may primarily improve the conditional probability of recalling information from one event upon recalling information from the other. This is because the context reinstated upon recalling one study event during free recall can prioritize the other study event for retrieval. In contrast, variability between states of context associated with two study events faciliates memory for either study event given an arbitrary contextual retrieval cue. A pair of distinct contextual states can be targetted by a broader range of possible contextual cues, making study items more accessible during free recall in general. Because these advantages are not mutually exclusive, the model

can predict that both contextual similarity and contextual variability can improve memory performance, but in different ways and under different conditions. This analysis matches with simulation results from Polyn et al. (2009), which found that simulating massed study of similarly encoded items in a free recall task weakened the overall probability of recalling applicable items but increased the conditional probability of recalling one grouped item given the recall of the other.

With the effects of contextual similarity and contextual variability understood this way, the inconsistent effects of spacing on the recall of repeated versus similar items can be reconciled. If the benefit of contextual similarity is to improve the conditional probability of recalling one item given the recall of another, then the advantage of massed presentation for semantically similar items can be explained by the increased likelihood of recalling one item upon recalling the other. In contrast, an increased conditional probability of recalling one item given the recall of another does not necessarily benefit memory for repeated items, as once one instance of the repeated item is recalled, there is no further need to recall the other. This dynamic explains why massed presentation can be beneficial for semantically similar items, which are distinct and therefore both need to be recalled during free recall, whereas spaced repetition is more effective for repeated items, where the goal is to retrieve as many unique items as possible.

According to this framework, contextual similarity driven by the study-phase retrieval mechanism primarily benefits memory for item repetitions by making neighboring items better cues for the repeated item via the shared contextual features reinstated by the repetition. However, study-phase retrieval may also benefit memory for semantically similar items, even when not presented in immediate succession. For instance, in research by Jacoby (1974), participants were asked to identify whether a current item belonged to the same semantic category as previously studied items. Results indicated that recall was significantly higher for items recognized as belonging to the same category as an earlier item, even when separated by various lags. This finding supports the idea that semantically similar items can trigger the reinstatement of earlier contextual states, reinforcing their association in memory and providing additional support for

the continuum hypothesis. Together, these results suggest that despite distinct consequences for memory performance, the mechanisms of encoding variability and study-phase retrieval in CMR can be applied to both repetition and similarity effects. The proposed experiment will directly test this unified account of how these factors shape memory search in free recall and offers strong support for the continuum hypothesis that underpins this research.

The Separation of Contextual Layers for Unique and Shared Features

While the CMR model can explain both repetition and semantic similarity effects through similar contextual dynamics, its dual-layer structure introduces a critical distinction that differentiates the mechanisms behind these effects. The contextual features reinstated by an item in context's base layer are largely orthogonal to those associated with other items, capturing the unique aspects of each study event. These features primarily support the temporal organization of memory search, particularly the serial position effect and the lag-contiguity effect. In contrast, the secondary layer of context in CMR encodes the shared features among related items, such as those within the same semantic category or processed with the same encoding task (Polyn et al., 2009). The evolution of these layers is typically parametrized separately, allowing independent adjustment of the integration rate of retrieved contextual features as well as of the strength of associations between items and features in either layer (Morton & Polyn, 2016; Polyn et al., 2009).

This distinction between item-specific and distributional contextual layers addresses key findings on how semantic and temporal contiguity interact in memory search. For instance, variability in the extent to which participants cluster items by semantic similarity or by temporal position in free recall tasks can be accounted for by encoding these associations in separate contextual layers and adjusting the strength of these associations independently. Another key finding is that reinstated features from earlier recall events may persist and continue to influence the organization of later recall events, even as new features associated with other items

are integrated into context over time. This persistence is observed as an increased probability of transitioning to temporal or semantic neighbors of items recalled earlier in a sequence, such as neighbors of the second-to-last item recalled rather than neighbors of the most recently recalled item (L. J. Lohnas & Kahana, 2014). Pivotally, this persistent influence has been found to facilitate temporal clustering more strongly than semantic clustering, suggesting that context's secondary layer may evolve more rapidly than its base layer (Morton & Polyn, 2016). By using different drift rates for different layers of context, the CMR model can capture the interplay between temporal and semantic contiguity effects in memory search, providing a flexible account of how these factors interact to structure recall responses.

In the retrieved context framework, repetition effects in free recall are typically explained in terms of the base layer of context, where the context associated with each presentation of an item is largely orthogonal to that of other items (L. L. Lohnas & Kahana, 2014). Repetition reinstates the context of its initial presentation, deviating from the usual orthogonality of reinstated context, with a range of consequences for memory performance. In contrast, semantic similarity effects are often explained through a secondary layer, where shared features among items can reinstate overlapping contextual states, facilitating their joint retrieval (Polyn et al., 2009). The proposed research provides an opportunity to clarify whether repetition and similarity effects are indeed best understood as supported by distinct contextual layers, or whether they might reflect a more integrated process that unifies these phenomena under a single framework.

Motivation for New Experiment

The Context Maintenance and Retrieval (CMR) model provides a comprehensive framework for understanding how repetition and semantic similarity influence memory search in free recall (Polyn et al., 2009). It asserts that memory performance is mediated by a representation of temporal context that evolves by reinstating and gradually integrating the context associated with items as they are studied. Similar items are assumed to reinstate similar contexts,

which can facilitate the retrieval of related items during recall. Item repetition engage the same mechanisms more robustly, with repeated items reinstating context from their initial presentation and enhancing the likelihood of recalling neighboring items in the study list. These mechanisms implement a form the continuum hypothesis, which posits that the same processes underlie how repetition and similarity affect memory, differing only in the degree of similarity between items. However, while CMR's major components have been applied to account for both repetition and semantic effects in memory search, these phenomena have rarely been evaluated together under a unified theoretical framework. This research gap motivates the proposed experiment, which aims to directly compare the effects of repetition and semantic similarity on memory search within a single experimental framework, providing a critical test of the continuum hypothesis and clarifying ambiguities in the retrieved context account of semantic and repetition effects. I seek a more integrated model of memory search that reinterprets measures of repetition, spacing, and similarity effects in free recall as probes of a shared set of mechanisms and structures, rather than distinct processes.

The proposed experiment is designed to address this gap by directly comparing the effects of item repetition and semantic similarity on free recall performance. This design will allow for a more rigorous test of the continuum hypothesis by examining how these effects manifest under identical study conditions. Specifically, by manipulating the spacing and temporal arrangement of repeated and similar items, the experiment will test whether the contextual dynamics that support contextual variability and study-phase retrieval can account for both effects. In the proposed experiment, participants will study lists that interleave unrelated items with either repeated items or pairs of highly semantically similar items, matching repeated and paired items to the same study positions. Key manipulations include varying the spacing between presentations of these critical items (massed vs. spaced) and their positions within the list (beginning, middle, or end). The primary dependent measures will include recall accuracy and the organization of recalled items, particularly the transitions between them. We thus examine in parallel how repetition and similarity effects influence the temporal organization of memory search, assessing the extent to which they share common underlying mechanisms.

To interpret the results of this experiment, I will combine benchmark analyses with extensions of traditional measures of repetition effects to study non-repeating but similar word pairs. To investigate the extent to which encoding of similar or repeated items triggers the reinstatement of similar context, I will extend the neighbor contiguity analysis developed by L. L. Lohnas & Kahana (2014) to study the transition rates between +1/+2 neighbors of items with high semantic similarity compared to repeated and unrelated items. If similar items reinstate similar contextual states, I would expect to observe a higher probability of transitioning between neighbors of highly similar items than between neighbors of unrelated items. The continuum hypothesis further predicts that this effect should be stronger for repeated items than for similar items, reflecting the greater similarity between contexts associated with repeated items.

I will also examine whether similarity or repetition can modulate the benefit of spaced over massed study for memory performance. Adapting the OR score analysis of recall rates for either of two once-presented items (L. J. Lohnas et al., 2011; Ross & Landauer, 1978), I will compare the effect of spacing on the recall of semantic pairs to the recall of repeated items and to pairs of unrelated items. Our analysis of how contextual similarity and variability processes interact with spacing suggests a complex effect of spacing for memory of similar items, with potential benefits for both massed and spaced presentation. In contrast, we expect a more straightforward benefit of spaced over massed presentation for repeated and even unrelated items, consistent with spacing and OR-score effect observed in previous research (Cepeda et al., 2006; Ebbinghaus, 1885; L. J. Lohnas et al., 2011). Combined with modeling, these analyses will enable us to examine whether the same mechanisms that drive the superadditive benefits of spaced repetition can also explain the enhanced recall of massed, semantically similar items.

Finally, I will investigate how semantic similarity and repetition interact with serial position to structure response organization. To this end, I will apply variations of the lag and distance conditional response probability (lag-CRP and dist-CRP) analyses to assess how semantic and temporal contiguity effects manifest and interact in free recall for repeated and similar items. In particular, I will explore how contextual evolution may differentially shape recall sequences

for repeated and similar items by examining the predictive power of earlier recalls—not just the immediately preceding one—on the current recall, with respect to both temporal and semantic contiguity. These analyses together will help evaluate competing computational models of the relationship between semantic and repetition effects in memory search, deciding whether these effects are best understood as reflecting distinct or overlapping processes.

Proposed Experiment

Methods

The proposed research aims to address these questions by adding a parallel experimental condition to the proposed experiment described in the first chapter of this proposal which itself extends the L. L. Lohnas & Kahana (2014) study of repetition effects in free recall. In the experiment described in my last chapter for studying repetition effects, I proposed that participants would engage in a free recall task where study lists of length 32 are constructed to include either 32 unique items or a mix of 20 unique items and 6 repeated items. I additionally proposed controlling the spacing between repeated items to vary 2 to 8 positions apart and also the initial position of the repeated item in the list to occur near the beginning, middle, or end of the list. The proposed experiment for this chapter adds two new conditions to the design where participants will study lists of length 40 that include pairs of items with high or moderate semantic similarity. Analyzing these conditions together will allow for a direct comparison of how repetition and similarity effects interact with serial position and other sources of temporal organization to support free recallcompared to a control condition of unique, semantically unrelated items.

Participants

The additional conditions for the experiment will involve approximately 35 participants, a sample size consistent with that used in the study by L. L. Lohnas & Kahana (2014), which

forms the basis for this research. This sample size was found to be sufficient in detecting significant effects in free recall tasks involving similar manipulations. By using a comparable sample size, I aim to ensure that my study is adequately powered to detect the effects of item repetition and semantic similarity on memory search, while also allowing for meaningful comparisons with the findings from previous research. Where possible, the same participants will be recruited for both the repetition and similarity conditions to facilitate within-subject comparison of the effects of these factors on memory search. If so, participants will complete the four conditions across the two proposed experiments in two separate 90-minute sessions, with the order of the conditions counterbalanced across participants.

Materials

To the extent possible, word stimuli will be selected from the PEERS word pool, which consists of 1,638 words derived from the University of South Florida free association norms. These are the same stimuli used in the PEERS study, which will allow for direct comparison of the results of the proposed experiment with the findings from the archival dataset. Within a session, no word will occur in more than one list. Words will be presented on a screen one at a time for 2 seconds each, with an 800ms interstimulus interval during which a fixation cross will be displayed This follows established procedures for free recall tasks and will allow for direct comparison of the results of the proposed experiment with the findings from archival datasets. After the study phase, participants will perform immediate free recall of the list, with 60 seconds to recall as many items as possible in any order. Participants will be encouraged to continue recalling until the time limit is reached, promoting thorough retrieval.

Semantic relatedness between all words in the full word pool will be determined using the GloVe embedding similarity metric (Pennington et al., 2014). This metric calculates the cosine similarity between the word embeddings of each pair of words in the pool, with similarity scores ranging from 0 (no semantic similarity) to 1 (identical meaning). I sort similarity scores into bins representing low, moderate, and high semantic similarity:

- Low similarity : 0.0 to 0.3
- Moderate similarity: 0.4 to 0.6
- High similarity: 0.8 to 1.0

Bins may be rescaled to ensure that each bin contains a sufficient number of words for the proposed experiment. If necessary, the word pool will be supplemented using a thesaurus to identify synonyms, or a larger word pool will be used to ensure that each bin contains an adequate number of words. To do this, I will percentile rank the similarity scores in the word pool and define discriminable thresholds for low, moderate, and high similarity based on the distribution of scores. Any additional words will be subject to the same GloVe embedding analysis to ensure they meet the specified similarity thresholds.

Design

To comprise lists in the repetition condition, I defined two types of items: singletons (items presented once) and repeaters (items presented twice). Here I also define moderate- and high-similarity semantic pairs, which consist of two items with moderate or high semantic similarity. Outside semantic pairings, items will be controlled for low semantic similarity to other items in the list (similarity score < 0.3) to minimize unintended semantic associations.

In my proposed repetition experiment, I defined two types of study lists. Control lists consisted entirely of singletons across 32 study positions. Repeater-mixed lists contained a mix of 20 singletons and 6 repeaters across 32 study positions. The spacing between repeaters ranged from 2 to 8 positions apart, with equal probability of each lag within a session. In each list, two primacy repeaters were defined such that the initial presentation of the item always occurred in one of the first five positions of the list. Two recency repeaters were defined such that the second presentation of the item always occurred in one of the last five positions of the list. Finally, two mid-list repeaters were defined such that the second presentation of the item always occurred in one of the middle ten positions of the list. As such, each list contained 2 primacy repeaters, 2 recency repeaters, and 2 mid-list repeaters.

Parallel to this design, I propose two additional conditions to study the effects of semantic similarity on free recall in comparison to the effects of repetition. I define high-similarity mixed lists and moderate-similarity mixed lists, which contain a mix of 20 singletons and 6 high- or moderate-similarity semantic pairs across 40 study positions. Following the structure of repeater-mixed lists, each list will contain 2 primacy semantic pairs, 2 recency semantic pairs, and 2 mid-list semantic pairs, with application positions randomized within these constraints.

If participants are completing both the repetition and similarity conditions, study positions for repeaters and semantic pairs will be matched across conditions. Spanning control, repeater-mixed, high-similarity mixed, and moderate-similarity mixed lists, the proposed experiment will involve 48 lists per participant across two 90-minute sessions. This matching ensures that each trial in the repetition condition has a corresponding trial in the similarity condition with the same study positions for associated study events. If participants only complete one session, for the current research question, I will focus data collection on the high-similarity mixed lists and the repeater-mixed lists, with 24 lists per participant presented in a single 90-minute session.

Archival Dataset

The proposed experiment will be conducted alongside analyses of an archival dataset from the Penn Electrophysiology of Encoding and Retrieval Study (PEERS). In the applicable experiment, participants performed a delayed free recall task over 24 experimental sessions, each containing 24 lists of 24 words sampled from a pool of 1638 used across all PEERS experiments and derived from the University of South Florida Free Association Norms. Across sessions, lists were constructed such that varying degrees of semantic similarity occurred at adjacent and non-adjacent positions within lists. The word lists were constructed to include 16 semantically controlled words per list, derived from four similarity bins (low, medium-low, medium-high, high) using Word Association Space (WAS) scores. Each bin contributed two pairs of words:

one pair presented adjacently, and the other pair separated by at least two intervening items. The remaining eight words in each list were randomly selected and positioned. This dataset enables the investigation of how semantic similarity and temporal spacing interact in free recall, providing a foundation for the proposed experiment. By more systematically manipulating spacing and similarity between item pairs in the proposed experiment and pairing the design with a control condition, the proposed experiment enables more highly powered and controlled analyses of the effects of these factors on memory search.

Proposed Analyses

The goal of this project is to examine the extent to which semantic similarity effects in free recall are distinct from repetition effects or reflect a continuum over a single underlying process. To address this, I will extend core analyses of repetition effects to non-repeating but semantically similar word pairs, allowing a direct comparison of how different types of item relationships influence recall performance and the organization of recall responses. I will perform the same analyses as described in the previous chapter for the repetition condition, including recall performance, lag-contiguity, and pair-neighbor lag-contiguity analyses. These analyses can be extended to study memory for arbitrary pairs of items by treating the pairs as the points of reference for analyses that normally focus on repeated items. Rather than recapitulating the details of all analyses presented in the previous chapter, this section will focus on the most critical and novel analyses that directly inform the proposed research question.

Spacing Effects for Repeated, Similar, and Unrelated Items

To investigate how spacing influences the recall of repeated versus semantically similar items, I will conduct a detailed analysis of recall probability as a function of lag between item presentations. This analysis will focus on comparing the spacing effect across three conditions: repeated items, moderate-similarity item pairs, high-similarity pairs, and unrelated pairs.

For repeated items, I will calculate the recall rate as a function of the lag between their presentations, providing a direct measure of the spacing effect in this condition. For pairs of distinct items (moderate- and high-similarity pairs), I will use the OR score, which measures the probability of recalling either item in a pair as a function of the lag between their presentations. In control lists, unrelated items will be paired to match the study positions of the semantic pairs in the similarity conditions. Following the approach proposed in this previous chapter, for 100 iterations, matchings between control list trials and study positions from an experimental condition can be repeatedly sampled with replacement. This approach controls for the effect of serial position in our comparison of the effects of spacing across conditions, improving the interpretability of the results.

To explore how serial position influences these effects, I will conduct the recall probability by lag analysis separately for pairs presented at different positions within the list (primacy, middle, and recency). This will determine whether the impact of spacing on recall varies depending on the pair's position in the study sequence, offering insight into the interaction between serial position and spacing in influencing memory performance. This approach will also test whether these interactions are consistent across different types of item relationships.

If the data for any variation of these analyses are insufficient to estimate recall rates for specific lags, I will aggregate lags into broader intervals (e.g., 1-2, 3-4, 5-6) to ensure each lag bin contains a sufficient number of observations for reliable estimation.

Effects of Context Reinstatement for Repeated and Similar Items

To examine the extent to which encoding of repeated or semantically similar items triggers the reinstatement of similar contextual states, I will employ the neighbor contiguity analysis developed by L. L. Lohnas & Kahana (2014), extending it to examine transitions between neighbors of semantically similar items. For two items at study positions i and j, this analysis focuses on the probability of transitioning between the items at $i+1/i+2$ to items at $j+1/j+2$ during recall. For repeated items, the analysis measures the likelihood of recalling a neighbor

of the first presentation followed by a neighbor of the second presentation, and vice versa. The extension of this analysis to semantically similar item pairs involves measuring transitions between neighbors of each item in the pair. Critically, this analysis does not directly examine recalls of semantically paired items but instead tests a prediction about how semantic similarity shapes the evolution of context during the study phases of list learning experiments to influence transitions during free recall.

To establish a baseline for comparison, I will define matching item pairs in control lists, selecting unrelated items that align with the study positions of the repeated or similar pairs. The baseline transition rate between $i+1/i+2$ to $j+1/j+2$ neighbors of these control pairs allows for a more accurate assessment of the specific effects of context reinstatement on transitions within the experimental conditions, controlling for potential influences of serial position or general recall tendencies.

To explore how serial position influences these effects, I will conduct the neighbor contiguity analysis separately for pairs presented at different positions within the list (primacy, middle, and recency). I will also examine the interaction of spacing with the neighbor contiguity effect by comparing the transition rates between neighbors of paired study events presented at different lags. Similar to the goals of our spacing analysis, this approach will provide insight into how the effects of context reinstatement vary depending on the spacing and serial position of the paired study events.

Measuring Semantic Clustering in Free Recall

To investigate how semantic similarity influences the organization of recall responses, I will employ the Distance-Conditional Response Probability (Distance-CRP) analysis. This analysis measures the probability of transitioning from one recalled item to another based on their semantic similarity, providing a detailed examination of similarity-based clustering in free recall. The Distance-CRP parallels the lag-CRP analysis, which measures the temporal

organization of recall by evaluating transitions based on the temporal lag between study positions of recalled items. While the lag-CRP analysis has been well established in previous work to measure the lag-contiguity effect, the Distance-CRP specifically focuses on semantic contiguity, providing complementary insights into how semantic similarity guides memory search during free recall.

The Distance-CRP is calculated by first determining the semantic distance between each pair of items in a recall sequence. These distances are then assigned to predefined distance bins, such as 0 to 0.3, based on the range within which the semantic distance falls. For each transition in a recall sequence, the analysis tracks the actual distance bin transitioned to as the numerator value and the possible distance bins that were available based on items not yet recalled as the denominator value. This approach enables the analysis to capture the relative likelihood of transitioning to items that are semantically close versus those that are more distant.

In this project, the Distance-CRP analysis will be used to assess the extent of semantic contiguity in the data, which is critical for testing the continuum hypothesis. The baseline result of observing a semantic contiguity effect – where participants are more likely to transition to semantically similar items – will be essential to achieve in pilot testing. Demonstrating this effect in our data is important, as it sets the stage for exploring a unified framework that can account for both semantic contiguity and traditional repetition effects within the same model of memory search. Other analysis of semantic organization in free recall largely extend or specialize the Distance-CRP analysis to examine the effects of different conditions on semantic clustering in recall responses.

Measuring the Persistence of Contextual Influence in Memory Search

To investigate how the context reinstated by earlier recall events influences the organization of subsequent recall events, I will perform shifted lag-CRP and shifted dist-CRP analyses. These analyses extend traditional conditional response probability measures by examining the predictive power of earlier recalls—specifically, the next-to-last recalled item—on the current

recall. In the shifted lag-CRP analysis, the temporal lag between the next-to-last item and the current recall is used to determine the likelihood of transitioning based on temporal proximity. Similarly, the shifted dist-CRP analysis evaluates transitions based on the semantic similarity between the next-to-last recalled item and the current recall, assigning these transitions to predefined semantic distance bins.

This approach is particularly valuable for measuring the persistence of contextual influence during memory search. While the neighbor contiguity analysis, discussed previously, tests the persistence of contextual features during the encoding phase by examining transitions between the neighbors of paired study events, the shifted lag-CRP and dist-CRP analyses focus on the retrieval phase. They specifically measure whether and how the context reinstated by earlier recall events—beyond the most recent recall—continues to shape subsequent recall behavior.

Expected Outcomes and Significance

Our experiment seeks to explore the continuum hypothesis by systematically examining the extent to which repetition and semantic similarity effects in memory search can be understood as points along a single underlying process. The proposed analyses will dissect these effects by focusing on how different types of item relationships – repetition, high similarity, moderate similarity, and unrelated – interact with the temporal organization of memory search, as well as how these effects are modulated by factors such as spacing and serial position.

First, the lag-CRP and OR score analyses will provide a detailed characterization of how spacing influences the recall probability for repeated and semantically similar items, allowing us to directly compare these effects and assess whether they follow similar or divergent patterns. Second, the neighbor contiguity analysis, extended to include similar pairs, will test the hypothesis that both repetition and similarity trigger the reinstatement of similar contextual states during encoding, thereby influencing the organization of recall responses in a comparable manner. Finally, the shifted lag-CRP and dist-CRP analyses will investigate whether the contextual influence established during earlier recall events persists in structuring subsequent recall events, and whether this persistence varies between repeated and semantically similar items. Together, this approach will help identify the extent to which repetition and similarity effects can be unified under a common theoretical framework, clarify the conditions under which these effects may diverge, and provide a critical evaluation of the CMR model's ability to account for these phenomena within a single integrated system.

In the following, we discuss more concretely the empirical and theoretical outcomes that we

expect from the proposed experiment. We anticipate that the analyses will reveal both commonalities and distinctions between repetition and similarity effects, providing support and challenges for the continuum hypothesis while also highlighting areas where modifications to the CMR model may be necessary. Specifically, we expect that similar contextual dynamics will account for both effects, but with important differences in how these dynamics manifest and interact with other factors such as spacing. The outcomes of these analyses will allow us to refine the CMR model's predictions, potentially leading to a more comprehensive and integrated account of how item and contextual relationships shape memory search in free recall.

Replication of Benchmark Effects

Before proceeding with the full experiment, it is essential to confirm that our experimental design successfully replicates key benchmark effects in memory search. These effects serve as foundational phenomena that our research seeks to clarify, and their replication in piloting will ensure that our design is robust and capable of addressing the proposed research questions.

A critical benchmark effect we aim to replicate is the spacing effect for item repetitions (Cepeda et al., 2006; Ebbinghaus, 1885). This effect demonstrates that memory performance improves when repetitions of the same item are spaced apart rather than massed together. In our experiment, we expect to observe a higher recall probability for items repeated with greater lags between presentations, confirming the robustness of this phenomenon in our study design and providing a basis for exploring its underlying mechanisms.

Another key effect we aim to replicate is the super-additive benefit of item repetition, where repeated items show enhanced recall performance compared to unrelated items. We will measure this by comparing the recall rate for repeated items to the OR scores (the rate of recall for at least one of two considered items) calculated for pairs of unrelated items, as in previous research (L. J. Lohnas et al., 2011; Ross & Landauer, 1978). We expect to observe that the

recall rate for repeated items exceeds the OR score for unrelated items across all lag conditions, indicating a higher likelihood of recalling repeated items together. This super-additive benefit may reflect an interaction between memory traces formed during the initial and subsequent presentations of the repeated item, and our study seeks to clarify how this interaction manifests under our experimental conditions.

The repetition neighbor contiguity effect, identified by L. L. Lohnas & Kahana (2014), is another key effect we aim to replicate. This effect shows that for a repeated item studied at positions i and j, the probability of transitioning from a $+1/+2$ neighbor of i to a $+1/+2$ neighbor of j is higher than if unrelated items are studied at these positions. Within the retrieved context framework, this enhanced transition rate reflects the reinstatement of similar contextual states during each encoding of the repeated item. By replicating this effect, we will confirm that our experimental design is sensitive to the contextual dynamics understood to underlie repetition effects in memory search.

We also aim to replicate the semantic contiguity effect, where participants are more likely to transition from one item to another if the items are semantically similar. This effect is central to our investigation of similarity effects in memory search, and its replication will validate our ability to measure and analyze how semantic relationships influence recall organization. Similarly, we predict that our design will replicate the finding that massed presentation of semantically similar items facilitates recall for those items and promotes clustered recall between them (Borges & Mangler, 1972; Glanzer, 1969; Puff, 1966). Specifically, we expect to observe higher transition rates between paired items in our high-similarity and moderate-similarity conditions compared to unrelated items in control lists. Replicating this effect will support our exploration of whether the distinct effects of spacing on repeated versus similar items can be reconciled within a unified framework of memory search.

Finally, we seek to replicate more general effects of serial position on memory search, such as the primacy and recency effects, which are well-established in the literature (Murdock Jr, 1962). These effects reflect the influence of item position within a study list on recall probability and are critical for understanding how temporal organization shapes memory search. Reproducing

both the primacy and recency effects is necessary to explore their interaction with repetition and similarity effects in our study. In the same vein, we expect to replicate the lag-contiguity effect, where participants are more likely to transition to items studied at adjacent positions in the list, reflecting the influence of temporal contiguity on recall organization. Clarifying how semantic and temporal contiguity effects interact in memory search is a key goal of our research, and replicating these benchmark effects will provide a strong foundation for this exploration.

Similar Items Reinstate Similar Contextual States

A key prediction of the continuum hypothesis is that the mechanisms underlying repetition and similarity effects in free recall should operate in a consistent manner across different types of item relationships. One way to test this prediction is through measurement of neighbor contiguity effect, originally identified by L. L. Lohnas & Kahana (2014), which demonstrates that for a repeated item studied at positions i and j, the probability of transitioning from a $+1/+2$ neighbor of i to a $+1/+2$ neighbor of j is higher than if unrelated items are studied at these positions. Within the retrieved context framework, this effect is interpreted as evidence that the contextual state reinstated by each presentation of the repeated item is similar, and that this similarity influences the organization of recall by promoting transitions between items that share similar contextual associations.

To test whether this mechanism extends to semantically similar items, I will adapt the neighbor contiguity analysis to examine transitions between the neighbors of paired items in two conditions where items are either moderately or highly semantically similar. In a third control condition, where pairs consist of unrelated items, I will conduct the same analysis to establish a baseline transition rate, matching the study positions of the unrelated pairs to those of the semantically similar pairs. According to the continuum hypothesis, the strength of the neighbor contiguity effect should scale with item similarity: the effect should be strongest for repeated items, present but weaker for semantically similar items, and absent for unrelated item pairs.

This outcome would provide empirical support for the idea that similar contextual dynamics underlie both repetition and similarity effects, validating the continuum hypothesis.

Additionally, I will explore whether the strength of the neighbor contiguity effect is correlated across conditions within subjects. Specifically, I predict that subjects who exhibit a strong neighbor contiguity effect in item repetition trials will also show a relatively strong effect in similarity pair trials. This analysis will provide insight into whether the mechanisms supporting repetition and similarity effects are truly integrated or whether they operate independently within the same individual. Finding a positive correlation would bolster the continuum hypothesis, suggesting that these effects share common underlying processes across different types of item relationships. Conversely, finding no correlation might challenge the hypothesis, implying that even if the mechanisms are similar, they might function separately for different types of item relationships. Differences in contextual drift rate or the learning rate for context-to-item associations for repeated versus similar items could explain such a discrepancy (Murdock Jr, 1962), and this analysis will help to clarify the extent to which these factors influence memory search.

By examining the neighbor contiguity effect across these conditions, the proposed research aims to clarify whether the reinstatement of similar contexts during encoding is a general mechanism that supports both repetition and similarity effects in free recall. The expected outcomes of these analyses will provide crucial evidence for or against the continuum hypothesis, helping to refine our understanding of how contextual dynamics shape memory search across different types of item relationships.

A U-shaped Spacing Effect for Semantic Similarity

A central aim of this research is to resolve the tension between reported results concerning the influence of spacing on memory for similar versus repeated items. The classic spacing effect, which applies to both repeated items and unrelated pairs, demonstrates that spaced study of two items or the same item typically benefits memory more than massed study (Cepeda

et al., 2006; Ebbinghaus, 1885). This effect is well-explained by the mechanism of encoding variability as implemented in CMR, where greater spacing between presentations leads to more distinct contextual states being associated with each presentation, thereby improving recall (L. L. Lohnas & Kahana, 2014). CMR also posits that the study-phase retrieval mechanism, which reinstates the context of an item's initial presentation when it is repeated, can enhance memory performance by promoting clustering between the repeated item and each of its study positions' neighbors (L. L. Lohnas & Kahana, 2014).

However, research on the interaction of temporal and semantic contiguity effects in free recall has demonstrated a contrasting pattern: semantically similar items often benefit from massed rather than spaced presentation (Borges & Mangler, 1972; Glanzer, 1969; Puff, 1966). This phenomenon is frequently observed in the form of enhanced recall and semantic clustering of same-category items when they are presented in temporal proximity compared to when they are spaced apart or distributed randomly in study lists. These differential effects of spacing on memory for repeated versus similar items challenge the notion that a single set of mechanisms underlies both types of effects, as posited by the continuum hypothesis.

In my review of the proposed research's theoretical framework, I explored how this apparent contradiction might be reconciled within the CMR model. The model suggests that contextual similarity and contextual variability benefit memory performance in different ways. Contextual variability benefits memory because it allows distinct contextual states to be associated with each presentation. This makes it so that a broader range of contextual states can serve as good retrieval cues for the item(s), increasing the likelihood of successful recall. This enacts a relatively broad-based enhancement to the accessibility of items during free recall. In contrast, similarity between contextual states associated with study events can benefit memory by improving the conditional probability of recalling one item given the recall of another. If two items are associated with similar contextual states, then recalling one item reinstates a context that is likely to support the recall of the other item. This enacts a relatively conditional enhancement to the accessibility of items during free recall: the benefit of recalling one item given the recall of another.

The study-phase retrieval mechanism that supports repetition effects in free recall is a specific instance of this general principle. A context similar an item's initial presentation is reinstated when the item is repeated, and persists in successive study events to increase the similarity between the contexts associated with neighbors of the repeated item's study positions. Massed presentation of similar items may benefit memory by promoting the reinstatement of similar contextual states in succession, causing the shared contextual information to dominate in the contextual representation associated with the similar items. This enhanced similarity between context representations and increase dissimilarity to other contexts in the list may facilitate clustering during recall between similar items. Since this benefit is conditional on the recall of one item given the recall of another, it is not relevant to the recall of repeated items, as participants will not need to recall the other study instance of a repeated item once one has been recalled.

This leads to a specific prediction for the proposed experiment: we expect to observe a U-shaped spacing effect for semantically similar items. In this pattern, both massed and highly spaced study conditions should outperform moderate spacing. Massed presentation will facilitate recall by maximizing contextual overlap, while highly spaced presentations may benefit from the same mechanisms that enhance recall for repeated items, such as encoding variability and study-phase retrieval. In contrast, moderate spacing might not provide sufficient contextual overlap to support clustered recall, nor sufficient contextual variability to benefit from spaced presentations.

By testing this prediction, the experiment will directly address whether the same mechanisms of context reinstatement and variability can account for both repetition and similarity effects in free recall. A U-shaped spacing effect would suggest that while these mechanisms are shared, their influence varies depending on the degree of item overlap and the spacing between study events. In combination with model simulations that predict the observed pattern of results, this outcome would provide strong evidence in favor of the continuum hypothesis, supporting the idea that repetition and similarity effects are not fundamentally distinct but rather reflect a continuum of contextual processes that can manifest differently depending on

study conditions.

Matched Consequences of Serial Position for Repetition and Similarity

The serial position effect, characterized by enhanced recall for items presented at the beginning (primacy effect) and end (recency effect) of a study list, offers a robust framework for examining how contextual dynamics influence memory search (Murdock Jr, 1962). In our proposed experiment, the strategic placement of paired study events – either early in the list, in the middle, or near the end – enables a direct investigation of how these components of the serial position effect modulate both similarity and repetition effects in free recall.

Primacy and recency effects have traditionally been explained through distinct mechanisms related to attention, rehearsal, and contextual dynamics. The primacy effect is thought to arise from increased attention or rehearsal for early items, leading to stronger learning rates and a subsequent advantage during recall (Morton & Polyn, 2016; Sederberg et al., 2008). In free recall tasks, participants may also initiate retrieval by reinstating the start-of-list context, thereby prioritizing early study instances for recall. On the other hand, the recency effect is primarily attributed to the persistence of contextual features from the last study event, which directly cue the recall of items presented near the end of the list. Mid-list items, lacking these advantages, typically exhibit lower recall probabilities, resulting in the characteristic U-shaped serial position curve (Murdock Jr, 1962).

When considering the continuum hypothesis, these serial position effects introduce important consequences for how repetition and similarity influence memory search. Early and late study instances are expected to benefit from the enhanced contextual cues provided by primacy and recency, which should not only boost the recallability of these items but also facilitate the retrieval of associated study instances. For semantically similar items, this means that recalling an early or late-presented item could promote the retrieval of its semantic associate, effectively

linking distinct study events across the list. Similarly, for item repetitions, recalling either presentation could enhance the likelihood of retrieving the neighbors of either presentation, due to the shared context reinstated during recall.

This conceptualization suggests that the traditional U-shaped serial position curve might be dampened in lists containing item repetitions or semantic structure. The presence of repetitions or semantic pairs creates additional retrieval pathways that could distribute recall across the list, reducing the relative disadvantage of mid-list items. For instance, recalling a mid-list item that is semantically associated with an early or late-presented item might enhance the retrieval of other mid-list items, mitigating the typical drop in recall performance seen in these positions. Through our experiment, we aim to explore these interactions by comparing how serial position effects manifest for repeated items, semantically similar pairs, and unrelated items.

Proposed Modeling

In this section, I will outline the strategy for evaluating a range of model variants within the Context Maintenance and Retrieval (CMR) framework to test the continuum hypothesis, which posits that repetition and similarity effects in memory search are driven by common underlying mechanisms rather than distinct processes. Rather than committing to a single model, the proposed research aims to explore and compare multiple variants of CMR, each representing different assumptions about how repetition and similarity effects are encoded, stored, and retrieved in memory. We will use the same techniques as in the previous chapter to fit these models to the data and evaluate their ability to account for the observed patterns of recall performance and organization.

Characterizing Contextual Dynamics in Memory Search

The CMR model provides a framework for understanding how temporal and semantic relationships among items influence memory retrieval. At its core, CMR posits that memory search is guided by an evolving state of context, which integrates information about the items as they are encoded. This context is represented in two distinct layers:

The primary layer – we'll term "temporal context" – captures item-specific information, with context reinstated upon encoding each item being orthogonal to that of other items. This layer is intended to capture the temporal relationships between items in a list, reflecting the order in which they were presented. The secondary layer – we'll term "semantic context", though it may also capture other forms of similarity – captures shared features among related items, such as semantic associations. Unlike the primary layer, this layer allows for overlapping contextual states, meaning that items with similar features are more likely to reinstate similar contexts during recall. This layer is intended to capture the semantic relationships between items in a list, reflecting the similarity between their meanings or associations. We use the similarity between items as measured by the GloVe embedding similarity metric to determine the strength of the semantic context between items (Pennington et al., 2014).

Specifications of CMR that use both layers tend to separately parametrize the drift rate of context in each layer, reflecting the rate at which context changes over time to reflect context reinstated by encoded or recalled items. When fitted, models with both layers tend to ascribe a faster drift rate to the semantic layer than to the temporal layer, reflecting the idea that reinstated semantic information is more fleeting and less stable than temporal information, especially during retrieval (Morton & Polyn, n.d., 2016). This could imply that effects of feature overlap between study events predicted in this proposal may be more transient and less robust if their overlap is primarily semantic in nature.

When applied to model repetition effects in free recall, a secondary layer is normally excluded from CMR, as repetition effects are typically understood to be driven by the reinstatement of temporal context during repeated item presentations. When an item is encoded for the second

time during simulation of a free recall task, the temporal context reinstated is a combination of the item's pre-experimentally associated context and the context associated with the first presentation of the item.

The proposed research will evaluate whether item repetition is best conceptualized as a special case of semantic similarity, where the similar context reinstated during repeated item presentations is represented in the semantic rather than temporal layer of context in CMR. According to this framework, in the temporal layer of context, each study event is always treated as reinstating a distinct context, regardless of the similarity between the items presented or whether an item is repeated. By contrast, in the semantic layer of context, the similarity between items determines the extent to which their contexts overlap, with more similar items reinstating more similar contexts during encoding and recall. This approach predicts that modulation of the contextual draft rate in the semantic layer of context will impact both repetition and similarity effects, as the shared context reinstated by repeated items or semantically similar pairs will influence the organization of recall responses. To evaluate competing integrative accounts of repetition and similarity effects, the proposed research will formally compare model variants that differ in how they represent the contextual dynamics underlying these effects.

Evaluating the Need for Separate Contextual Layers

Previous modeling efforts (Morton & Polyn, n.d.; e.g., Morton & Polyn, 2016; Polyn et al., 2009) have strongly argued for the need to maintain distinct contextual layers to account for the diverse range of effects observed in free recall tasks. These layers allow for the independent representation and manipulation of different types of contextual information – temporal in the primary layer and semantic in the secondary layer. This separation has been justified on the grounds that it enables the model to capture the variability in clustering behavior seen across different types of memory tasks. For example, participants may exhibit variability in whether they cluster items based on their temporal proximity in a study list or their semantic relatedness, and CMR's dual-layer structure accommodates this by allowing different forms of

context to exert independent influences on recall (Morton & Polyn, 2016). Participants may also exhibit variability in the extent to which context reinstated from earlier-than-most-recently recalled items influences the recall of subsequent items. The dual-layer structure of CMR allows for the independent representation of context from different study events, enabling the model to capture differences in the persistence of temporal and semantic contextual influence in memory search. The continuum hypothesis can be instantiated within this framework by representing repetition and similarity relationships between study events in the same secondary layer of context, but this approach leaves ambiguous just what kind of information is integrated by the primary layer of context. Rather than item-specific information, the layer might more abstractedly represent the temporal relationships between items. However, the focal claim in CMR's specification is that reinstated item-specific context itself – rather than general temporal information – is the primary source of temporal organization in memory search, suggesting a deeper tension than might be evident in the model's structure (Polyn et al., 2009).

Given the importance of this distinction in previous work, it is critical to evaluate whether maintaining separate contextual layers is indeed necessary or whether alternative approaches could provide a more unified understanding of memory search. One such approach is to explore a single-layer model that modulates the similarity of contextual states based on item similarity. In this model, rather than separating temporal and semantic contexts into distinct layers, a unified context would dynamically adjust the degree of similarity or orthogonality between contextual states, depending on the features of the items being encoded. This could be achieved by introducing a scaling factor that modulates the overlap in reinstated contextual features as a function of item similarity, thereby allowing the model to flexibly account for both repetition and similarity effects within a single framework. If this and related approaches fail to capture the full range of effects observed in the data, it would provide strong evidence in favor of maintaining separate contextual layers in CMR and prompt reconceptualization of what the primary layer of context represents.

Alternative Approaches

A related but distinct approach to characterization of the relationship between temporal and semantic contiguity effects in memory search is to explore how retrieval cues interact during the recall process rather than how they are encoded during study In an evaluation of competing specifications incorporating semantic organization in CMR, Morton & Polyn (2016) identified as the most successful model a variant where temporal and semantic cues act as separate but interacting sources of information that guide memory search. Their model suggests that while temporal contextual cues contact a context-to-item association memory to provide the organization of recall based on the sequence of study events, an additional cue based on the last recalled item contacts an item-to-item semantic association memory to help cluster items based on their relatedness or shared features. The semantic memory contacted under this specification is inert during encoding, held out from contextual dynamics and associative processes that occur during study, and only becomes active during retrieval to support the organization of recall responses. This variant of CMR conceptualized the simultaneous temporal and semantic organization of memory search as supported by a linear combination of these two sources of information, with the relative weighting of each cue parameterized by subject.

This approach is in tension with the continuum hypothesis about repetition and similarity effects because the retrieved context account of repetition effects typically centers on factors of contextual variability and reinstatement that are not applicable in a model that treats semantic organization as arising from a separate retrieval cue and memory structure. Nonetheless, it is a viable alternative to the dual-layer specification of CMR that has been successful in previous work, and it is important to evaluate whether it can provide a more parsimonious account of the data. Straightforward variants of the model can also alternatively accomodate the continuum hypothesis (e.g., by providing an account of repetition effects in terms of cues to semantic memory) or address tensions between repetition and semantic similarity effects in free recall. For example, rather than using a linear combination rule to integrate temporal

and semantic cues, the model could use a multiplicative rule or switching mechanism (Hills et al., 2012) to flexibly adjust the influence of each cue based on the features of the items being recalled. Such an approach could account for the effect of massed presentation on semantic clustering by exaggerating the influence of semantic cues when items are presented in close temporal proximity, thereby promoting the recall of semantically related items. The proposed experiment provides an opportunity to evaluate these and other alternative approaches to modeling memory search in the context of repetition and similarity effects, offering a comprehensive examination of how different assumptions about contextual dynamics influence the organization of recall responses.

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